

# **Basic Stochastic Processes**

Brzezniak & Zastawniak



## Contents

|                                                                 |    |
|-----------------------------------------------------------------|----|
| Chapter 1. Martingale Inequalities and Convergence              | 5  |
| 1. Doob's Martingale Inequalities                               | 5  |
| 2. Doob's Martingale Convergence Theorem                        | 7  |
| 3. Uniform Integrability and $L^1$ Convergence of Martingales   | 9  |
| Chapter 2. Markov Chains                                        | 17 |
| 1. First Examples and Definitions                               | 17 |
| 2. Classification of States                                     | 28 |
| 3. Long-Time Behaviour of Markov Chains: General Case           | 35 |
| 4. Long-Time Behaviour of Markov Chains with Finite State Space | 38 |
| Chapter 3. Stochastic Processes in Continuous Time              | 41 |
| 1. General Notions                                              | 41 |
| 2. Poisson Process                                              | 41 |
| 3. Brownian Motion                                              | 55 |
| Chapter 4. Itô Stochastic Calculus                              | 69 |
| 1. Itô Stochastic Integral: Definition                          | 69 |
| 2. Examples                                                     | 79 |
| 3. Properties of the Stochastic Integral                        | 82 |
| 4. Stochastic Differential and Itô Formula                      | 84 |
| 5. Stochastic Differential Equations                            | 94 |



## CHAPTER 1

# Martingale Inequalities and Convergence

### 1. Doob's Martingale Inequalities

**Proposition 1.1** (Doob's maximal inequality). *Suppose that  $\xi_n, n \in \mathbb{N}$ , is a non-negative submartingale (with respect to a filtration  $\mathcal{F}_n$ ). Then for any  $\lambda > 0$*

$$\lambda P \left( \max_{k \leq n} \xi_k \geq \lambda \right) \leq E \left( \xi_n 1_{\{\max_{k \leq n} \xi_k \geq \lambda\}} \right),$$

where  $1_A$  is the characteristic function of a set  $A$ .

**PROOF.** We put  $\xi_n^* = \max_{k \leq n} \xi_k$  for brevity. For  $\lambda > 0$  let us define

$$\tau = \min \{k \leq n : \xi_k \geq \lambda\},$$

if there is a  $k \leq n$  such that  $\xi_k \geq \lambda$ , and  $\tau = n$  otherwise. Then  $\tau$  is a stopping time such that  $\tau \leq n$  a.s. Since  $\xi_n$  is a submartingale,

$$E(\xi_n) \geq E(\xi_\tau).$$

But

$$E(\xi_\tau) = E(\xi_\tau 1_{\{\xi_n^* \geq \lambda\}}) + E(\xi_\tau 1_{\{\xi_n^* < \lambda\}}).$$

Observe that if  $\xi_n^* \geq \lambda$ , then  $\xi_\tau \geq \lambda$ . Moreover, if  $\xi_n^* < \lambda$ , then  $\tau = n$ , and so  $\xi_\tau = \xi_n$ . Therefore

$$E(\xi_n) \geq E(\xi_\tau) \geq \lambda P(\xi_n^* \geq \lambda) + E(\xi_n 1_{\{\xi_n^* < \lambda\}}),$$

It follows that

$$\lambda P(\xi_n^* \geq \lambda) \leq E(\xi_n) - E(\xi_n 1_{\{\xi_n^* < \lambda\}}) = E(\xi_n 1_{\{\xi_n^* \geq \lambda\}}),$$

completing the proof. □

**Theorem 1.2** (Doob's maximal  $L^2$  inequality). *If  $\xi_n, n \in \mathbb{N}$ , is a non-negative square integrable submartingale (with respect to a filtration  $\mathcal{F}_n$ ), then*

$$(4.2) \quad E \left| \max_{k \leq n} \xi_k \right|^2 \leq 4E|\xi_n|^2.$$

**PROOF.** Put  $\xi_n^* = \max_{k \leq n} \xi_k$ . By Exercise 1.9, Proposition 1.1, the Fubini theorem and finally the Cauchy-Schwarz inequality

$$\begin{aligned} E|\xi_n^*|^2 &= 2 \int_0^\infty t P(\xi_n^* > t) dt \leq 2 \int_0^\infty E(\xi_n 1_{\{\xi_n^* \geq t\}}) dt \\ &= 2 \int_0^\infty \left( \int_{\{\xi_n^* \geq t\}} \xi_n dP \right) dt = 2 \int_\Omega \xi_n \left( \int_0^{\xi_n^*} dt \right) dP \\ &= 2 \int_\Omega \xi_n \xi_n^* dP = 2E(\xi_n \xi_n^*) \leq 2(E|\xi_n|^2)^{1/2} (E|\xi_n^*|^2)^{1/2}. \end{aligned}$$

Dividing by  $(E|\xi_n|^2)^{1/2}$ , we get (4.2). □

The proof of Doob's Convergence Theorem in the next section hinges on an inequality involving the number of upcrossings.

DEFINITION 1.3. *Given an adapted sequence of random variables  $\xi_1, \xi_2, \dots$  and two real numbers  $a < b$ , we define a gambling strategy  $\alpha_1, \alpha_2, \dots$  by putting*

$$\alpha_1 = 0$$

and for  $n = 1, 2, \dots$

$$\alpha_{n+1} = \begin{cases} 1 & \text{if } \alpha_n = 0 \text{ and } \xi_n < a, \\ 1 & \text{if } \alpha_n = 1 \text{ and } \xi_n \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

*It will be called the **upcrossings strategy**. Each  $k = 1, 2, \dots$  such that  $\alpha_k = 1$  and  $\alpha_{k+1} = 0$  will be called an upcrossing of the interval  $[a, b]$ . The upcrossings form a (finite or infinite) increasing sequence*

$$u_1 < u_2 < \dots$$

*The number of upcrossings made up to time  $n$ , that is, the largest  $k$  such that  $u_k \leq n$  will be denoted by  $U_n[a, b]$  (we put  $U_n[a, b] = 0$  if no such  $k$  exists).*

The meaning of the above definition is this. Initially, we refrain from playing the game and wait until  $\xi_n$  becomes less than  $a$ . As soon as this happens, we start playing unit stakes at each round of the game and continue until  $\xi_n$  becomes greater than  $b$ . At this stage we refrain from playing again, wait until  $\xi_n$  becomes less than  $a$ , and so on. The strategy  $\alpha_n$  is defined in such a way that  $\alpha_n = 0$  whenever we refrain from playing the  $n$ -th game, and  $\alpha_n = 1$  otherwise. During each run of consecutive games with  $\alpha_n = 1$  the process  $\xi_n$  crosses the interval  $[a, b]$ , starting below  $a$  and finishing above  $b$ . This is what is meant by an upcrossing. Observe that each upcrossing will increase our total winnings by at least  $b - a$ . For convenience, we identify each upcrossing with its last step  $k$ , such that  $\alpha_k = 1$  and  $\alpha_{k+1} = 0$ . A typical sample path of the upcrossings strategy is shown in Figure 1.

FIGURE 1. Typical paths of  $\xi_n$ ,  $\alpha_n$  and  $\zeta_n$ ; upcrossings are indicated by bold lines

EXERCISE 1.1. *Verify that the upcrossings strategy  $\alpha_n$  is indeed a gambling strategy.*

*Hint: You want to prove that  $\alpha_n$  is  $\mathcal{F}_{n-1}$ -measurable for each  $n$ . Since the upcrossings strategy is defined by induction, a proof by induction on  $n$  may be your best bet.*

SOLUTION 1.1. *Because  $\alpha_1 = 0$  is constant, it is  $\mathcal{F}_0 = \{\emptyset, \Omega\}$ -measurable. Suppose that  $\alpha_n$  is  $\mathcal{F}_{n-1}$ -measurable for some  $n = 1, 2, \dots$ . Then*

$$\{\alpha_n = 0, \xi_n < a\} \cup \{\alpha_n = 1, \xi_n \leq b\} \in \mathcal{F}_n$$

*because  $\mathcal{F}_{n-1} \subset \mathcal{F}_n$  and  $\xi_n$  is  $\mathcal{F}_n$ -measurable. This means that*

$$\alpha_{n+1} = 1_{\{\alpha_n=0, \xi_n < a\} \cup \{\alpha_n=1, \xi_n \leq b\}}$$

*is  $\mathcal{F}_n$ -measurable. By induction it follows that  $\alpha_n$  is  $\mathcal{F}_{n-1}$ -measurable for each  $n = 1, 2, \dots$ , so  $\alpha_1, \alpha_2, \dots$  is a gambling strategy.*

**Lemma 1.4** (Upcrossings Inequality). *If  $\xi_1, \xi_2, \dots$  is a supermartingale and  $a < b$ , then*

$$(b - a)E(U_n[a, b]) \leq E((\xi_n - a)^-).$$

*By  $x^-$  we denote the negative part of a real number  $x$ , i.e.  $x^- = \max\{0, -x\}$ .*

PROOF. Let

$$\zeta_n = \alpha_1(\xi_1 - \xi_0) + \cdots + \alpha_n(\xi_n - \xi_{n-1})$$

be the total winnings at step  $n = 1, 2, \dots$  if the upcrossings strategy is followed, see Figure 1. It will be convenient to put  $\zeta_0 = 0$ . By Proposition 3.1 (one cannot beat the system using a gambling strategy)  $\zeta_n$  is a supermartingale.

Let us fix an  $n$  and put  $k = U_n[a, b]$ , so that

$$0 < u_1 < u_2 < \cdots < u_k \leq n.$$

Clearly, each upcrossing increases the total winnings by  $b - a$ ,

$$\zeta_{u_i} - \zeta_{u_{i-1}} \geq b - a$$

for  $i = 1, \dots, k$ . (We put  $u_0 = 0$  for simplicity.) Moreover,

$$\zeta_n - \zeta_{u_k} \geq -(\xi_n - a)^-.$$

It follows that

$$\zeta_n \geq (b - a)U_n[a, b] - (\xi_n - a)^-.$$

Taking the expectation on both sides, we get

$$E(\zeta_n) \geq (b - a)E(U_n[a, b]) - E((\xi_n - a)^-).$$

But  $\zeta_n$  is a supermartingale, so

$$0 = E(\zeta_1) \geq E(\zeta_n),$$

which proves the Upcrossings Inequality.  $\square$

## 2. Doob's Martingale Convergence Theorem

**Theorem 1.5** (Doob's Martingale Convergence Theorem). Suppose that  $\xi_1, \xi_2, \dots$  is a supermartingale (with respect to a filtration  $\mathcal{F}_1, \mathcal{F}_2, \dots$ ) such that

$$\sup_n E(|\xi_n|) < \infty.$$

Then there is an integrable random variable  $\xi$  such that

$$\lim_{n \rightarrow \infty} \xi_n = \xi \quad \text{a.s.}$$

REMARK 1.1. In particular, the theorem is valid for martingales because every martingale is a supermartingale. It is also valid for submartingales, since  $\xi_n$  is a submartingale if and only if  $-\xi_n$  is a supermartingale.

REMARK 1.2. Observe that even though all the  $\xi_n$  as well as the limit  $\xi$  are integrable random variables, it is claimed only that  $\xi_n$  trends to  $\xi$  a.s. Note that no convergence in  $L^1$  is asserted.

PROOF (Proof of Doob's Martingale Convergence Theorem). By the Upcrossings Inequality (Lemma 1.4)

$$E(U_n[a, b]) \leq \frac{E((\xi_n - a)^-)}{b - a} \leq \frac{M + |a|}{b - a} < \infty,$$

where

$$M = \sup_n E(|\xi_n|) < \infty.$$

Since  $U_n[a, b]$  is a non-decreasing sequence, it follows that

$$E\left(\lim_{n \rightarrow \infty} U_n[a, b]\right) = \lim_{n \rightarrow \infty} E(U_n[a, b]) \leq \frac{M + |a|}{b - a} < \infty.$$

This implies that

$$P\left\{\lim_{n \rightarrow \infty} U_n[a, b] < \infty\right\} = 1$$

for any  $a < b$ . Since the set of all pairs of rational numbers  $a < b$  is countable, the event

$$(4.3) \quad A = \bigcap_{a < b \text{ rational}} \left\{\lim_{n \rightarrow \infty} U_n[a, b] < \infty\right\}$$

has probability 1. (The intersection of countably many events has probability 1 if each of these events has probability 1.)

We claim that the sequence  $\xi_n$  converges a.s. to a limit  $\xi$ . Consider the set

$$B = \left\{\liminf_n \xi_n < \limsup_n \xi_n\right\} \subset \Omega$$

on which the sequence  $\xi_n$  fails to converge. Then for any  $\omega \in B$  there are rational numbers  $a, b$  such that

$$\liminf_n \xi_n(\omega) < a < b < \limsup_n \xi_n(\omega),$$

implying that  $\lim_{n \rightarrow \infty} U_n[a, b](\omega) = \infty$ . This means that  $B$  and the event  $A$  in (4.3) are disjoint, so  $P(B) = 0$ , since  $P(A) = 1$ , which proves the claim.

It remains to show that the limit  $\xi$  is an integrable random variable. By Fatou's lemma

$$\begin{aligned} E(|\xi|) &= E(\liminf_n |\xi_n|) \\ &\leq \liminf_n E(|\xi_n|) \\ &< \sup_n E(|\xi_n|) < \infty. \end{aligned}$$

This completes the proof. □

**EXERCISE 1.2.** Show that if  $\xi_n$  is a non-negative supermartingale, then it converges a.s. to an integrable random variable.

*Hint:* To apply Doob's Theorem all you need to verify is that the sequence  $\xi_n$  is bounded in  $L^1$ , i.e. the supremum of  $E(|\xi_n|)$  is less than  $\infty$ .

**SOLUTION 1.2.** For a non-negative supermartingale

$$\sup_n E(|\xi_n|) = \sup_n E(\xi_n) \leq E(\xi_1) = E(|\xi_1|) < \infty,$$

since

$$E(\xi_n) \leq E(\xi_1)$$

for each  $n = 1, 2, \dots$ . Thus Doob's Martingale Convergence Theorem implies that  $\xi_n$  converges a.s. to an integrable limit.



### 3. Uniform Integrability and $L^1$ Convergence of Martingales

The conditions of Doob's theorem imply pointwise (a.s.) convergence of martingales. In this section we shall study convergence in  $L^1$ . To this end we introduce a stronger condition called uniform integrability. Proposition 1.7 shows that it is a necessary condition for  $L^1$  convergence. In Theorem 1.9 we prove that uniform integrability is in fact sufficient for a martingale to converge in  $L^1$ . This enables us to show that each integrable martingale is of the form  $E(\xi|\mathcal{F}_n)$ . As an application we give a martingale proof of Kolmogorov's 0-1 law.

**EXERCISE 1.3.** *Show that a random variable  $\xi$  is integrable if and only if for every  $\varepsilon > 0$  there exists an  $M > 0$  such that*

$$\int_{\{|\xi| > M\}} |\xi| dP < \varepsilon.$$

*Hint: Split  $\Omega$  into two sets:  $\{|\xi| > M\}$  and  $\{|\xi| \leq M\}$ . The integrals of  $|\xi|$  over these sets must add up to  $E(|\xi|)$ . As  $M$  increases, one of the integrals increases, while the other one decreases. Investigate their limits as  $M \rightarrow \infty$ .*

**SOLUTION 1.3. Necessity.** Suppose that  $\xi$  is integrable. It follows that

$$P\{|\xi| < \infty\} = 1.$$

The sequence of random variables  $|\xi|1_{\{|\xi| > M\}}$  indexed by  $M = 1, 2, \dots$  is monotone and

$$|\xi|1_{\{|\xi| > M\}} \searrow 0 \quad \text{as } M \rightarrow \infty$$

on the set  $\{|\xi| < \infty\}$ , i.e. a.s. By the monotone convergence theorem for integrals

$$\int_{\{|\xi| > M\}} |\xi| dP \searrow 0 \quad \text{as } M \rightarrow \infty.$$

It follows that for every  $\varepsilon > 0$  there exists an  $M > 0$  such that

$$\int_{\{|\xi| > M\}} |\xi| dP < \varepsilon.$$

**Sufficiency.** Take  $\varepsilon = 1$ . There exists an  $M > 0$  such that

$$\int_{\{|\xi| > M\}} |\xi| dP < 1.$$

Then

$$\begin{aligned} E(|\xi|) &= \int_{\Omega} |\xi| dP \\ &= \int_{\{|\xi| > M\}} |\xi| dP + \int_{\{|\xi| \leq M\}} |\xi| dP \\ &< 1 + MP\{|\xi| \leq M\} \\ &\leq 1 + M < \infty. \end{aligned}$$

Thus, for any sequence  $\xi_n$  of integrable random variables and any  $\varepsilon > 0$  there is a sequence of numbers  $M_n > 0$  such that

$$\int_{\{|\xi_n| > M_n\}} |\xi_n| dP < \varepsilon.$$

If the  $M_n$  are independent of  $n$ , then we say that the sequence  $\xi_n$  is uniformly integrable.

**DEFINITION 1.6.** A sequence  $\xi_1, \xi_2, \dots$  of random variables is called **uniformly integrable** if for every  $\varepsilon > 0$  there exists an  $M > 0$  such that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < \varepsilon$$

for all  $n = 1, 2, \dots$

**EXERCISE 1.4.** Let  $\Omega = [0, 1]$  with the  $\sigma$ -field of Borel sets and Lebesgue measure. Take

$$\xi_n = n1_{(0, \frac{1}{n})}.$$

Show that the sequence  $\xi_1, \xi_2, \dots$  is not uniformly integrable.

*Hint:* What is the integral of  $\xi_n$  over  $\{\xi_n > M\}$  if  $n > M$ ?

**SOLUTION 1.4.** For any  $M > 0$  and any  $n > M$  we have

$$(0, \frac{1}{n}) = \{\xi_n > M\},$$

so

$$\int_{\{\xi_n > M\}} \xi_n dP = \int_{(0, \frac{1}{n})} n dP = 1.$$

This means that there is no  $M > 0$  such that for all  $n$

$$\int_{\{\xi_n > M\}} \xi_n dP < \frac{1}{2}.$$

The sequence  $\xi_n$  is not uniformly integrable.

**Proposition 1.7.** Uniform integrability is a necessary condition for a sequence  $\xi_1, \xi_2, \dots$  of integrable random variables to converge in  $L^1$ .

**Lemma 1.8.** If  $\xi$  is integrable, then for every  $\varepsilon > 0$  there is a  $\delta > 0$  such that

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon.$$

**PROOF** (Proof of Lemma 1.8). Let  $\varepsilon > 0$ . Since  $\xi$  is integrable, by Exercise 1.3 there is an  $M > 0$  such that

$$\int_{\{|\xi| > M\}} |\xi| dP < \frac{\varepsilon}{2}.$$

Now

$$\begin{aligned} \int_A |\xi| dP &= \int_{A \cap \{|\xi| \leq M\}} |\xi| dP + \int_{A \cap \{|\xi| > M\}} |\xi| dP \\ &\leq \int_A M dP + \int_{\{|\xi| > M\}} |\xi| dP \\ &< MP(A) + \frac{\varepsilon}{2}. \end{aligned}$$

Let  $\delta = \frac{\varepsilon}{2M}$ . Then

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon,$$

as required. □

**EXERCISE 1.5.** Let  $\xi$  be an integrable random variable and  $\mathcal{F}_1, \mathcal{F}_2, \dots$  a filtration. Show that  $E(\xi | \mathcal{F}_n)$  is a uniformly integrable martingale.

*Hint:* Use Lemma 1.8.

SOLUTION 1.5. In Example 3.4 it was verified that  $\xi_n = E(\xi|\mathcal{F}_n)$  is a martingale. Let  $\varepsilon > 0$ . By Lemma 1.8 there is a  $\delta > 0$  such that

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon.$$

By Jensen's inequality  $|\xi_n| \leq E(|\xi||\mathcal{F}_n)$  a.s., so

$$E(|\xi|) \geq E(|\xi_n|) \geq \int_{\{|\xi_n| \geq M\}} |\xi_n| dP \geq MP\{|\xi_n| > M\}.$$

If we take  $M > E(|\xi|)/\delta$ , then

$$P\{|\xi_n| > M\} < \delta.$$

Since  $\{|\xi_n| > M\} \in \mathcal{F}_n$ , it follows that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP \leq \int_{\{|\xi_n| > M\}} E(|\xi||\mathcal{F}_n) dP = \int_{\{|\xi_n| > M\}} |\xi| dP < \varepsilon,$$

proving that  $\xi_n = E(\xi|\mathcal{F}_n)$  is a uniformly integrable sequence.

PROOF (Proof of Proposition 1.7). Suppose that  $\xi_n \rightarrow \xi$  in  $L^1$ , i.e.  $E|\xi_n - \xi| \rightarrow 0$ . We take any  $\varepsilon > 0$ . There is an integer  $N$  such that

$$n \geq N \implies E|\xi_n - \xi| < \frac{\varepsilon}{2}.$$

By Lemma 1.8 there is a  $\delta > 0$  such that

$$P(A) < \delta \implies \int_A |\xi| dP < \frac{\varepsilon}{2}.$$

Taking a smaller  $\delta > 0$  if necessary, we also have

$$P(A) < \delta \implies \int_A |\xi_n| dP < \varepsilon \quad \text{for } n = 1, \dots, N.$$

We claim that there is an  $M > 0$  such that

$$P\{|\xi_n| > M\} < \delta$$

for all  $n$ . Indeed, since

$$E(|\xi_n|) \geq \int_{\{|\xi_n| > M\}} |\xi_n| dP \geq MP\{|\xi_n| > M\},$$

it suffices to take

$$M = \frac{1}{\delta} \sup_n E(|\xi_n|).$$

(Because the sequence  $\xi_n$  converges in  $L^1$ , it is bounded in  $L^1$ , so the supremum is  $< \infty$ .)

Now, since  $P\{|\xi_n| > M\} < \delta$ ,

$$\begin{aligned} \int_{\{|\xi_n| > M\}} |\xi_n| dP &\leq \int_{\{|\xi_n| > M\}} |\xi| dP + \int_{\{|\xi_n| > M\}} |\xi_n - \xi| dP \\ &\leq \int_{\{|\xi_n| > M\}} |\xi| dP + E(|\xi_n - \xi|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

for any  $n > N$  and

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < \varepsilon$$

for any  $n = 1, \dots, N$ , completing the proof.

EXERCISE 1.6. Show that a uniformly integrable sequence of random variables is bounded in  $L^1$ , i.e.

$$\sup_n E(|\xi_n|) < \infty.$$

Hint: Write  $E(|\xi_n|)$  as the sum of the integrals of  $|\xi_n|$  over  $\{|\xi_n| > M\}$  and  $\{|\xi_n| \leq M\}$ .

SOLUTION 1.6. Because  $\xi_n$  is a uniformly integrable sequence, there is an  $M > 0$  such that for all  $n$

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < 1.$$

It follows that

$$\begin{aligned} E(|\xi_n|) &= \int_{\{|\xi_n| > M\}} |\xi_n| dP + \int_{\{|\xi_n| \leq M\}} |\xi_n| dP \\ &< 1 + MP \{|\xi_n| \leq M\} \\ &< 1 + M < \infty \end{aligned}$$

for all  $n$ , proving that  $\xi_n$  is a bounded sequence in  $L^1$ .

Exercise 1.6 implies that each uniformly integrable martingale satisfies the conditions of Doob's theorem. Therefore it converges a.s. to an integrable random variable. We shall show that in fact it converges in  $L^1$ .

**Theorem 1.9.** Every uniformly integrable supermartingale (submartingale)  $\xi_n$  converges in  $L^1$ .

PROOF. By Exercise 1.6 the sequence  $\xi_n$  is bounded in  $L^1$ , so it satisfies the conditions of Theorem 1.5 (Doob's Martingale Convergence Theorem). Therefore, there is an integrable random variable  $\xi$  such that  $\xi_n \rightarrow \xi$  a.s. We can assume without loss of generality that  $\xi = 0$  (since  $\xi_n - \xi$  can be taken in place of  $\xi_n$ ). That is to say,

$$P \left\{ \lim_n \xi_n = 0 \right\} = 1.$$

It follows that  $\xi_n \rightarrow 0$  in probability, i.e. for any  $\varepsilon > 0$

$$P \{ |\xi_n| > \varepsilon \} \rightarrow 0$$

as  $n \rightarrow \infty$ . This is because by Fatou's lemma

$$\begin{aligned} \limsup_n P \{ |\xi_n| > \varepsilon \} &\leq P (\limsup_n \{ |\xi_n| > \varepsilon \}) \\ &\leq P (\Omega \setminus \{ \lim_n \xi_n = 0 \}) \\ &= 0. \end{aligned}$$

Let  $\varepsilon > 0$ . By uniform integrability there is an  $M > 0$  such that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP \leq \frac{\varepsilon}{3}$$

for all  $n$ . Since  $\xi_n \rightarrow 0$  in probability, there is an integer  $N$  such that if  $n > N$ , then

$$P \left\{ |\xi_n| > \frac{\varepsilon}{3} \right\} < \frac{\varepsilon}{3M}.$$

We can assume without loss of generality that  $M > \frac{\varepsilon}{3}$ . Then

$$\begin{aligned} E(|\xi_n|) &= \int_{\{|\xi_n| > M\}} |\xi_n| dP + \int_{\{M \geq |\xi_n| > \frac{\varepsilon}{3}\}} |\xi_n| dP \\ &\quad + \int_{\{\frac{\varepsilon}{3} \geq |\xi_n|\}} |\xi_n| dP \\ &\leq \frac{\varepsilon}{3} + MP\left\{|\xi_n| > \frac{\varepsilon}{3}\right\} + \frac{\varepsilon}{3}P\left\{\frac{\varepsilon}{3} \geq |\xi_n|\right\} \\ &< \varepsilon \end{aligned}$$

for all  $n > N$ . This proves that  $E(|\xi_n|) \rightarrow 0$ , that is,  $\xi_n \rightarrow 0$  in  $L^1$ .  $\square$

**Theorem 1.10.** Let  $\xi_n$  be a uniformly integrable martingale. Then

$$\xi_n = E(\xi | \mathcal{F}_n),$$

where  $\xi = \lim_n \xi_n$  is the limit of  $\xi_n$  in  $L^1$  and  $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$  is the filtration generated by  $\xi_n$ .

PROOF. For any  $m > n$

$$E(\xi_m | \mathcal{F}_n) = \xi_n,$$

i.e. for any  $A \in \mathcal{F}_n$

$$\int_A \xi_m dP = \int_A \xi_n dP.$$

Let  $n$  be an arbitrary integer and let  $A \in \mathcal{F}_n$ . For any  $m > n$

$$\begin{aligned} \left| \int_A (\xi_n - \xi) dP \right| &= \left| \int_A (\xi_m - \xi) dP \right| \\ &\leq \int_A |\xi_m - \xi| dP \\ &\leq E(|\xi_m - \xi|) \rightarrow 0 \end{aligned}$$

as  $m \rightarrow \infty$ . It follows that

$$\int_A \xi_n dP = \int_A \xi dP$$

for any  $A \in \mathcal{F}_n$ , so  $\xi_n = E(\xi | \mathcal{F}_n)$ .  $\square$

**EXERCISE 1.7.** Show that if  $\xi_n$  is a martingale and  $\xi_n \rightarrow a$  in  $L^1$  for some  $a \in \mathbb{R}$ , then  $\xi_n = a$  a.s. for each  $n$ .

Hint: Apply Theorem 1.10.

**SOLUTION 1.7.** By Theorem 1.10,  $\xi_n = E(a | \mathcal{F}_n)$  a.s. But  $E(a | \mathcal{F}_n) = a$  a.s., which proves that  $\xi_n = a$  a.s.

**Theorem 1.11** (Kolmogorov's 0–1 Law). Let  $\eta_1, \eta_2, \dots$  be a sequence of independent random variables. We define the tail  $\sigma$ -field

$$\mathcal{T} = \mathcal{T}_1 \cap \mathcal{T}_2 \cap \dots,$$

where  $\mathcal{T}_n = \sigma(\eta_n, \eta_{n+1}, \dots)$ . Then

$$P(A) = 0 \text{ or } 1$$

for any  $A \in \mathcal{T}$ .

PROOF. Take any  $A \in \mathcal{T}$  and define

$$\xi_n = E(1_A | \mathcal{F}_n),$$

where  $\mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n)$ . By Exercise 1.5  $\xi_n$  is a uniformly integrable martingale, so  $\xi_n \rightarrow \xi$  in  $L^1$ . By Theorem 1.10

$$E(\xi | \mathcal{F}_n) = E(1_A | \mathcal{F}_n)$$

for all  $n$ . Both  $\xi = \lim_n \xi_n$  and  $1_A$  are measurable with respect to the  $\sigma$ -field

$$\mathcal{F}_\infty = \sigma(\eta_1, \eta_2, \dots).$$

The family  $\mathcal{G}$  consisting of all sets  $B \in \mathcal{F}$  such that  $\int_B \xi dP = \int_B 1_A dP$  is a  $\sigma$ -field containing  $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots$ . As a result,  $\mathcal{G}$  contains the  $\sigma$ -field  $\mathcal{F}_\infty$  generated by the family  $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots$ . By Lemma 2.1 it follows that  $\xi = 1_A$  a.s.

Since  $\eta_n$  is a sequence of independent random variables, the  $\sigma$ -fields  $\mathcal{F}_n$  and  $\mathcal{T}_{n+1}$  are independent. Because  $\mathcal{T} \subset \mathcal{T}_{n+1}$ , the  $\sigma$ -fields  $\mathcal{F}_n$  and  $\mathcal{T}$  are independent. Being  $\mathcal{T}$ -measurable,  $1_A$  is therefore independent of  $\mathcal{F}_n$  for any  $n$ . This means that

$$\xi_n = E(1_A | \mathcal{F}_n) = E(1_A) = P(A) \quad \text{a.s.}$$

Therefore the limit  $\lim_{n \rightarrow \infty} \xi_n = \xi$  is also constant and equal to  $P(A)$  a.s. This means that  $P(A) = 1_A$  a.s., so  $P(A) = 0$  or  $1$ .  $\square$

EXERCISE 1.8. Show that if  $A_n \in \sigma(\xi_n)$  for each  $n$ , then the events

$$\limsup_n A_n = \bigcap_{j \geq 1} \bigcup_{i \geq j} A_i$$

and

$$\liminf_n A_n = \bigcup_{j \geq 1} \bigcap_{i \geq j} A_i$$

belong to the tail  $\sigma$ -field  $\mathcal{T}$ .

Hint: You need to write  $\limsup_n A_n$  and  $\liminf_n A_n$  in terms of  $A_k, A_{k+1}, \dots$  for any  $k$ , that is, to show that  $\limsup_n A_n$  and  $\liminf_n A_n$  will not be affected if any finite number of sets are removed from the sequence  $A_1, A_2, \dots$

SOLUTION 1.8. Observe that

$$\limsup_n A_n = \bigcap_{j \geq k} \bigcup_{i \geq j} A_i$$

for any  $k$ . Since

$$A_i \in \sigma(\xi_i) \subset \mathcal{T}_k$$

for every  $i \geq k$ , it follows that

$$\limsup_n A_n = \bigcap_{j \geq k} \bigcup_{i \geq j} A_i \in \mathcal{T}_k$$

for every  $k$ . Therefore

$$\sup_n A_n \in \mathcal{T}.$$

The argument for  $\liminf_n A_n$  is similar.

EXERCISE 1.9. Use Kolmogorov's 0–1 law to show that in a sequence of coin tosses there are a.s. infinitely many heads.

Hint: Show that the event

$$\{\eta_1, \eta_2, \dots \text{ contains infinitely many heads} \}$$

belongs to the  $\sigma$ -field  $\mathcal{T}$ . Can the probability of this event be 0? The probability of the event

$$\{\eta_1, \eta_2, \dots \text{ contains infinitely many tails} \}$$

should be the same. Can both probabilities be equal to 0? Can you simultaneously have finitely many heads and finitely many tails in the sequence  $\eta_1, \eta_2, \dots$ ?

SOLUTION 1.9. Let  $\eta_1, \eta_2, \dots$  be a sequence of coin tosses, i.e. independent random variables with values  $+1, -1$  (heads or tails) taken with probability  $\frac{1}{2}$  each. Consider the following event:

$$A = \{ \eta_1, \eta_2, \dots \text{ contains infinitely many heads} \}.$$

This event belongs to the tail  $\sigma$ -field  $\mathcal{T}$  because

$$A = \limsup_n A_n,$$

where

$$A_n = \{ \eta_n = \text{heads} \} \in \sigma(\eta_n)$$

(see Exercise 1.8). Thus, by Kolmogorov's 0-1 law  $P(A) = 0$  or  $1$ . However, it cannot be  $0$  because the event

$$B = \{ \eta_1, \eta_2, \dots \text{ contains infinitely many tails} \}$$

has the same probability by symmetry and  $\Omega = A \cup B$  (there must be infinitely many heads or infinitely many tails).





## CHAPTER 2

### Markov Chains

#### 1. First Examples and Definitions

**1.1. Example 5.1.** In some homes the use of the telephone can become quite a sensitive issue. Suppose that if the phone is free during some period of time, say the  $n$ th minute, then with probability  $p$ , where  $0 < p < 1$ , it will be busy during the next minute. If the phone has been busy during the  $n$ th minute, it will become free during the next minute with probability  $q$ , where  $0 < q < 1$ . Assume that the phone is free in the 0th minute. We would like to answer the following two questions.

- (1) What is the probability  $x_n$  that the telephone will be free in the  $n$ th minute?
- (2) What is  $\lim_{n \rightarrow \infty} x_n$ , if it exists?

Denote by  $A_n$  the event that the phone is free during the  $n$ th minute and let  $B_n = \Omega \setminus A_n$  be its complement, i.e. the event that the phone is busy during the  $n$ th minute. The conditions of the example give us

$$(5.1) \quad P(B_{n+1}|A_n) = p,$$

$$(5.2) \quad P(A_{n+1}|B_n) = q.$$

We also assume that  $P(A_0) = 1$ , i.e.  $x_0 = 1$ . Using this notation, we have  $x_n = P(A_n)$ . Then the total probability formula, see Exercise 1.10, together with (5.1)-(5.2) imply that

$$(5.3) \quad x_{n+1} = P(A_{n+1})$$

$$(1) \quad = P(A_{n+1}|A_n)P(A_n) + P(A_{n+1}|B_n)P(B_n)$$

$$(2) \quad = (1-p)x_n + q(1-x_n) = q + (1-p-q)x_n.$$

It's a bit tricky to find an explicit formula for  $x_n$ . To do so we suppose first that the sequence  $\{x_n\}$  is convergent, i.e.

$$(5.4) \quad \lim_{n \rightarrow \infty} x_n = x.$$

The elementary properties of limits and equation (5.3), i.e.  $x_{n+1} = q + (1-p-q)x_n$ , yield

$$(5.5) \quad x = q + (1-p-q)x.$$

The unique solution to the last equation is

$$(5.6) \quad x = \frac{q}{q+p}.$$

In particular,

$$(5.7) \quad \frac{q}{q+p} = q + (1-p-q) \frac{q}{q+p}.$$

Subtracting (5.7) from (5.3), we infer that

$$(5.8) \quad x_{n+1} - \frac{q}{q+p} = (1-p-q) \left( x_n - \frac{q}{q+p} \right).$$

Thus,  $\{x_n - \frac{q}{q+p}\}$  is a geometric sequence and therefore, for all  $n \in \mathbb{N}$ ,

$$(3) \quad x_n - \frac{q}{q+p} = (1-p-q)^n \left( x_0 - \frac{q}{q+p} \right).$$

Hence, by taking into account the initial condition  $x_0 = 1$ , we have

$$(5.9) \quad x_n = \frac{q}{q+p} + \left( x_0 - \frac{q}{q+p} \right) (1-p-q)^n$$

$$(4) \quad = \frac{q}{q+p} + \frac{p}{q+p} (1-p-q)^n.$$

Let us point out that although we have used the assumption (5.4) to derive (5.8), the proof of the latter is now complete. Indeed, having proven (5.9), we can show that the assumption (5.4) is indeed satisfied. This is because the conditions  $0 < p, q < 1$  imply that  $|1-p-q| < 1$ , and so  $(1-p-q)^n \rightarrow 0$  as  $n \rightarrow \infty$ . Thus, (5.4) holds. This provides an answer to the second part of the example, i.e.  $\lim_{n \rightarrow \infty} x_n = \frac{q}{p+q}$ .

**EXERCISE 2.1.** *In the framework of Example 1.1, let  $y_n$  denote the probability that the telephone is busy in the  $n$ th minute. Supposing that  $y_0 = 1$ , find an explicit formula for  $y_n$  and, if it exists,  $\lim_{n \rightarrow \infty} y_n$ .*

**REMARK 2.1. *Hint:*** *This exercise can be solved directly by repeating the above argument, or indirectly by using some of the results in Example 1.1.*

**SOLUTION 2.1.** *Since the telephone must be either free or busy at any time  $n$ , we have  $x_n + y_n = 1$ , where  $x_n$  is the probability that the telephone is free. If  $y_0 = 1$ , then  $x_0 = 0$ . Substituting  $x_0 = 0$  into the formula (5.9) for  $x_n$  derived in Example 1.1, we obtain*

$$x_n = \frac{q}{q+p} + \left( 0 - \frac{q}{q+p} \right) (1-p-q)^n = \frac{q}{q+p} - \frac{q}{q+p} (1-p-q)^n.$$

Therefore,

$$y_n = 1 - x_n = 1 - \frac{q}{q+p} + \frac{q}{q+p} (1-p-q)^n = \frac{p}{q+p} + \frac{q}{q+p} (1-p-q)^n.$$

As  $n \rightarrow \infty$ , since  $|1-p-q| < 1$ , the term  $(1-p-q)^n$  tends to zero, so

$$\lim_{n \rightarrow \infty} y_n = \frac{p}{q+p}.$$

REMARK 2.2. The formulae (5.3) and (5.9) can be written collectively in a compact form by using vector and matrix notation. First of all, since  $x_n + y_n = 1$ , we get

$$(5) \quad x_{n+1} = (1 - p)x_n + qy_n,$$

$$(6) \quad y_{n+1} = px_n + (1 - q)y_n.$$

Hence, the matrix version takes the form

$$(7) \quad \begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} 1 - p & q \\ p & 1 - q \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix}.$$

The situation described in Example 1.1 is quite typical. Often the probability of a certain event at time  $n + 1$  depends only on what happens at time  $n$ , but not further into the past. Example 1.1 provides us with a simple case of a Markov chain. See also the following definition and exercises.

DEFINITION 2.1 (Markov chain). Suppose that  $S$  is a finite or a countable set. Suppose also that a probability space  $(\Omega, \mathcal{F}, P)$  is given. An  $S$ -valued sequence of random variables  $\xi_n$ ,  $n \in \mathbb{N}$ , is called an  $S$ -valued Markov chain or a Markov chain on  $S$  if for all  $n \in \mathbb{N}$  and all  $s \in S$

$$(5.10) \quad P(\xi_{n+1} = s \mid \xi_0, \dots, \xi_n) = P(\xi_{n+1} = s \mid \xi_n).$$

Here  $P(\xi_{n+1} = s \mid \xi_n)$  is the conditional probability of the event  $\{\xi_{n+1} = s\}$  with respect to random variable  $\xi_n$ , or equivalently, with respect to the  $\sigma$ -field  $\sigma(\xi_n)$  generated by  $\xi_n$ . Similarly,  $P(\xi_{n+1} = s \mid \xi_0, \dots, \xi_n)$  is the conditional probability of  $\{\xi_{n+1} = s\}$  with respect to the  $\sigma$ -field  $\sigma(\xi_0, \dots, \xi_n)$  generated by the random variables  $\xi_0, \dots, \xi_n$ .

Property (5.10) will usually be referred to as the Markov property of the Markov chain  $\xi_n, n \in \mathbb{N}$ . The set  $S$  is called the state space and the elements of  $S$  are called states.

**Proposition 2.2.** The model in Example 1.1 and Exercise 2.1 is a Markov chain.

PROOF. Let  $S = \{0, 1\}$ , where 0 and 1 represent the states of the phone being free or busy. First we need to construct an appropriate probability space. Let  $\Omega$  be the set of all sequences  $\omega_0, \omega_1, \dots$  with values in  $S$ . Let  $\mu_0$  be any probability measure on  $S$ . For example,  $\mu = \delta_0$  corresponds to the case when the phone is free at time 0. We shall define  $P$  by induction. For any  $S$ -valued sequence  $s_0, s_1, \dots$  we put

$$(5.11) \quad P(\{\omega \in \Omega : \omega_0 = s_0\}) = \mu_0(\{s_0\})$$

and

$$(5.12) \quad P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n + 1\})$$

$$(8) \quad = p(s_{n+1} \mid s_n)P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n\}),$$

where  $p(s \mid r)$  are the entries of the  $2 \times 2$  matrix

$$(9) \quad \begin{bmatrix} p(0 | 0) & p(0 | 1) \\ p(1 | 0) & p(1 | 1) \end{bmatrix} = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}.$$

*It seems reasonable to expect  $P$  to be a probability measure (with respect to the trivial  $\sigma$ -field of all subsets of  $\Omega$ ). Take this for granted, check only that  $P(\Omega) = 1$ .*

*How would you define the process  $\xi_n, n \in \mathbb{N}$ ? We shall do it in the standard way, i.e.*

$$(5.13) \quad \xi_n(\omega) = \omega_n, \omega \in \Omega.$$

*First we shall show that the transition probabilities of  $\xi_n$  are what they should be, i.e.*

$$(5.14) \quad P(\xi_{n+1} = 1 \mid \xi_n = 0) = p,$$

$$(5.15) \quad P(\xi_{n+1} = 0 \mid \xi_n = 1) = q.$$

*The definition of conditional probability yields*

$$(10) \quad P(\xi_{n+1} = 1 \mid \xi_n = 0) = \frac{P(\xi_{n+1} = 1, \xi_n = 0)}{P(\xi_n = 0)}.$$

*Next, the definition of  $P$  gives*

$$(11) \quad \begin{aligned} P(\xi_{n+1} = 1, \xi_n = 0) &= P(\{\omega \in \Omega : \omega_n = 0, \omega_{n+1} = 1\}) \\ &= \sum_{s_0, \dots, s_{n-1} \in S} P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n-1, \omega_n = 0, \omega_{n+1} = 1\}) \\ &= \sum_{s_0, \dots, s_{n-1} \in S} pP(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n-1, \omega_n = 0\}) \\ &= pP(\xi_n = 0), \end{aligned}$$

*by (5.12). We have proven (5.14). Moreover, (5.15) follows by the same argument. A similar line of reasoning shows that  $\xi_n$  is indeed a Markov chain. For this we need to verify that for any  $n \in \mathbb{N}$  and any  $s_0, s_1, \dots, s_{n+1} \in S$*

$$(12) \quad P(\xi_{n+1} = s_{n+1} \mid \xi_0 = s_0, \dots, \xi_n = s_n) = P(\xi_{n+1} = s_{n+1} \mid \xi_n = s_n).$$

*We have*

$$(13) \quad \begin{aligned} &P(\xi_0 = s_0, \dots, \xi_n = s_n, \xi_{n+1} = s_{n+1}) \\ &= P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n+1\}) \\ &= p(s_{n+1} \mid s_n)P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n\}) \\ &= p(s_{n+1} \mid s_n)P(\xi_0 = s_0, \dots, \xi_n = s_n) \end{aligned}$$

*which, in view of the definition of conditional probability, gives*

$$(14) \quad P(\xi_{n+1} = s_{n+1} \mid \xi_0 = s_0, \dots, \xi_n = s_n) = p(s_{n+1} \mid s_n).$$

On the other hand, by an easy generalization of (5.14) and (5.15)

$$(15) \quad P(\xi_{n+1} = s_{n+1} \mid \xi_n = s_n) = p(s_{n+1} \mid s_n),$$

which proves (5.10).  $\square$

In Example 1.1 the transition probabilities from state  $i$  to state  $j$  do not depend on the time  $n$ . This is an important class of Markov chains.

**DEFINITION 2.3** (Time-homogeneous Markov chain). *An  $S$ -valued Markov chain  $\xi_n, n \in \mathbb{N}$ , is called time-homogeneous or homogeneous if for all  $n \in \mathbb{N}$  and all  $i, j \in S$*

$$(5.16) \quad P(\xi_{n+1} = j \mid \xi_n = i) = P(\xi_1 = j \mid \xi_0 = i).$$

The number  $P(\xi_1 = j \mid \xi_0 = i)$  is denoted by  $p(j \mid i)$  and called the transition probability from state  $i$  to state  $j$ . The matrix  $P = [p(j \mid i)]_{j,i \in S}$  is called the transition matrix of the chain  $\xi_n$ .

**EXERCISE 2.2.** In the discussion so far we have seen an example of a transition matrix,  $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$ . Obviously the sum of the entries in each column of  $P$  is equal to 1. Prove that this is true in general.

**REMARK 2.3. *Hint:*** Remember that  $P(\Omega \mid A) = 1$  for any event  $A$ .

**SOLUTION 2.2.** Let  $P = [p(j \mid i)]_{j,i \in S}$  be a transition matrix, where  $p(j \mid i) = P(\xi_{n+1} = j \mid \xi_n = i)$ . The sum of the entries in the  $i$ -th column is given by

$$\sum_{j \in S} p(j \mid i) = \sum_{j \in S} P(\xi_{n+1} = j \mid \xi_n = i) = P(\xi_{n+1} \in S \mid \xi_n = i).$$

Since the random variable  $\xi_{n+1}$  must take some value in the state space  $S$ , the event  $\{\xi_{n+1} \in S\}$  is the entire sample space  $\Omega$ . Thus,

$$\sum_{j \in S} p(j \mid i) = P(\Omega \mid \xi_n = i) = 1$$

for any  $i \in S$ .

**DEFINITION 2.4** (Stochastic matrix). (1) the sum of the entries in each column is 1, i.e.  $\sum_{j \in S} a_{ji} = 1$  for any  $i \in S$ .

$A$  is called a double stochastic matrix if both  $A$  and its transpose  $A^t$  are stochastic matrices.

**Proposition 2.5.** Show that a stochastic matrix is doubly stochastic if and only if the sum of the entries in each row is 1, i.e.  $\sum_{i \in S} a_{ji} = 1$  for any  $j \in S$ .

**SOLUTION 2.3.** By definition, a stochastic matrix  $A$  is doubly stochastic if  $A^t$  is also a stochastic matrix. Let  $A = [a_{ji}]$ . The sum of the entries in the  $j$ -th column of  $A^t$  is equal to the sum of the entries in the  $j$ -th row of  $A$ :

$$\sum_{i \in S} (A^t)_{ij} = \sum_{i \in S} a_{ji}.$$

If  $A$  is doubly stochastic, this sum must be 1 for all  $j \in S$ . Conversely, if  $\sum_{i \in S} a_{ji} = 1$  for all  $j \in S$ , then  $A^t$  is a stochastic matrix (since its columns sum to 1 and its entries are non-negative), which means  $A$  is doubly stochastic.

EXERCISE 2.3. Show that if  $P = [p_{ji}]_{j,i \in S}$  is a stochastic matrix, then any natural power  $P^n$  of  $P$  is a stochastic matrix. Is the corresponding result true for a double stochastic matrix?

REMARK 2.4. **Hint:** Show that if  $A$  and  $B$  are two stochastic matrices, then so is  $BA$ . For the second problem, recall that  $(BA)^t = A^t B^t$ .

SOLUTION 2.4. Let  $A$  and  $B$  be two stochastic matrices. Let  $C = BA$ . The entries of  $C$  are  $c_{ji} = \sum_k b_{jk} a_{ki}$ . Since  $a_{ki} \geq 0$  and  $b_{jk} \geq 0$ , it follows that  $c_{ji} \geq 0$ . The sum of the entries in each column of  $C$  is

$$\sum_j c_{ji} = \sum_j \sum_k b_{jk} a_{ki} = \sum_k a_{ki} \left( \sum_j b_{jk} \right) = \sum_k a_{ki} \cdot 1 = 1,$$

where we used the fact that  $B$  is stochastic ( $\sum_j b_{jk} = 1$ ) and  $A$  is stochastic ( $\sum_k a_{ki} = 1$ ). Thus  $BA$  is stochastic. By induction, if  $P$  is stochastic, then  $P^n = P \cdot P^{n-1}$  is stochastic for all  $n \in \mathbb{N}$ .

If  $P$  is doubly stochastic, then  $P$  and  $P^t$  are both stochastic. From the result above,  $P^n$  is stochastic. Also,  $(P^n)^t = (P^t)^n$  is a product of stochastic matrices  $(P^t)$ , so  $(P^n)^t$  is also stochastic. Thus  $P^n$  is doubly stochastic.

EXERCISE 2.4. Let  $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$ . Show that

$$(16) \quad P^2 = \begin{bmatrix} 1+p^2-2p+pq & 2q-pq-q^2 \\ 2p-pq-p^2 & 1+q^2-2q+pq \end{bmatrix}.$$

REMARK 2.5. **Hint:** This is just simple matrix multiplication.

SOLUTION 2.5. Using matrix multiplication, we have

$$\begin{aligned} P^2 &= \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix} \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix} \\ &= \begin{bmatrix} (1-p)^2 + qp & (1-p)q + q(1-q) \\ p(1-p) + (1-q)p & pq + (1-q)^2 \end{bmatrix} \\ &= \begin{bmatrix} 1-2p+p^2+pq & q-pq+q-q^2 \\ p-p^2+p-pq & pq+1-2q+q^2 \end{bmatrix} \\ &= \begin{bmatrix} 1+p^2-2p+pq & 2q-pq-q^2 \\ 2p-pq-p^2 & 1+q^2-2q+pq \end{bmatrix}. \end{aligned}$$

We see that there is a problem with finding higher powers of the matrix  $P$ . When multiplying  $P^2$  by  $P$ ,  $P^3$ , and so on, we obtain more and more complicated expressions.

DEFINITION 2.6 ( $n$ -step transition matrix). The  $n$ -step transition matrix of a Markov chain  $\xi_n$  with transition probabilities  $p(j | i)$ ,  $j, i \in S$  is the matrix  $P_n$  with entries

$$(5.17) \quad p_n(j | i) = P(\xi_n = j | \xi_0 = i).$$

EXERCISE 2.5. Find an exact formula for  $P_n$  for the matrix  $P$  from Exercise 2.4.

REMARK 2.6. **Hint:** Put  $x_n = P(\xi_n = 0 \mid \xi_0 = 0)$  and  $y_n = P(\xi_n = 1 \mid \xi_0 = 1)$ . Is it correct to suppose that  $p_n(0 \mid 0) = x_n$  and  $p_n(1 \mid 1) = y_n$ ? If yes, you may be able use Example 1.1 and Exercise 2.1.

SOLUTION 2.6. The entries of  $P_n$  are  $p_n(j \mid i) = P(\xi_n = j \mid \xi_0 = i)$ . From Example 1.1, if  $\xi_0 = 0$  (phone is free), the probability that  $\xi_n = 0$  is  $x_n$  given by (5.9):

$$p_n(0 \mid 0) = \frac{q}{q+p} + \frac{p}{q+p}(1-p-q)^n.$$

Then  $p_n(1 \mid 0) = 1 - p_n(0 \mid 0) = \frac{p}{q+p} - \frac{p}{q+p}(1-p-q)^n$ . From Exercise 2.1, if  $\xi_0 = 1$  (phone is busy), the probability that  $\xi_n = 1$  is  $y_n$ :

$$p_n(1 \mid 1) = \frac{p}{q+p} + \frac{q}{q+p}(1-p-q)^n.$$

Then  $p_n(0 \mid 1) = 1 - p_n(1 \mid 1) = \frac{q}{q+p} - \frac{q}{q+p}(1-p-q)^n$ . Thus, the  $n$ -step transition matrix is

$$P_n = \frac{1}{q+p} \begin{bmatrix} q+p(1-p-q)^n & q-q(1-p-q)^n \\ p-p(1-p-q)^n & p+q(1-p-q)^n \end{bmatrix}.$$

EXERCISE 2.6. You may suspect that  $P_n$  equals  $P^n$ , the  $n$ th power of the matrix  $P$ . This holds for  $n = 1$ . Check if it is true for  $n = 2$ . If this is the case, try to prove that  $P_n = P^n$  for all  $n \in \mathbb{N}$ .

REMARK 2.7. **Hint:** Once again, this is an exercise in matrix multiplication.

SOLUTION 2.7. For  $n = 2$ ,  $p_2(j \mid i) = P(\xi_2 = j \mid \xi_0 = i)$ . By the law of total probability,

$$p_2(j \mid i) = \sum_{k \in S} P(\xi_2 = j \mid \xi_1 = k, \xi_0 = i) P(\xi_1 = k \mid \xi_0 = i).$$

By the Markov property,  $P(\xi_2 = j \mid \xi_1 = k, \xi_0 = i) = P(\xi_2 = j \mid \xi_1 = k) = p(j \mid k)$ . Thus,

$$p_2(j \mid i) = \sum_{k \in S} p(j \mid k) p(k \mid i),$$

which are the entries of  $P^2$ . So  $P_2 = P^2$ . By induction, assume  $P_n = P^n$ . Then

$$p_{n+1}(j \mid i) = \sum_{k \in S} P(\xi_{n+1} = j \mid \xi_n = k, \xi_0 = i) P(\xi_n = k \mid \xi_0 = i) = \sum_{k \in S} p(j \mid k) p_n(k \mid i).$$

In matrix notation, this is  $P_{n+1} = PP_n = PP^n = P^{n+1}$ .

The following is a generalization of Exercise 2.6.

**Proposition 2.7** (Chapman–Kolmogorov equation). Suppose that  $\xi_n, n \in \mathbb{N}$ , is an  $S$ -valued Markov chain with  $n$ -step transition probabilities  $p_n(j \mid i)$ . Then for all  $k, n \in \mathbb{N}$

$$(5.18) \quad p_{n+k}(j \mid i) = \sum_{s \in S} p_n(j \mid s) p_k(s \mid i), \quad i, j \in S.$$

EXERCISE 2.7. Prove Proposition 2.7.

REMARK 2.8. **Hint:**  $p_{n+k}(j \mid i)$  are the entries of the matrix  $P_{n+k} = P^{n+k}$ .

SOLUTION 2.8. The  $n$ -step transition probabilities  $p_m(j \mid i)$  are the entries of the matrix  $P_m = P^m$ . The Chapman-Kolmogorov equation

$$p_{n+k}(j \mid i) = \sum_{s \in S} p_n(j \mid s) p_k(s \mid i)$$

is equivalent to the matrix identity

$$P_{n+k} = P_n P_k.$$

Since  $P_m = P^m$ , we have  $P_{n+k} = P^{n+k} = P^n P^k = P_n P_k$ , which proves the proposition.

EXERCISE 2.8. Suppose that  $S = \mathbb{Z}$ . Let  $\eta_n, n \geq 1$  be a sequence of independent identically distributed random variables with  $P(\eta_1 = 1) = p$  and  $P(\eta_1 = -1) = q = 1 - p$ . Define  $\xi_n = \sum_{i=1}^n \eta_i$  for  $n \geq 1$  and  $\xi_0 = 0$ . Show that  $\xi_n$  is a Markov chain with transition probabilities

$$(17) \quad p(j \mid i) = \begin{cases} p, & \text{if } j = i + 1, \\ q, & \text{if } j = i - 1, \\ 0, & \text{otherwise.} \end{cases}$$

$\xi_n, n \geq 0$ , is called a random walk starting at 0. Replacing  $\xi_0 = 0$  with  $\xi_0 = i$ , we get a random walk starting at  $i$ .

REMARK 2.9. **Hint:**  $\xi_{n+1} = \xi_n + \eta_{n+1}$ . Are  $\xi_n$  and  $\eta_{n+1}$  independent?

SOLUTION 2.9. We have  $\xi_{n+1} = \xi_n + \eta_{n+1}$ . Since  $\eta_{n+1}$  is independent of  $\eta_1, \dots, \eta_n$ , it is independent of  $\xi_0, \dots, \xi_n$ . Thus,

$$\begin{aligned} P(\xi_{n+1} = j \mid \xi_n = i, \xi_{n-1} = s_{n-1}, \dots, \xi_0 = s_0) &= P(\xi_n + \eta_{n+1} = j \mid \xi_n = i, \dots, \xi_0 = s_0) \\ &= P(i + \eta_{n+1} = j \mid \xi_n = i, \dots, \xi_0 = s_0) \\ &= P(\eta_{n+1} = j - i). \end{aligned}$$

Since this depends only on  $i$ , the chain is Markov. The transition probabilities are

$$p(j \mid i) = P(\eta_{n+1} = j - i) = \begin{cases} p & \text{if } j - i = 1 \Rightarrow j = i + 1 \\ q & \text{if } j - i = -1 \Rightarrow j = i - 1 \\ 0 & \text{otherwise} \end{cases}$$

which is what we wanted to show.

EXERCISE 2.9. For the random walk  $\xi_n$  defined in Exercise 2.8 prove that

$$(5.19) \quad P(\xi_n = j \mid \xi_0 = i) = \binom{n}{\frac{n+j-i}{2}} p^{\frac{n+j-i}{2}} q^{\frac{n-j+i}{2}}$$

if  $n + j - i$  is an even non-negative integer, and  $P(\xi_n = j \mid \xi_0 = i) = 0$  otherwise.

REMARK 2.10. **Hint:** Use induction. Note that  $\binom{n}{\frac{n+j-i}{2}} p^{\frac{n+j-i}{2}} q^{\frac{n-j+i}{2}}$  equals 0 if  $|j - i| \geq n + 1$ .

SOLUTION 2.10. Let  $k$  be the number of steps to the right (+1) and  $m$  be the number of steps to the left (-1) in  $n$  trials. Then  $k + m = n$ . The displacement is  $\xi_n - \xi_0 = k - m = k - (n - k) = 2k - n$ . We want to find  $P(\xi_n = j \mid \xi_0 = i)$ , which is the probability that  $2k - n = j - i$ , or  $2k = n + j - i$ . This gives  $k = \frac{n+j-i}{2}$  and  $m = n - k = \frac{n-j+i}{2}$ . For



such  $k$  and  $m$  to exist,  $n + j - i$  must be an even integer between 0 and  $2n$ . Since the steps are independent and have the same distribution, the number of steps to the right follows a binomial distribution  $B(n, p)$ . Thus,

$$P(\xi_n = j \mid \xi_0 = i) = P(k \text{ steps to the right}) = \binom{n}{k} p^k q^{n-k} = \binom{n}{\frac{n+j-i}{2}} p^{\frac{n+j-i}{2}} q^{\frac{n-j+i}{2}}.$$

If  $n + j - i$  is odd, or if  $k < 0$  or  $k > n$ , it is impossible to reach state  $j$  from  $i$  in  $n$  steps, so the probability is 0.

**Proposition 2.8.** ...

EXERCISE 2.10. Prove formula (5.24).

REMARK 2.11. **Hint:** Use the Taylor formula to expand the right-hand side of (5.24) into a power series.

SOLUTION 2.11. Assuming (5.24) refers to the generating function for the return probabilities  $P_{ii}(x) = (1 - 4pqx^2)^{-1/2}$ , we use the binomial series expansion:

$$(1 - z)^{-1/2} = \sum_{k=0}^{\infty} \binom{-1/2}{k} (-z)^k = \sum_{k=0}^{\infty} \frac{(-1/2)(-3/2) \cdots (-(2k-1)/2)}{k!} (-z)^k.$$

This simplifies to

$$(1 - z)^{-1/2} = \sum_{k=0}^{\infty} \frac{1 \cdot 3 \cdots (2k-1)}{2^k k!} z^k = \sum_{k=0}^{\infty} \frac{(2k)!}{2^{2k} (k!)^2} z^k = \sum_{k=0}^{\infty} \binom{2k}{k} \left(\frac{z}{4}\right)^k.$$

Setting  $z = 4pqx^2$ , we obtain

$$P_{ii}(x) = \sum_{k=0}^{\infty} \binom{2k}{k} (pqx^2)^k = \sum_{k=0}^{\infty} \binom{2k}{k} p^k q^k x^{2k}.$$

Comparing this with  $P_{ii}(x) = \sum_n p_n(i \mid i) x^n$ , we see that  $p_{2k}(i \mid i) = \binom{2k}{k} p^k q^k$  and  $p_{2k+1}(i \mid i) = 0$ , which is consistent with the formula derived in Exercise 2.9.

EXERCISE 2.11 (Branching process). On the island Elschenbieden there lives an almost extinct species called Vugiel. Vugiel's males can produce zero, one, two,  $\dots$  male offspring with probability  $p_0, p_1, p_2, \dots$  respectively, where  $p_i \geq 0$  and  $\sum_{i=0}^{\infty} p_i = 1$ . A challenging problem would be to find the Vugiel's chances of survival assuming that each individual lives exactly one year. At this moment, we ask you only to rewrite the problem in the language of Markov chains.

REMARK 2.12. **Hint:** The number of descendants of each male has the same distribution.

SOLUTION 2.12. Let  $\xi_n$  be the number of males in the  $n$ -th generation. Since each individual lives for exactly one year and produces offspring at the end of that year, the population at time  $n + 1$  consists entirely of the offspring of the population at time  $n$ . Thus,

$$\xi_{n+1} = \sum_{j=1}^{\xi_n} X_j^{(n)},$$

where  $X_j^{(n)}$  is the number of offspring produced by the  $j$ -th individual of the  $n$ -th generation. The random variables  $X_j^{(n)}$  are independent and identically distributed with  $P(X_j^{(n)} = k) = p_k$ .

If  $\xi_n = 0$ , the sum is empty and we define  $\xi_{n+1} = 0$ , meaning the species is extinct. The process  $\xi_n$  is a Markov chain on the state space  $S = \{0, 1, 2, \dots\}$  with transition probabilities

$$p(j | i) = P\left(\sum_{k=1}^i X_k = j\right),$$

where  $X_k$  are independent random variables each with the offspring distribution  $\{p_k\}$ .

EXERCISE 2.12. Consider the following two cases:

(1) In Exercise 2.11 suppose that

$$(18) \quad p_m = \binom{N}{m} p^m (1-p)^{N-m}.$$

for some  $p \in (0, 1)$  and  $N \in \mathbb{N}^*$ , where  $\mathbb{N}^* = \{1, 2, 3, \dots\}$ . (Note that  $p_m = 0$  if  $m > N$ .) Show that

$$(5.26) \quad p(j | i) = \binom{Ni}{j} p^j (1-p)^{Ni-j}.$$

Deduce that, in particular,  $p(j | i) = 0$  if  $j > Ni$ .

(2) Suppose that

$$(19) \quad p_m = \frac{\lambda^m}{m!} e^{-\lambda}, m \in \mathbb{N},$$

for some  $\lambda > 0$ . In other words, assume that each  $X_j$  has the Poisson distribution with mean  $\lambda$ . Show that

$$(5.27) \quad p(j | i) = \frac{(\lambda i)^j}{j!} e^{-\lambda i}, j, i \geq 0.$$

REMARK 2.13. **Hint:** If  $X_1$  has the binomial distribution  $P(X_1 = m) = \binom{N}{m} p^m (1-p)^{N-m}$ ,  $m \in \mathbb{N}$ , then there exists a finite sequence  $\eta_1^1, \dots, \eta_N^1$  of independent identically distributed random variables such that  $P(\eta_j^1 = 1) = p$ ,  $P(\eta_j^1 = 0) = 1 - p$  and  $X_j = \eta_1^j + \dots + \eta_N^j = m$ . Hence,  $X_1 + \dots + X_i = \sum_{r=1}^i \sum_{s=1}^N \eta_s^r$ , i.e. the sum of  $Ni$  independent identically distributed random variables with distribution as above. Hence we infer (5.26).

SOLUTION 2.13. (1) The number of offspring  $X_k$  of each individual has the binomial distribution  $B(N, p)$ . The transition probability  $p(j | i)$  is the probability that the sum of  $i$  such independent variables is  $j$ . Since the sum of  $i$  independent  $B(N, p)$  variables is a  $B(Ni, p)$  variable, we have

$$p(j | i) = P\left(\sum_{k=1}^i X_k = j\right) = \binom{Ni}{j} p^j (1-p)^{Ni-j}.$$

Clearly, if  $j > Ni$ , then  $p(j | i) = 0$ .

- (2) If each  $X_k$  has the Poisson distribution with mean  $\lambda$ , then the sum of  $i$  independent such variables has the Poisson distribution with mean  $i\lambda$ . Thus,

$$p(j \mid i) = P\left(\sum_{k=1}^i X_k = j\right) = \frac{(i\lambda)^j}{j!} e^{-i\lambda}.$$

**Proposition 2.9.** The probability of survival in Exercise 2.12, part 2) equals 0 if  $\lambda \leq 1$ , and  $1 - \hat{r}^k$  if  $\lambda > 1$ , where  $k$  is the initial Vugiel population and  $\hat{r} \in (0, 1)$  is a solution to

$$(5.28) \quad r = e^{(r-1)\lambda}.$$

REMARK 2.14. The method presented in the last solution works for any distribution of the variables  $X_j$ . It turns out that the mean value  $\lambda$  of  $X_1$  plays the same role as above. One can show that if  $\lambda \leq 1$ , then the population will become extinct with probability 1, while for  $\lambda > 1$  the probability of extinction is larger than 0 and smaller than 1.

EXERCISE 2.13. On the, now familiar, island of Elschenbieden the question of survival of the Vugiel is a hot political issue. The (human) population of the island is  $N$ . Those who believe that action should be taken in order to help the animals preach their conviction quite convincingly. For if a supporter discusses the issue with a non-supporter, the latter will change his mind with probability one. However, they do so only in face-to-face encounters. Suppose that the probability of an encounter of exactly one pair of humans during one day is  $p$  and that with probability  $q$  this pair is a supporter-non-supporter one. Write down a Markov chain model of this situation. Neglect the probability of two or more encounters during one day.

REMARK 2.15. **Hint:** On each day the number of supporters can either increase by 1 or remain unchanged. What is the probability of the former?

SOLUTION 2.14. Let  $\xi_n$  be the number of supporters on day  $n$ . The state space is  $S = \{0, 1, \dots, N\}$ . On each day, the number of supporters increases by 1 if there is an encounter between a supporter and a non-supporter. Otherwise, the number of supporters remains the same. According to the problem, the probability of an encounter is  $p$ , and given that an encounter occurs, the probability that it is between a supporter and a non-supporter is  $q$ . Thus, the probability that the number of supporters increases by 1 is  $pq$ . The transition probabilities are:

$$\begin{aligned} p(i+1 \mid i) &= pq \quad \text{for } 0 \leq i < N, \\ p(i \mid i) &= 1 - pq \quad \text{for } 0 \leq i < N, \\ p(N \mid N) &= 1. \end{aligned}$$

All other transition probabilities are 0.

EXERCISE 2.14 (Queuing model). A car wash machine can serve at most one customer at a time. With probability  $p$ ,  $0 < p < 1$ , the machine can finish serving a customer in a unit time. If this happens, the next waiting car (if any) can be served at the beginning of the next unit of time. During the time interval between the  $n$ th and  $(n+1)$ st unit of time the number of cars arriving has the Poisson distribution with parameter  $\lambda > 0$ . Let  $\xi_n$  denote the number of cars being served or waiting to be served at the beginning of unit  $n$ . Show that  $\xi_n, n \in \mathbb{N}$ , is a Markov chain and find its transition probabilities.

**REMARK 2.16. *Hint:*** Let  $Z_n, n = 0, 1, 2, \dots$  be a sequence of independent identically distributed random variables, each having the Poisson distribution with parameter  $\lambda$ . Then  $\xi_{n+1} - \xi_n - Z_n$  equals  $-1$  or  $0$ .

**SOLUTION 2.15.** Let  $Z_n$  be the number of cars arriving between time  $n$  and  $n + 1$ .  $Z_n$  are i.i.d.  $\text{Poisson}(\lambda)$ . Let  $Y_n$  be a Bernoulli random variable such that  $Y_n = 1$  if a service is completed (with probability  $p$ ) and  $Y_n = 0$  otherwise. If  $\xi_n = i > 0$ , then one car is being served. At the end of the unit time, this car leaves if  $Y_n = 1$ , and remains if  $Y_n = 0$ . Then  $Z_n$  new cars arrive. So

$$\xi_{n+1} = i - Y_n + Z_n.$$

If  $\xi_n = 0$ , no car is being served, so no car can leave. Thus

$$\xi_{n+1} = Z_n.$$

Since  $Z_n$  and  $Y_n$  are independent of the past,  $\xi_n$  is a Markov chain. The transition probabilities for  $i > 0$  are:

$$\begin{aligned} p(j | i) &= P(i - Y_n + Z_n = j) \\ &= P(Y_n = 1)P(Z_n = j - i + 1) + P(Y_n = 0)P(Z_n = j - i) \\ &= p \frac{\lambda^{j-i+1}}{(j-i+1)!} e^{-\lambda} + (1-p) \frac{\lambda^{j-i}}{(j-i)!} e^{-\lambda} \quad \text{for } j \geq i-1, \end{aligned}$$

and  $p(j | i) = 0$  for  $j < i - 1$ . For  $i = 0$ :

$$p(j | 0) = P(Z_n = j) = \frac{\lambda^j}{j!} e^{-\lambda} \quad \text{for } j \geq 0.$$

**REMARK 2.17.** In the last model we are interested in the behaviour of  $\xi_n$  for large values of  $n$ . In particular, it is interesting to determine whether the limit of  $\xi_n$  or that of  $\mathbb{E}\xi_n$  (as  $n \rightarrow \infty$ ) exists. In Exercise 2.31 we shall find conditions which guarantee the existence of a unique invariant measure and imply that the Markov chain in question is ergodic.

## 2. Classification of States

In what follows we fix an  $S$ -valued Markov chain with transition matrix  $P = [p(j | i)]_{j,i \in S}$ , where  $S$  is a non-empty and at most countable set.

**DEFINITION 2.10 (Recurrent and transient states).** A state  $i$  is called recurrent if the process  $\xi_n$  will eventually return to  $i$  given that it starts at  $i$ , i.e.

$$(5.36) \quad P(\xi_n = i \text{ for some } n \geq 1 \mid \xi_0 = i) = 1.$$

If the condition (5.36) is not satisfied, then the state  $i$  is called transient.

**Theorem 2.11.** Show that for a random walk on  $\mathbb{Z}$  with parameter  $p \in (0, 1)$ , the state  $0$  is recurrent if and only if  $p = 1/2$ . Show that the same holds if  $0$  is replaced by any other state  $i \in \mathbb{Z}$ .

**DEFINITION 2.12 (Communicating states).** We say that a state  $i$  communicates with a state  $j$  if with positive probability the chain will visit the state  $j$  having started at  $i$ , i.e.

$$(5.37) \quad P(\xi_n = j \text{ for some } n \geq 0 \mid \xi_0 = i) > 0.$$

If  $i$  communicates with  $j$ , then we shall write  $i \rightarrow j$ . We say that the state  $i$  intercommunicates with a state  $j$ , and write  $i \leftrightarrow j$ , if  $i \rightarrow j$  and  $j \rightarrow i$ .

EXERCISE 2.15. Show that  $i \rightarrow j$  if and only if  $p_k(j | i) > 0$  for some  $k \geq 1$ .

REMARK 2.18. **Hint:** Recall that  $p_k(j | i) = P(\xi_k = j | \xi_0 = i)$ .

SOLUTION 2.16. By definition,  $i \rightarrow j$  if  $P(\xi_n = j \text{ for some } n \geq 1 | \xi_0 = i) > 0$ . If  $p_k(j | i) > 0$  for some  $k \geq 1$ , then

$$P(\xi_n = j \text{ for some } n \geq 1 | \xi_0 = i) = P\left(\bigcup_{n=1}^{\infty} \{\xi_n = j\} | \xi_0 = i\right) \geq P(\xi_k = j | \xi_0 = i) = p_k(j | i) > 0,$$

so  $i \rightarrow j$ . Conversely, if  $i \rightarrow j$ , then by the subadditivity of probability,

$$0 < P\left(\bigcup_{n=1}^{\infty} \{\xi_n = j\} | \xi_0 = i\right) \leq \sum_{n=1}^{\infty} P(\xi_n = j | \xi_0 = i) = \sum_{n=1}^{\infty} p_n(j | i).$$

For the sum to be positive, there must exist at least one  $k \geq 1$  such that  $p_k(j | i) > 0$ .

EXERCISE 2.16. Show that

- (1)  $i \leftrightarrow i$ ,
- (2) if  $i \leftrightarrow j$  then  $j \leftrightarrow i$ , and
- (3) if  $i \leftrightarrow j, j \leftrightarrow k$  then  $i \leftrightarrow k$ .

In other words, show that  $\leftrightarrow$  is an equivalence relation on  $S$ .

REMARK 2.19. **Hint:** 1) and 2) are obvious. For 3) use the Chapman-Kolmogorov equations.

SOLUTION 2.17. (1)  $i \leftrightarrow i$ : Since  $p_0(i | i) = 1 > 0$ , we have  $i \rightarrow i$ , hence  $i \leftrightarrow i$ .

(2) Symmetry: If  $i \leftrightarrow j$ , then  $i \rightarrow j$  and  $j \rightarrow i$  by definition. This is equivalent to saying  $j \rightarrow i$  and  $i \rightarrow j$ , which means  $j \leftrightarrow i$ .

(3) Transitivity: Suppose  $i \leftrightarrow j$  and  $j \leftrightarrow k$ . Then  $i \rightarrow j$  and  $j \rightarrow k$ . There exist  $n, m \geq 0$  such that  $p_n(j | i) > 0$  and  $p_m(k | j) > 0$ . By the Chapman-Kolmogorov equation,

$$p_{n+m}(k | i) = \sum_{s \in S} p_m(k | s) p_n(s | i) \geq p_m(k | j) p_n(j | i) > 0.$$

Thus  $i \rightarrow k$ . Similarly, since  $k \rightarrow j$  and  $j \rightarrow i$ , we have  $k \rightarrow i$ . Thus  $i \leftrightarrow k$ .

EXERCISE 2.17. For  $|x| < 1$  and  $j, i \in S$  define

$$(5.38) \quad P_{ji}(x) = \sum_{n=0}^{\infty} p_n(j | i) x^n,$$

$$(5.39) \quad F_{ji}(x) = \sum_{n=1}^{\infty} f_n(j | i) x^n,$$

where  $f_n(j | i) = P(\xi_n = j, \xi_k \neq j, k = 1, \dots, n-1 | \xi_0 = i)$ . Show that the power series in (5.38)-(5.39) are absolutely convergent for  $|x| < 1$  and that

$$(5.40) \quad P_{ji}(x) = F_{ji}(x)P_{jj}(x), \text{ if } j \neq i,$$

$$(5.41) \quad P_{ii}(x) = 1 + F_{ii}(x)P_{ii}(x).$$

REMARK 2.20. **Hint:** Note that  $|p_n(j | i)| \leq 1$ , so the radius of convergence of the power series (5.38) is  $\geq 1$ .

SOLUTION 2.18. Since  $|p_n(j | i)| \leq 1$  and  $|f_n(j | i)| \leq 1$ , the power series converge absolutely for  $|x| < 1$  by comparison with the geometric series  $\sum x^n$ . The probability  $p_n(j | i)$  can be decomposed based on the time  $k$  of the first visit to state  $j$ :

$$p_n(j | i) = \sum_{k=1}^n P(\xi_n = j, \text{ first visit to } j \text{ at time } k | \xi_0 = i).$$

By the Markov property,

$$P(\xi_n = j, \text{ first visit to } j \text{ at time } k | \xi_0 = i) = P(\xi_n = j | \xi_k = j)P(\text{first visit to } j \text{ at time } k | \xi_0 = i) = p_{n-k}(j | i).$$

So  $p_n(j | i) = \sum_{k=1}^n f_k(j | i)p_{n-k}(j | j)$  for  $n \geq 1$ . Multiply by  $x^n$  and sum over  $n \geq 1$ :

$$\begin{aligned} \sum_{n=1}^{\infty} p_n(j | i)x^n &= \sum_{n=1}^{\infty} \sum_{k=1}^n f_k(j | i)p_{n-k}(j | j)x^n \\ &= \left( \sum_{k=1}^{\infty} f_k(j | i)x^k \right) \left( \sum_{m=0}^{\infty} p_m(j | j)x^m \right) \\ &= F_{ji}(x)P_{jj}(x). \end{aligned}$$

If  $j \neq i$ , then  $p_0(j | i) = 0$ , so  $P_{ji}(x) = \sum_{n=1}^{\infty} p_n(j | i)x^n = F_{ji}(x)P_{jj}(x)$ . If  $j = i$ , then  $p_0(i | i) = 1$ , so  $P_{ii}(x) = 1 + \sum_{n=1}^{\infty} p_n(i | i)x^n = 1 + F_{ii}(x)P_{ii}(x)$ .

EXERCISE 2.18. Show that  $\lim_{x \rightarrow 1} P_{jj}(x) = \sum_{n=0}^{\infty} p_n(j | j)$  and  $\lim_{x \rightarrow 1} F_{jj}(x) = \sum_{n=0}^{\infty} f_n(j | j)$ .

REMARK 2.21. **Hint:** Apply Abel's lemma: If  $a_k \geq 0$  for all  $k \geq 0$  and  $\limsup_{k \rightarrow \infty} \sqrt[k]{a_k} \leq 1$ , then  $\lim_{x \rightarrow 1} \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k$ , no matter whether this sum is finite or infinite.

SOLUTION 2.19. For the power series  $P_{jj}(x) = \sum p_n(j | j)x^n$  and  $F_{jj}(x) = \sum f_n(j | j)x^n$ , all coefficients are non-negative. Since  $p_n(j | j) \leq 1$  and  $f_n(j | j) \leq 1$ , the radius of convergence is at least 1. Applying Abel's Lemma as stated in the hint directly yields the desired results.

EXERCISE 2.19. Show that a state  $j$  is recurrent if and only if  $\sum_n p_n(j | j) = \infty$ . Deduce that the state  $j$  is transient if and only if

$$(5.42) \quad \sum_n p_n(j | j) < \infty.$$

Show that if  $j$  is transient, then for each  $i \in S$

$$(5.43) \quad \sum_n p_n(j | i) < \infty.$$

REMARK 2.22. **Hint:** If  $j$  is recurrent, then  $F_{jj}(x) \rightarrow \sum_n f_n(j | j) = 1$  as  $x \nearrow 1$ . Use (5.41) in conjunction with Abel's Lemma.

SOLUTION 2.20. From (5.41), we have  $P_{ii}(x) = 1 + F_{ii}(x)P_{ii}(x)$ , which implies  $P_{ii}(x) = \frac{1}{1-F_{ii}(x)}$ . By Abel's Lemma,  $\lim_{x \nearrow 1} P_{ii}(x) = \sum_{n=0}^{\infty} p_n(i | i)$  and  $\lim_{x \nearrow 1} F_{ii}(x) = \sum_{n=1}^{\infty} f_n(i | i) = P(\text{ever return to } i | \xi_0 = i)$ . If  $i$  is recurrent, then  $\sum f_n(i | i) = 1$ . Then as  $x \nearrow 1$ ,  $1 - F_{ii}(x) \rightarrow 0$ , so  $P_{ii}(x) \rightarrow \infty$ . Thus  $\sum_n p_n(i | i) = \infty$ . If  $i$  is transient, then  $\sum f_n(i | i) = f < 1$ . Then  $P_{ii}(x) \rightarrow \frac{1}{1-f} < \infty$ . Thus  $\sum_n p_n(i | i) < \infty$ . For (5.43), if  $j \neq i$ , from (5.40) we have  $P_{ji}(x) = F_{ji}(x)P_{jj}(x)$ . As  $x \nearrow 1$ ,  $F_{ji}(x) \rightarrow P(\text{ever visit } j | \xi_0 = i) \leq 1$ . Since  $j$  is transient,  $P_{jj}(x) \rightarrow \sum_n p_n(j | j) < \infty$ . Thus  $\lim_{x \nearrow 1} P_{ji}(x) = (\lim_{x \nearrow 1} F_{ji}(x))(\lim_{x \nearrow 1} P_{jj}(x)) < \infty$ , so  $\sum_n p_n(j | i) < \infty$ .

EXERCISE 2.20. For a Markov chain  $\xi_n$  with transition matrix  $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$  show that both states are recurrent.

REMARK 2.23. **Hint:** Use Exercise 2.19 and Exercise 2.5.

SOLUTION 2.21. From Exercise 2.5, the  $n$ -step transition probability for state 0 is

$$p_n(0 | 0) = \frac{q}{q+p} + \frac{p}{q+p}(1-p-q)^n.$$

Since  $0 < p, q < 1$ , we have  $|1-p-q| < 1$ . As  $n \rightarrow \infty$ ,  $p_n(0 | 0) \rightarrow \frac{q}{q+p} > 0$ . The series  $\sum_n p_n(0 | 0)$  diverges because the general term does not tend to zero. By Exercise 2.19, state 0 is recurrent. Similarly,  $p_n(1 | 1) = \frac{p}{q+p} + \frac{q}{q+p}(1-p-q)^n \rightarrow \frac{p}{q+p} > 0$ , so state 1 is also recurrent.

One may suspect that if the state space  $S$  is finite, then there must exist at least one recurrent state. For otherwise, if all states were transient and  $S = \{1, 2, \dots, N\}$ , then with positive probability a chain starting from 1 would visit 1 only a finite number of times. Thus, after visiting that state for the last time, the chain would move to a different state, say  $i_2$ , in which it would stay for a finite time only with positive probability. Thus, in finite time, with positive probability, the chain will never return to states 1 and  $i_2$ . By induction, in finite time, with positive probability, the chain will never return to any of the states. This is impossible. The following exercise will give precision to this argument.

EXERCISE 2.21. Show that if  $\xi_n$  is a Markov chain with finite state space  $S$ , then there exists at least one recurrent state  $i \in S$ .

REMARK 2.24. **Hint:** Argue by contradiction and use (5.43).

SOLUTION 2.22. Suppose that all states  $j \in S$  are transient. Then by (5.43), for any  $i, j \in S$ ,  $\sum_{n=0}^{\infty} p_n(j | i) < \infty$ . This implies that  $\lim_{n \rightarrow \infty} p_n(j | i) = 0$  for all  $j \in S$ . However, since  $S$  is finite, we have  $\sum_{j \in S} p_n(j | i) = 1$  for all  $n$ . Taking the limit as  $n \rightarrow \infty$ :

$$1 = \lim_{n \rightarrow \infty} \sum_{j \in S} p_n(j | i) = \sum_{j \in S} \lim_{n \rightarrow \infty} p_n(j | i) = \sum_{j \in S} 0 = 0,$$

which is a contradiction. The interchange of limit and sum is justified because the sum is finite. Thus, there must be at least one recurrent state.

The following result is quoted here for reference. The proof is surprisingly difficult and falls beyond the scope of this book.

**Theorem 2.13.** *A state  $j \in S$  is recurrent if and only if*

$$(20) \quad P(\xi_n = j \text{ for infinitely many } n \mid \xi_0 = j) = 1,$$

*and is transient if and only if*

$$(21) \quad P(\xi_n = j \text{ for infinitely many } n \mid \xi_0 = j) = 0.$$

**DEFINITION 2.14** (Null-recurrent and positive-recurrent states). *For an  $S$ -valued Markov chain  $\xi_n, n \in \mathbb{N}$ , a state  $i \in S$  is called null-recurrent if it is recurrent and its mean recurrence time  $m_i$  defined by*

$$(5.44) \quad m_i := \sum_{n=0}^{\infty} n f_n(i \mid i)$$

*equals  $\infty$ . A state  $i \in S$  is called positive-recurrent if it is recurrent and its mean recurrence time  $m_i$  is finite.*

**REMARK 2.25.** *One can show that a recurrent state  $i$  is null-recurrent if and only if  $p_n(i \mid i) \rightarrow 0$ .*

We already know that for a random walk on  $\mathbb{Z}$  the state 0 is recurrent if and only if  $p = 1/2$ , i.e. if and only if the random walk is symmetric. In the following problem we shall try to answer if 0 is a null-recurrent or positive-recurrent state (when  $p = 1/2$ ).

**EXERCISE 2.22.** *Consider a symmetric random walk on  $\mathbb{Z}$ . Show that 0 is a null-recurrent state. Can you deduce whether other states are positive-recurrent or null-recurrent?*

**REMARK 2.26. *Hint:*** *State 0 is null-recurrent if and only if  $\sum_n n f_n(0 \mid 0) = \infty$ . As in Exercise 2.18,  $\sum_n n f_n(0 \mid 0) = \lim_{x \rightarrow 1} F'_{00}(x)$ , where  $F_{00}$  is defined by (5.39).*

**SOLUTION 2.23.** *For a symmetric random walk ( $p = q = 1/2$ ), the generating function for return probabilities is  $P_{00}(x) = (1 - x^2)^{-1/2}$ . From  $P_{00}(x) = \frac{1}{1 - F_{00}(x)}$ , we have  $F_{00}(x) = 1 - \frac{1}{P_{00}(x)} = 1 - (1 - x^2)^{1/2}$ . To check if the state is null-recurrent, we compute the mean recurrence time  $m_0 = F'_{00}(1)$ . Differentiating  $F_{00}(x)$ :*

$$F'_{00}(x) = -\frac{1}{2}(1 - x^2)^{-1/2}(-2x) = \frac{x}{\sqrt{1 - x^2}}.$$

*As  $x \nearrow 1$ ,  $F'_{00}(x) \rightarrow \infty$ . Thus  $m_0 = \infty$ , so state 0 is null-recurrent. Since all states in a random walk communicate, and null-recurrence is a class property (Proposition 2.16), all other states are also null-recurrent.*

**EXERCISE 2.23.** *For the Markov chain  $\xi_n$  from Exercise 2.20 show that not only are all states recurrent, but they are positive-recurrent.*

**REMARK 2.27. *Hint:*** *Calculate  $f_n(0 \mid 0)$  directly.*

**SOLUTION 2.24.** *For state 0 to be visited for the first time at time  $n \geq 1$ : -  $n = 1$ :  $P(\xi_1 = 0 \mid \xi_0 = 0) = 1 - p$ . So  $f_1(0 \mid 0) = 1 - p$ . -  $n \geq 2$ : The path must be  $0 \rightarrow 1 \rightarrow 1 \rightarrow \dots \rightarrow$*



$1 \rightarrow 0$ . The probability is  $P(\xi_1 = 1, \xi_2 = 1, \dots, \xi_{n-1} = 1, \xi_n = 0 \mid \xi_0 = 0) = p(1 - q)^{n-2}q$ . The mean recurrence time  $m_0$  is:

$$m_0 = \sum_{n=1}^{\infty} n f_n(0 \mid 0) = 1(1 - p) + \sum_{n=2}^{\infty} n p q (1 - q)^{n-2}.$$

Let  $k = n - 2$ , then  $n = k + 2$ :

$$\sum_{k=0}^{\infty} (k + 2) p q (1 - q)^k = p q \sum_{k=0}^{\infty} k (1 - q)^k + 2 p q \sum_{k=0}^{\infty} (1 - q)^k.$$

Using  $\sum k r^k = \frac{r}{(1-r)^2}$  and  $\sum r^k = \frac{1}{1-r}$ :

$$p q \frac{1 - q}{q^2} + 2 p q \frac{1}{q} = \frac{p(1 - q)}{q} + 2p = \frac{p - p q + 2 p q}{q} = \frac{p + p q}{q} = \frac{p}{q} + p.$$

So  $m_0 = 1 - p + \frac{p}{q} + p = 1 + \frac{p}{q} = \frac{p+q}{q}$ . Since  $p, q > 0$ ,  $m_0 < \infty$ , so state 0 is positive-recurrent. By symmetry, state 1 is also positive-recurrent.

The last two exercises suggest that the type of a state  $i \in S$ , i.e. whether it is transient, null-recurrent or positive-recurrent is invariant under the equivalence  $\leftrightarrow$ . We shall investigate this question in more detail below, but even before doing so we need one more notion: that of a periodic state.

**DEFINITION 2.15** (Periodic and aperiodic states). Suppose that  $\xi_n, n \in \mathbb{N}$ , is a Markov chain on a state space  $S$ . Let  $i \in S$ . We say that  $i$  is a periodic state if and only if the greatest common divisor (gcd) of all  $n \in \mathbb{N}^*$ , where  $\mathbb{N}^* = \{1, 2, 3, \dots\}$ , such that  $p_n(i \mid i) > 0$  is  $\geq 2$ . Otherwise, the state  $i$  is called aperiodic. In both cases, the gcd is denoted by  $d(i)$  and is called the period of the state  $i$ . Thus,  $i$  is periodic if and only if  $d(i) \geq 2$ . A state  $i$  which is positive recurrent and aperiodic is called ergodic.

**EXERCISE 2.24.** Is this claim that  $p_{d(i)}(i \mid i) > 0$  true or not?

**REMARK 2.28. Hint:** Think of a Markov chain in which it is possible to return to the starting point by two different routes. One route with four steps, the other one with six steps.

**SOLUTION 2.25.** The claim is false. Consider a state  $i$  where the possible return times are  $\{4, 6, 8, \dots\}$ . The gcd is  $d(i) = 2$ . However,  $p_2(i \mid i) = 0$ . An example is a chain with states  $\{i, a, b, c, e, f, g\}$  and transitions  $i \rightarrow a \rightarrow b \rightarrow c \rightarrow i$  (length 4) and  $i \rightarrow e \rightarrow f \rightarrow g \rightarrow h \rightarrow j \rightarrow i$  (length 6). The return times are  $4n + 6m$ , all of which are even. The gcd is 2, but there is no way to return in exactly 2 steps.

One of the by-products of the following exercise is another example of the type asked for in Exercise 2.24.

**EXERCISE 2.25.** Consider a Markov chain on  $S = \{1, 2\}$  with transition probability matrix  $P = \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix}$ . This chain can also be described by the graph in Figure 1. Find  $d(1)$  and  $d(2)$ .

**REMARK 2.29. Hint:** Calculate  $P^2$  and  $P^3$ . This can be done in two different ways: either algebraically or, probabilistically.

SOLUTION 2.26. Calculating powers of  $P$ :  $P = \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix}$   $P^2 = \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/4 \\ 1/2 & 3/4 \end{bmatrix}$   $P^3 = \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix} \begin{bmatrix} 1/2 & 1/4 \\ 1/2 & 3/4 \end{bmatrix} = \begin{bmatrix} 1/4 & 3/8 \\ 3/4 & 5/8 \end{bmatrix}$  For state 1,  $p_2(1 | 1) = 1/2 > 0$  and  $p_3(1 | 1) = 1/4 > 0$ . The gcd of  $\{2, 3, \dots\}$  is 1. Thus  $d(1) = 1$ . For state 2,  $p_1(2 | 2) = 1/2 > 0$ , so  $d(2) = 1$ . Both states are aperiodic.

FIGURE 1. Transition probabilities of the Markov chain in Exercise 2.25

**Proposition 2.16.** Suppose that  $i, j \in S$  and  $i \leftrightarrow j$ . Show that

- (1)  $i$  is transient if and only if  $j$  is;
- (2)  $i$  is recurrent if and only if  $j$  is;
- (3)  $i$  is null-recurrent if and only if  $j$  is;
- (4)  $i$  is positive-recurrent if and only if  $j$  is;
- (5)  $i$  is periodic if and only if  $j$  is, in which case  $d(i) = d(j)$ ;
- (6)  $i$  is ergodic if and only if  $j$  is.

EXERCISE 2.26. Show that the following modification of 2) above is true: If  $i$  is recurrent and  $i \rightarrow j$ , then  $j \rightarrow i$ . Deduce that if  $i$  is recurrent and  $i \rightarrow j$ , then  $j$  is recurrent and  $j \leftrightarrow i$ .

REMARK 2.30. **Hint:** Is it possible for a chain starting from  $i$  to visit  $j$  and then never return to  $i$ ? Is such a situation possible when  $i$  is a recurrent state?

SOLUTION 2.27. Suppose  $i$  is recurrent and  $i \rightarrow j$ , but  $j \not\rightarrow i$ . This means there is a positive probability of reaching  $j$  from  $i$ , but once at  $j$ , the probability of returning to  $i$  is zero. Let  $A$  be the event that the chain reaches  $j$  and never returns to  $i$ . Since  $i \rightarrow j$  and  $j \not\rightarrow i$ ,  $P(A | \xi_0 = i) > 0$ . However, if  $i$  is recurrent, the probability of never returning to  $i$  must be zero, i.e.,  $P(\text{no return to } i | \xi_0 = i) = 0$ . This is a contradiction. Thus we must have  $j \rightarrow i$ , so  $i \leftrightarrow j$ . By Proposition 2.16, since  $i$  is recurrent and  $i \leftrightarrow j$ ,  $j$  must also be recurrent.

The following result describes how the state space  $S$  can be partitioned into a countable sum of classes. One of these classes consists of all transient states. Each of the other classes consists of interconnecting recurrent states. If the chain enters one of the classes of second type, it will never leave it. However, if the chain enters the class of transient states, it will eventually leave it (and so never return to it). We begin with a definition.

DEFINITION 2.17 (Closed and irreducible sets). Suppose that  $\xi_n, n \in \mathbb{N}$ , is a Markov chain on a countable state space  $S$ .

- (1) A set  $C \subset S$  is called closed if once the chain enters  $C$  it will never leave it, i.e.

$$(5.45) \quad P(\xi_k \in S \setminus C \text{ for some } k \geq n | \xi_n \in C) = 0.$$

- (2) A set  $C \subset S$  is called irreducible if any two elements  $i, j$  of  $C$  intercommunicate, i.e. for all  $i, j \in C$  there exists an  $n \in \mathbb{N}$  such that  $p_n(j | i) > 0$ .

**Theorem 2.18.** Suppose that  $\xi_n, n \in \mathbb{N}$ , is a Markov chain on a countable state space  $S$ . Then

$$(5.46) \quad S = T \cup \bigcup_{j=1}^N C_j, \text{ (disjoint sum),}$$

where  $T$  is the set of all transient states in  $S$  and each  $C_j$  is a closed irreducible set of recurrent states.

**EXERCISE 2.27.** Suppose that  $\xi_n, n \in \mathbb{N}$ , is a Markov chain on a countable state space  $S$ . Show that a set  $C \subset S$  is closed if and only if  $p(j | i) = 0$  for all  $i \in C$  and  $j \in S \setminus C$ .

**REMARK 2.31. Hint:** One implication is trivial. For the other one use the countable additivity of the measure  $P$ .

**SOLUTION 2.28.** ( $\Rightarrow$ ) Suppose  $C$  is closed. By (5.45),  $P(\xi_1 \in S \setminus C | \xi_0 = i) = 0$  for any  $i \in C$ . This can be written as  $\sum_{j \in S \setminus C} p(j | i) = 0$ . Since  $p(j | i) \geq 0$ , this implies  $p(j | i) = 0$  for all  $j \in S \setminus C$  and  $i \in C$ . ( $\Leftarrow$ ) Suppose  $p(j | i) = 0$  for all  $i \in C, j \in S \setminus C$ . We want to show that if  $\xi_0 \in C$ , then  $P(\xi_k \in S \setminus C \text{ for some } k \geq 1 | \xi_0 \in C) = 0$ . By subadditivity, it suffices to show  $P(\xi_k \in S \setminus C | \xi_0 = i) = 0$  for all  $k \geq 1, i \in C$ . For  $k = 1$ ,  $P(\xi_1 \in S \setminus C | \xi_0 = i) = \sum_{j \in S \setminus C} p(j | i) = 0$ . Assume it holds for  $k$ . Then

$$P(\xi_{k+1} \in S \setminus C | \xi_0 = i) = \sum_{s \in S} P(\xi_{k+1} \in S \setminus C | \xi_k = s) P(\xi_k = s | \xi_0 = i).$$

If  $s \in C$ ,  $P(\xi_{k+1} \in S \setminus C | \xi_k = s) = 0$ . If  $s \in S \setminus C$ ,  $P(\xi_k = s | \xi_0 = i) = 0$  by the induction hypothesis. Thus the sum is zero for all  $k$ .

### 3. Long-Time Behaviour of Markov Chains: General Case

For convenience we shall denote the countable state space  $S$  by  $\{1, 2, 3, \dots\}$  when  $S$  is an infinite set and by  $\{1, 2, \dots, n\}$  when  $S$  is finite.

**Proposition 2.19.** Let  $P = [p(j | i)]$  be the transition matrix of a Markov chain with state space  $S$ . Suppose that for all  $i, j \in S$

$$(5.47) \quad \lim_{n \rightarrow \infty} p_n(j | i) =: \pi_j.$$

(In particular, the limit is independent of  $i$ .) Then

- (1)  $\sum_j \pi_j \leq 1$ ;
- (2)  $\sum_i p(j | i) \pi_i = \pi_j$ ;
- (3) either  $\sum_j \pi_j = 1$ , or  $\pi_j = 0$  for all  $j \in S$ .

**Proposition 2.20.** Let  $P = [p(j | i)]$  be the transition matrix of a Markov chain with state space  $S$ . Suppose that for all  $i, j \in S$

$$(5.47) \quad \lim_{n \rightarrow \infty} p_n(j | i) =: \pi_j.$$

Then the vector  $\pi = (\pi_j)_{j \in S}$  satisfies  $\pi = P^T \pi$ , i.e.,

$$(5.65) \quad \sum_{i \in S} p(j | i) \pi_i = \pi_j \quad \text{for all } j \in S.$$

**Theorem 2.21.** Suppose that  $a \geq 1$  is a natural number. Consider a random walk on  $S = \{0, 1, \dots, a\}$  with absorbing barriers at 0 and  $a$ , and with probability  $p$  of moving to the right and probability  $q = 1 - p$  of moving to the left from any of the states  $1, \dots, a - 1$ . Hence, our random walk is a Markov chain with transition probabilities

$$(22) \quad p(j | i) = \begin{cases} p, & \text{if } 1 \leq i \leq a - 1, j = i + 1, \\ q, & \text{if } 1 \leq i \leq a - 1, j = i - 1, \\ 1, & \text{if } i = j = 0 \text{ or } i = j = a, \\ 0, & \text{otherwise.} \end{cases}$$

Find, (a) all invariant measures (there may be just one), (b) the probability of hitting the right-hand barrier prior to hitting the left-hand one.

REMARK 2.32. **Hint:** For (a) recall Exercise 2.33 and for (b) Exercise 2.12.

EXERCISE 2.28. Ian plays a fair game of dice. After the  $n$ th roll of a die he writes down the maximum outcome  $\xi_n$  obtained so far. Show that  $\xi_n$  is a Markov chain and find its transition probabilities.

REMARK 2.33. **Hint:**  $\xi_{n+1} = \max\{\xi_n, X_{n+1}\}$ , where  $X_k$  is the outcome of the  $k$ th roll.

SOLUTION 2.29. Let  $X_n$  be the outcome of the  $n$ -th roll.  $X_n$  are i.i.d. random variables with  $P(X_n = k) = 1/6$  for  $k \in \{1, \dots, 6\}$ . The maximum outcome after  $n + 1$  rolls is  $\xi_{n+1} = \max\{\xi_n, X_{n+1}\}$ . Since  $X_{n+1}$  is independent of  $X_1, \dots, X_n$ , it is also independent of  $\xi_n$ . Thus, the value of  $\xi_{n+1}$  depends only on  $\xi_n$  and the independent roll  $X_{n+1}$ , which proves that  $\xi_n$  is a Markov chain. The state space is  $S = \{1, \dots, 6\}$ . The transition probabilities  $p(j | i) = P(\xi_{n+1} = j | \xi_n = i)$  are: - if  $j < i$ ,  $p(j | i) = 0$  (the maximum cannot decrease). - if  $j = i$ , then  $X_{n+1} \leq i$ , so  $p(i | i) = P(X_{n+1} \leq i) = i/6$ . - if  $j > i$ , then  $X_{n+1} = j$ , so  $p(j | i) = P(X_{n+1} = j) = 1/6$ . In summary,

$$p(j | i) = \begin{cases} i/6 & \text{if } j = i \\ 1/6 & \text{if } j > i \\ 0 & \text{otherwise} \end{cases}$$

for  $i, j \in \{1, \dots, 6\}$ .

EXERCISE 2.29. Analyse the Markov chain described in Exercise 2.28, but with fair die replaced by a fair pyramid.

REMARK 2.34. **Hint:** A pyramid has four faces only.

SOLUTION 2.30. The analysis is identical to Exercise 2.28, but with a state space  $S = \{1, 2, 3, 4\}$  and probabilities based on 4 faces. The transition probabilities are

$$p(j | i) = \begin{cases} i/4 & \text{if } j = i \\ 1/4 & \text{if } j > i \\ 0 & \text{otherwise} \end{cases}$$

for  $i, j \in \{1, 2, 3, 4\}$ .

**EXERCISE 2.30.** Consider a random walk on  $S = \{0, 1, \dots, a\}$  with absorbing barriers at 0 and  $a$ , and with probability  $p$  of moving to the right and probability  $q = 1 - p$  of moving to the left from any of the states  $1, \dots, a - 1$ . Find the probability of hitting the right-hand barrier before the left-hand one.

**REMARK 2.35. Hint:** Use the result from Proposition 2.22 or solve directly using recurrence relations.

**SOLUTION 2.31.** Let  $h_i$  be the probability of hitting  $a$  before 0 starting from state  $i$ . By the law of total probability,

$$h_i = ph_{i+1} + qh_{i-1} \quad \text{for } 1 \leq i \leq a - 1,$$

with boundary conditions  $h_0 = 0$  and  $h_a = 1$ . The general solution to this linear difference equation depends on whether  $p = q$ . If  $p \neq q$ , the solution is  $h_i = A + B(q/p)^i$ . Applying the boundary conditions:  $h_0 = A + B = 0 \Rightarrow B = -A$ .  $h_a = A(1 - (q/p)^a) = 1 \Rightarrow A = \frac{1}{1 - (q/p)^a}$ . Thus  $h_i = \frac{1 - (q/p)^i}{1 - (q/p)^a}$ . If  $p = q = 1/2$ , the solution is  $h_i = A + Bi$ .  $h_0 = A = 0$ .  $h_a = Ba = 1 \Rightarrow B = 1/a$ . Thus  $h_i = i/a$ . These results match Proposition 2.22.

**Proposition 2.22** (Hitting Probability for Random Walk). Consider a random walk on  $S = \{0, 1, \dots, a\}$  with absorbing barriers at 0 and  $a$ , and with probability  $p$  of moving to the right and probability  $q = 1 - p$  of moving to the left from any of the states  $1, \dots, a - 1$ . Let  $h_i$  be the probability of hitting  $a$  before 0 starting from state  $i$ .

Then for  $p \neq q$ :

$$h_i = \frac{1 - (q/p)^i}{1 - (q/p)^a},$$

and for  $p = q = 1/2$ :

$$h_i = \frac{i}{a}.$$

**EXERCISE 2.31.** Prove that if there exists an invariant measure for the Markov chain in Exercise 2.14, then  $\lambda' := \sum_{j=0}^{\infty} jq'_j < 1$ . Assuming that the converse is also true, conclude that the chain is ergodic if and only if  $\lambda' < 1$ . Show that if such an invariant measure exists, then it is unique.

**REMARK 2.36. Hint:** Suppose that  $\pi = \sum_{j=0}^{\infty} \pi_j \delta_j$  is an invariant measure. Write down an infinite system of linear equations for  $\pi_j$ . If you don't know how to follow, look at the solution.

**SOLUTION 2.32.** Let  $\pi = (\pi_0, \pi_1, \dots)$  be an invariant distribution. Then  $\pi = P^T \pi$ . For the queuing model, the drift at state  $i > 0$  is  $E[\xi_{n+1} - \xi_n \mid \xi_n = i] = E[Z_n - Y_n] = \lambda - p$ . If an invariant distribution exists, the average drift must be zero:

$$0 = \sum_{i=0}^{\infty} \pi_i E[\xi_{n+1} - \xi_n \mid \xi_n = i] = \pi_0 E[Z_n] + \sum_{i=1}^{\infty} \pi_i (\lambda - p) = \pi_0 \lambda + (1 - \pi_0)(\lambda - p).$$

This simplifies to  $0 = \lambda - (1 - \pi_0)p$ , or  $\lambda/p = 1 - \pi_0$ . Since  $\pi_0 > 0$  for a stable queue, we must have  $\lambda/p < 1$ , or  $\lambda < p$ . If we identify  $\lambda' = \lambda/p$  (assuming the unit time is normalized), the condition is  $\lambda' < 1$ . The uniqueness follows from the fact that for an irreducible Markov chain, if an invariant distribution exists, it is unique.

#### 4. Long-Time Behaviour of Markov Chains with Finite State Space

...

EXERCISE 2.32. Show that  $p_n(j | i) \rightarrow \pi_j$  at an exponential rate.

REMARK 2.37. **Hint:** Recall that  $m_n(j) \leq \pi_j \leq M_n(j)$  and use (5.62).

SOLUTION 2.33. From the proof of Theorem 2.23, it is shown that the difference between the maximum and minimum values of  $p_n(j | \cdot)$  decreases geometrically. Specifically,

$$M_{kn_0}(j) - m_{kn_0}(j) \leq (1 - \varepsilon)^k$$

for some  $\varepsilon > 0$ . Since  $m_n(j) \leq p_n(j | i) \leq M_n(j)$  and  $m_n(j) \leq \pi_j \leq M_n(j)$ , we have

$$|p_{kn_0}(j | i) - \pi_j| \leq (1 - \varepsilon)^k = [(1 - \varepsilon)^{1/n_0}]^{kn_0}.$$

Setting  $\alpha = (1 - \varepsilon)^{1/n_0} < 1$ , we see that  $|p_n(j | i) - \pi_j| \leq C\alpha^n$  for some constant  $C$ , which demonstrates an exponential rate of convergence.

The above proves the following important result.

**Theorem 2.23** (Finite State Space Convergence). Suppose that  $S$  is finite and the transition matrix  $P = [p(j | i)]$  of a Markov chain on  $S$  satisfies the condition

$$(5.54) \quad \exists n_0 \in \mathbb{N} \exists \varepsilon > 0 : p_{n_0}(j | i) \geq \varepsilon, \quad i, j \in S.$$

Then, the following limit exists for all  $i, j \in S$  and is independent of  $i$ :

$$(5.55) \quad \lim_{n \rightarrow \infty} p_n(j | i) = \pi_j.$$

The numbers  $\pi_j$  satisfy

$$(5.56) \quad \pi_j > 0, \quad j \in S \quad \text{and} \quad \sum_{j \in S} \pi_j = 1.$$

Conversely, if a sequence of numbers  $\pi_j, j \in S$  satisfies conditions (5.55)-(5.56), then assumption (5.54) is also satisfied.

**Theorem 2.24** (Ergodic Finite State Space). Suppose that  $\xi_n, n \in \mathbb{N}$ , is an irreducible aperiodic Markov chain with finite state space  $S$ . Then  $\xi_n$  is ergodic if and only if it has a unique invariant measure.

EXERCISE 2.33. Investigate the existence and uniqueness of an invariant measure for the Markov chain in Proposition 2.2.

REMARK 2.38. **Hint:** Are the assumptions of Theorem 2.23 satisfied?

SOLUTION 2.34. The transition matrix is  $P = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}$ . Since  $0 < p, q < 1$ , all entries of  $P$  are strictly positive. Thus, condition (5.54) is satisfied with  $n_0 = 1$  and  $\varepsilon = \min\{1-p, q, p, 1-q\} > 0$ . By Theorem 2.24, there exists a unique invariant measure for this Markov chain.

REMARK 2.39. The solution to Exercise 2.33 allows us to find the unique invariant measure by direct methods, i.e. by solving the linear equations (5.79)-(5.80).

EXERCISE 2.34. Find the invariant measure from Exercise 2.33 by calculating the limits (5.55).

REMARK 2.40. **Hint:** Refer to Solution 5.5.

SOLUTION 2.35. From Exercise 2.5, the  $n$ -step transition probabilities are

$$p_n(0 | 0) = \frac{q}{q+p} + \frac{p}{q+p}(1-p-q)^n \rightarrow \frac{q}{q+p},$$

$$p_n(0 | 1) = \frac{q}{q+p} - \frac{q}{q+p}(1-p-q)^n \rightarrow \frac{q}{q+p}.$$

Thus  $\pi_0 = \frac{q}{q+p}$ . Similarly,  $\pi_1 = \frac{p}{q+p}$ . The invariant measure is  $\pi = \left(\frac{q}{q+p}, \frac{p}{q+p}\right)$ .

EXERCISE 2.35. For each of the following transition matrices, determine: (a) whether the chain is irreducible, (b) whether it is aperiodic, (c) the invariant distribution (if it exists), (d) whether the chain is ergodic.

$$(1) P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

$$(2) P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$(3) P = \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}$$

REMARK 2.41. **Hint:** Check the definitions of irreducibility, aperiodicity, and use Theorem 2.24.

SOLUTION 2.36. (1)  $P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ : (a) Irreducible: All entries are positive, so any state can reach any other. (b) Aperiodic:  $p(1 | 1) > 0$ . (c) Invariant distribution:  $\pi = (0.5, 0.5)$ . (d) Ergodic: Yes, it is irreducible, aperiodic, and positive recurrent (finite state).

(2)  $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ : (a) Irreducible: Yes. (b) Periodic:  $p_1(1 | 1) = 0$ ,  $p_2(1 | 1) = 1$ ,  $p_3(1 | 1) = 0$ . Period  $d = 2$ . (c) Invariant distribution:  $\pi = (0.5, 0.5)$ . (d) Ergodic: No (not aperiodic).

(3)  $P = \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}$ : (a) Not irreducible: State 3 is absorbing and cannot reach states 1 or 2. (b) Aperiodic: All diagonal entries are positive. (c) Invariant distribution:  $\pi = (0, 0, 1)$ . (d) Not ergodic: Not irreducible.





## CHAPTER 3

# Stochastic Processes in Continuous Time

### 1. General Notions

The following definitions are straightforward extensions of those introduced earlier for sequences of random variables, the underlying idea being that of a family of random variables depending on time.

**DEFINITION 3.1** (Stochastic Process). *A stochastic process is a family of random variables  $\xi(t)$  parametrized by  $t \in T$ , where  $T \subset \mathbb{R}$ . When  $T = \{1, 2, \dots\}$ , we shall say that  $\xi(t)$  is a stochastic process in discrete time (i.e. a sequence of random variables). When  $T$  is an interval in  $\mathbb{R}$  (typically  $T = [0, \infty)$ ), we shall say that  $\xi(t)$  is a stochastic process in continuous time.*

For every  $\omega \in \Omega$  the function

$$T \ni t \mapsto \xi(t, \omega)$$

is called a path (or sample path) of  $\xi(t)$ .

**DEFINITION 3.2** (Filtration). *A family  $\mathcal{F}_t$  of  $\sigma$ -fields on  $\Omega$  parametrized by  $t \in T$ , where  $T \subset \mathbb{R}$ , is called a filtration if*

$$\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$$

for any  $s, t \in T$  such that  $s \leq t$ .

**DEFINITION 3.3** (Martingale). *A stochastic process  $\xi(t)$  parametrized by  $t \in T$  is called a martingale (submartingale, supermartingale) with respect to a filtration  $\mathcal{F}_t$  if*

- (1)  $\xi(t)$  is integrable for each  $t \in T$ ;
- (2)  $\xi(t)$  is  $\mathcal{F}_t$ -measurable for each  $t \in T$  (in which case we say that  $\xi(t)$  is adapted to  $\mathcal{F}_t$ );
- (3)  $\xi(s) = E(\xi(t) \mid \mathcal{F}_s)$  (respectively,  $\leq$  or  $\geq$ ) for every  $s, t \in T$  such that  $s \leq t$ .

In earlier chapters we have seen various stochastic processes, in particular, martingales in discrete time such as the symmetric random walk, for example. In what follows we shall study in some detail two processes in continuous time, namely, the Poisson process and Brownian motion.

### 2. Poisson Process

#### 2.1. Exponential Distribution and Lack of Memory.

**DEFINITION 3.4** (Exponential Distribution). *We say that a random variable  $\eta$  has the exponential distribution of rate  $\lambda > 0$  if*

$$(23) \quad P\{\eta > t\} = e^{-\lambda t}$$

for all  $t \geq 0$ .

For example, the emissions of particles by a sample of radioactive material (or calls made at a telephone exchange) occur at random times. The probability that no particle is emitted (no call is made) up to time  $t$  is known to decay exponentially as  $t$  increases. That is to say, the time  $\eta$  of the first emission has the exponential distribution,  $P\{\eta > t\} = e^{-\lambda t}$ .

**EXERCISE 3.1.** *What is the distribution function of a random variable  $\eta$  with exponential distribution? Does it have a density? If so, find the density.*

**Hint:** *What is the probability that  $\eta > 0$ ? What is the probability that  $\eta > t$  for any given  $t < 0$ ? Can you express the distribution function in terms of  $P\{\eta > t\}$ ? Is the distribution function differentiable?*

**SOLUTION 3.1.** *Suppose that  $\eta$  is a random variable with exponential distribution of rate  $\lambda$ . The distribution function of  $\eta$  is*

$$(24) \quad F(t) = P\{\eta \leq t\} = 1 - P\{\eta > t\} = \begin{cases} 0 & \text{if } t < 0, \\ 1 - e^{-\lambda t} & \text{if } t \geq 0. \end{cases}$$

*Therefore  $\eta$  has density*

$$(25) \quad f(t) = \frac{d}{dt}F(t) = \begin{cases} 0 & \text{if } t < 0, \\ \lambda e^{-\lambda t} & \text{if } t \geq 0. \end{cases}$$

*The distribution function  $F(t)$  and density  $f(t)$  are shown in fig. 1.*

FIGURE 1. The distribution function  $F(t)$  and density  $f(t)$  of a random variable with exponential distribution of rate  $\lambda = 2$

**EXERCISE 3.2.** *Compute the expectation and variance of a random variable having the exponential distribution.*

**Hint:** *Use the density found in Exercise 3.1.*

**SOLUTION 3.2.** *Using the density  $f(t) = \lambda e^{-\lambda t}$  found in Exercise 3.1 and integrating by parts, we obtain*

$$(26) \quad \begin{aligned} E(\eta) &= \int_{-\infty}^{\infty} t f(t) dt = \int_0^{\infty} t \lambda e^{-\lambda t} dt = - \int_0^{\infty} t \frac{d}{dt} e^{-\lambda t} dt \\ &= -t e^{-\lambda t} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda t} dt = 0 - \frac{1}{\lambda} e^{-\lambda t} \Big|_0^{\infty} = \frac{1}{\lambda}. \end{aligned}$$

*In a similar way we compute*

$$(27) \quad \begin{aligned} E(\eta^2) &= \int_{-\infty}^{\infty} t^2 f(t) dt = \int_0^{\infty} t^2 \lambda e^{-\lambda t} dt = - \int_0^{\infty} t^2 \frac{d}{dt} e^{-\lambda t} dt \\ &= -t^2 e^{-\lambda t} \Big|_0^{\infty} + 2 \int_0^{\infty} t e^{-\lambda t} dt = 0 + \frac{2}{\lambda} \int_0^{\infty} t f(t) dt = \frac{2}{\lambda^2}. \end{aligned}$$

*It follows that the variance is equal to*

$$(28) \quad \text{var}(\eta) = E(\eta^2) - (E(\eta))^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

EXERCISE 3.3. Show that a random variable  $\eta$  with exponential distribution satisfies

$$(6.1) \quad P\{\eta > t + s\} = P\{\eta > t\}P\{\eta > s\}$$

for any  $s, t \geq 0$ .

**Hint:** When the probabilities are replaced by exponents, the equality should become obvious.

SOLUTION 3.3. By the definition of a random variable with exponential distribution

$$P\{\eta > s + t\} = e^{-\lambda(s+t)} = e^{-\lambda s}e^{-\lambda t} = P\{\eta > s\}P\{\eta > t\}.$$

EXERCISE 3.4. Show that the equality in Exercise 3.3 is equivalent to

$$(6.2) \quad P\{\eta > t + s \mid \eta > s\} = P\{\eta > t\}$$

for any  $s, t \geq 0$ .

**Hint:** Recall how to compute conditional probability. Observe that  $\eta > s + t$  implies  $\eta > s$ .

SOLUTION 3.4. By the definition of conditional probability

$$(29) \quad \begin{aligned} P\{\eta > t + s \mid \eta > s\} &= \frac{P\{\eta > t + s, \eta > s\}}{P\{\eta > s\}} \\ &= \frac{P\{\eta > t + s\}}{P\{\eta > s\}}, \end{aligned}$$

since  $\{\eta > t + s, \eta > s\} = \{\eta > t + s\}$  (because  $\eta > s + t$  implies that  $\eta > s$ ). Now it is clear that ?? is equivalent to eq. (6.2).

The equality eq. (6.2) (or, equivalently, ??) is known as the **lack of memory property**. The odds that no particle will be emitted (no call will be made) in the next time interval of length  $t$  are not affected by the length of time  $s$  it has already taken to wait, given that no emission (no call) has occurred yet.

EXERCISE 3.5. Show that the exponential distribution is the only probability distribution satisfying the lack of memory property.

**Hint:** The lack of memory property means that  $g(t) = P\{\eta > t\}$  satisfies the functional equation

$$g(t + s) = g(t)g(s)$$

for any  $s, t > 0$ . Find all non-negative non-increasing solutions of this functional equation.

SOLUTION 3.5. The exponential distribution is the only probability distribution satisfying the lack of memory property because the only non-negative non-increasing solutions of the functional equation

$$g(t + s) = g(t)g(s)$$

are of the form  $g(t) = a^t$  for some  $0 \leq a \leq 1$ .

To verify this observe that  $[g(m/n)]^n = g(m) = [g(1)]^m$  for any integers  $m$  and  $n \neq 0$ . Let  $a := g(1)$ . It follows that

$$g(q) = a^q \quad \text{for any } q \in \mathbb{Q}.$$

Since  $g$  is non-increasing,  $0 \leq a \leq 1$  and

$$a^t = \inf_{t > q \in \mathbb{Q}} g(q) \geq g(t) \geq \sup_{t < q \in \mathbb{Q}} g(q) = a^t,$$

so indeed

$$g(t) = a^t \quad \text{for any } t \in \mathbb{R}.$$

As a result,  $P\{\eta > t\} = a^t$  for some  $0 \leq a \leq 1$ . But the distribution function of a random variable cannot be constant, so  $0 \neq a \neq 1$ . Hence  $a = e^{-\lambda}$  for some  $\lambda > 0$ , completing the argument.

**2.2. Construction of the Poisson Process.** Let  $\eta_1, \eta_2, \dots$  be a sequence of independent random variables, each having the same exponential distribution of rate  $\lambda$ . For example, the times between the emissions of radioactive particles (or between calls made at a telephone exchange) have this property. We put

$$\xi_n = \eta_1 + \dots + \eta_n,$$

which can be thought of as the time of the  $n$ -th emission (the  $n$ -th call). We also put  $\xi_0 = 0$  for convenience. The number of emissions (calls) up to time  $t \geq 0$  is an  $n$  such that  $\xi_{n+1} > t \geq \xi_n$ . In other words, the number of emissions (calls) up to time  $t \geq 0$  is equal to  $\max\{n : t \geq \xi_n\}$ .

**DEFINITION 3.5 (Poisson Process).** We say that  $N(t)$ , where  $t \geq 0$ , is a Poisson process if

$$(30) \quad N(t) = \max\{n : t \geq \xi_n\}.$$

Thus,  $N(t)$  can be regarded as the number of particles emitted (calls made) up to time  $t$ . It is an example of a stochastic process in continuous time. A typical path of  $N(t)$  is shown in fig. 2. It begins at the origin,  $N(0) = 0$  (no particles emitted at time 0), and is right-continuous, non-decreasing and piecewise constant with jumps of size 1 at the times  $\xi_n$ .

FIGURE 2. A typical path of  $N(t)$  and the jump times  $\xi_n$

What is the distribution of  $N(t)$ ? To answer this question we need to recall the definition of the Poisson distribution.

**DEFINITION 3.6 (Poisson Distribution).** A random variable  $\nu$  has the Poisson distribution with parameter  $\alpha > 0$  if

$$(31) \quad P\{\nu = n\} = e^{-\alpha} \frac{\alpha^n}{n!}$$

for any  $n = 0, 1, 2, \dots$

The probabilities  $P\{\nu = n\}$  for various values of  $\alpha$  are shown in fig. 3.

FIGURE 3. Poisson distribution with parameter  $\alpha$

**Proposition 3.7 (Poisson Process Distribution).**  $N(t)$  has the Poisson distribution with parameter  $\lambda t$ ,

$$(32) \quad P\{N(t) = n\} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

PROOF. First of all, observe that

$$(33) \quad \{N(t) < n\} = \{\xi_n > t\}.$$

It suffices to compute the probability of this event for any  $n$  because

$$(6.3) \quad \begin{aligned} P\{N(t) = n\} &= P\{N(t) < n+1\} - P\{N(t) < n\} \\ &= P\{\xi_{n+1} > t\} - P\{\xi_n > t\}. \end{aligned}$$

We shall prove by induction on  $n$  that

$$(6.4) \quad P\{\xi_n > t\} = e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}.$$

For  $n = 1$ :

$$P\{\xi_1 > t\} = P\{\eta_1 > t\} = e^{-\lambda t}.$$

Next, suppose that ?? holds for some  $n$ . Then, expressing  $\xi_{n+1}$  as the sum of the independent random variables  $\xi_n$  and  $\eta_{n+1}$ , we compute

$$(34) \quad \begin{aligned} P\{\xi_{n+1} > t\} &= P\{\xi_n + \eta_{n+1} > t\} \\ &= P\{\eta_{n+1} > t\} + P\{\xi_n > t - \eta_{n+1}, t \geq \eta_{n+1} > 0\} \\ &= e^{-\lambda t} + \int_0^t P\{\xi_n > t - s\} f_{\eta_{n+1}}(s) ds \\ &= e^{-\lambda t} + \int_0^t e^{-\lambda(t-s)} \sum_{k=0}^{n-1} \frac{(\lambda(t-s))^k}{k!} \lambda e^{-\lambda s} ds \\ &= e^{-\lambda t} + e^{-\lambda t} \sum_{k=0}^{n-1} \frac{\lambda^{k+1}}{k!} \int_0^t (t-s)^k ds \\ &= e^{-\lambda t} \sum_{k=0}^n \frac{(\lambda t)^k}{k!}, \end{aligned}$$

where  $f_{\eta_{n+1}}(s)$  is the density of  $\eta_{n+1}$ . By induction, ?? holds for any  $n$ . Now apply ?? to complete the proof.

EXERCISE 3.6. What is the expectation of  $N(t)$ ?

**Hint:** What are the possible values of  $N(t)$ ? What are the corresponding probabilities? Can you compute the expectation from these? To simplify the result use the Taylor expansion of  $e^x$ .

SOLUTION 3.6. Since  $N(t)$  has the Poisson distribution with parameter  $\lambda t$  we have  $E(N(t)) = \lambda t$ . Indeed

$$(35) \quad \begin{aligned} E(N(t)) &= \sum_{n=0}^{\infty} n P\{N(t) = n\} = \sum_{n=0}^{\infty} n e^{-\lambda t} \frac{(\lambda t)^n}{n!} \\ &= \lambda t e^{-\lambda t} \sum_{n=1}^{\infty} \frac{(\lambda t)^{n-1}}{(n-1)!} = \lambda t e^{-\lambda t} e^{\lambda t} = \lambda t. \end{aligned}$$

EXERCISE 3.7. Compute  $P\{N(s) = 1, N(t) = 2\}$  for any  $0 \leq s < t$ .

**Hint:** Express  $\{N(s) = 1, N(t) = 2\}$  as  $\{\eta_1 \leq s < \eta_1 + \eta_2 \leq t < \eta_1 + \eta_2 + \eta_3\}$ . You can compute the probability of the latter, since  $\eta_1, \eta_2, \eta_3$  are exponentially distributed and independent.

SOLUTION 3.7. Using the fact that  $\eta_1, \eta_2, \dots$  are independent and exponentially distributed, we obtain

$$\begin{aligned}
 P\{N(s) = 1, N(t) = 2\} &= P\{\xi_1 \leq s < \xi_2 \leq t < \xi_3\} \\
 &= P\{\eta_1 \leq s < \eta_1 + \eta_2 \leq t < \eta_1 + \eta_2 + \eta_3\} \\
 &= \int_0^s P\{s < u + \eta_2 \leq t < u + \eta_2 + \eta_3\} \lambda e^{-\lambda u} du \\
 (36) \quad &= \int_0^s \left( \int_{s-u}^{t-u} P\{t < u + v + \eta_3\} \lambda e^{-\lambda v} dv \right) \lambda e^{-\lambda u} du \\
 &= \int_0^s \left( \int_{s-u}^{t-u} e^{-\lambda(t-u-v)} \lambda e^{-\lambda v} dv \right) \lambda e^{-\lambda u} du \\
 &= \lambda^2 e^{-\lambda t} s(t-s).
 \end{aligned}$$

**2.3. Poisson Process Starts from Scratch at Time  $t$ .** Imagine that you are to take part in an experiment to count the emissions of a radioactive particle. Unfortunately, in the excitement of proving the lack of memory property you forget about the engagement and arrive late to find that the experiment has already been running for time  $t$  and you have missed the first  $N(t)$  emissions. Determined to make the best of it, you start counting right away, so at time  $t + s$  you will have registered  $N(t + s) - N(t)$  emissions. It will now be necessary to discuss  $N(t + s) - N(t)$  instead of  $N(s)$  in your report.

What are the properties of  $N(t + s) - N(t)$ ? Perhaps you can guess something from the physical picture? After all, a sample of radioactive material will keep emitting particles no matter whether anyone cares to count them or not. So the moment when someone starts counting does not seem important. You can expect  $N(t + s) - N(t)$  to behave in a similar way as  $N(s)$ . And because radioactive emissions have no memory of the past,  $N(t + s) - N(t)$  should be independent of  $N(t)$ .

To study this conjecture recall the construction of a Poisson process  $N(t)$  based on a sequence of independent random variables  $\eta_1, \eta_2, \dots$ , all having the same exponential distribution. We shall try to represent  $N(t + s) - N(t)$  in a similar way.

Let us put

$$(37) \quad \eta_1^t := \xi_{N(t)+1} - t, \quad \eta_n^t := \eta_{N(t)+n}, \quad n = 2, 3, \dots,$$

see fig. 4. These are the times between the jumps of  $N(t + s) - N(t)$ . Then we define

$$(38) \quad \xi_n^t = \eta_1^t + \dots + \eta_n^t,$$

$$(39) \quad N^t(s) = \max\{n : \xi_n^t \leq s\}.$$

FIGURE 4. The random variables  $\eta_1^t, \eta_2^t, \eta_3^t, \dots$

EXERCISE 3.8. *Show that*

$$(40) \quad N^t(s) = N(t+s) - N(t).$$

**Hint:** *First show that  $\xi_n^t = \xi_{N(t)+n} - t$ .*

SOLUTION 3.8. *Since*

$$(41) \quad \begin{aligned} \xi_n^t &= \eta_1^t + \cdots + \eta_n^t \\ &= \xi_{N(t)+1} - t + \eta_{N(t)+2} + \cdots + \eta_{N(t)+n} \\ &= \xi_{N(t)+n} - t, \end{aligned}$$

*it follows that*

$$(42) \quad \begin{aligned} N^t(s) &= \max\{n : \xi_n^t \leq s\} \\ &= \max\{n : \xi_{N(t)+n} \leq t+s\} \\ &= \max\{n : \xi_n \leq t+s\} - N(t) \\ &= N(t+s) - N(t). \end{aligned}$$

If we can show that  $\eta_1^t, \eta_2^t, \dots$  are independent random variables having the same exponential distribution as  $\eta_1, \eta_2, \dots$ , it will mean that  $N(t+s) - N(t)$  is a Poisson process with the same probability distribution as  $N(s)$ . Moreover, if the  $\eta_n^t$  turn out to be independent of  $N(t)$ , it will imply that  $N(t+s) - N(t)$  is also independent of  $N(t)$ .

Before setting about this task beware of one common mistake. It is sometimes claimed that the times  $\eta_1^t, \eta_2^t, \eta_3^t, \dots$  between the jumps of  $N(t+s) - N(t)$  are equal to  $\xi_{n+1} - t, \eta_{n+2}, \eta_{n+3}, \dots$  for some  $n$ . Hence  $\eta_1^t, \eta_2^t, \eta_3^t, \dots$  are independent because the random variables  $\xi_{n+1}, \eta_{n+2}, \eta_{n+3}, \dots$  are. The flaw in this is that, in fact,  $\eta_1^t, \eta_2^t, \eta_3^t, \dots$  are equal to  $\xi_{n+1} - t, \eta_{n+2}, \eta_{n+3}, \dots$  only on the set  $\{N(t) = n\}$ . However, the argument can be saved by conditioning with respect to  $N(t)$ . Our task becomes an exercise in computing conditional probabilities.

EXERCISE 3.9. *Show that*

$$(43) \quad P\{\eta_1^t > s \mid N(t)\} = P\{\eta_1 > s\}.$$

**Hint:** *It suffices (why?) to verify that*

$$P\{\eta_1^t > s, N(t) = n\} = P\{\eta_1 > s\}P\{N(t) = n\}$$

*for any  $n$ . To this end, write the sets  $\{N(t) = n\}$  and  $\{\eta_1^t > s, N(t) = n\}$  in terms of  $\xi_n$  and  $\eta_{n+1}$ , which are independent random variables, and use the lack of memory property for  $\eta_{n+1}$ .*

SOLUTION 3.9. *It is easily verified that*

$$\begin{aligned} \{N(t) = n\} &= \{\eta_{n+1} > t - \xi_n, t \geq \xi_n\} \\ \{\eta_1^t > s, N(t) = n\} &= \{\eta_{n+1} > s + t - \xi_n, t \geq \xi_n\} \end{aligned}$$

Since  $\xi_n, \eta_{n+1}$  are independent and  $\eta_{n+1}$  satisfies the lack of memory property from Exercise 3.3,

$$\begin{aligned}
 P\{\eta_1^t > s, N(t) = n\} &= P\{\eta_{n+1} > s + t - \xi_n, t \geq \xi_n\} \\
 &= \int_{-\infty}^t P\{\eta_{n+1} > s + t - u\} P_{\xi_n}(du) \\
 (44) \quad &= P\{\eta_{n+1} > s\} \int_{-\infty}^t P\{\eta_{n+1} > t - u\} P_{\xi_n}(du) \\
 &= P\{\eta_{n+1} > s\} P\{\eta_{n+1} > t - \xi_n, t \geq \xi_n\} \\
 &= P\{\eta_{n+1} > s\} P\{N(t) = n\} \\
 &= P\{\eta_1 > s\} P\{N(t) = n\}.
 \end{aligned}$$

The last equality holds because  $\eta_{n+1}$  has the same distribution as  $\eta_1$ . Now divide both sides by  $P\{N(t) = n\}$  to get

$$P\{\eta_1^t > s \mid N(t) = n\} = P\{\eta_1 > s\}$$

for any  $n = 0, 1, 2, \dots$ . As a result,

$$P\{\eta_1^t > s \mid N(t)\} = P\{\eta_1 > s\}$$

because  $N(t)$  is a discrete random variable with values  $0, 1, 2, \dots$ .

EXERCISE 3.10. Show that

$$(45) \quad P\{\eta_1^t > s_1, \dots, \eta_k^t > s_k \mid N(t)\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\}.$$

**Hint:** Verify that

$$P\{\eta_1^t > s_1, \eta_2^t > s_2, \dots, \eta_k^t > s_k, N(t) = n\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\} P\{N(t) = n\}$$

for any  $n$ . This is done in Exercise 3.9 for  $k = 1$ .

SOLUTION 3.10. As in Solution 3.9,

$$\{N(t) = n\} = \{\eta_{n+1} > t - \xi_n, t \geq \xi_n\},$$

$$\{\eta_1^t > s_1, N(t) = n\} = \{\eta_{n+1} > s_1 + t - \xi_n, t \geq \xi_n\},$$

and, more generally,

$$\begin{aligned}
 (46) \quad &\{\eta_1^t > s_1, \eta_2^t > s_2, \dots, \eta_k^t > s_k, N(t) = n\} \\
 &= \{\eta_{n+1} > s_1 + t - \xi_n, t \geq \xi_n\} \cap \{\eta_{n+2} > s_2\} \cap \cdots \cap \{\eta_{n+k} > s_k\}.
 \end{aligned}$$

Since  $\xi_n, \eta_{n+1}, \dots, \eta_{n+k}$  are independent and  $\eta_{n+2}, \dots, \eta_{n+k}$  have the same distribution as  $\eta_2, \dots, \eta_k$ , using Exercise 3.9 we find that

$$\begin{aligned}
 (47) \quad &P\{\eta_1^t > s_1, \eta_2^t > s_2, \dots, \eta_k^t > s_k, N(t) = n\} \\
 &= P\{\eta_{n+1} > s_1 + t - \xi_n, \xi_n \leq t\} P\{\eta_{n+2} > s_2\} \cdots P\{\eta_{n+k} > s_k\} \\
 &= P\{\eta_1^t > s_1, N(t) = n\} P\{\eta_2 > s_2\} \cdots P\{\eta_k > s_k\} \\
 &= P\{\eta_1^t > s_1 \mid N(t) = n\} P\{\eta_2 > s_2\} \cdots P\{\eta_k > s_k\} P\{N(t) = n\} \\
 &= P\{\eta_1 > s_1\} P\{\eta_2 > s_2\} \cdots P\{\eta_k > s_k\} P\{N(t) = n\}.
 \end{aligned}$$

As in Solution 3.9, this implies the desired equality.



EXERCISE 3.11. From the formula in Exercise 3.10 deduce that the random variables  $\eta_n^t$  and  $N(t)$  are independent, and the  $\eta_n^t$  have the same probability distribution as  $\eta_n$ , i.e. exponential with parameter  $\lambda$ .

**Hint:** What do you get if you take the expectation on both sides of the equality in Exercise 3.10? Can you deduce the probability distribution of  $\eta_n^t$ ? Can you see that the  $\eta_n^t$  are independent?

To prove that the  $\eta_n^t$  are independent of  $N(t)$  you need to be a little more careful and integrate over  $\{N(t) = n\}$  instead of taking the expectation.

SOLUTION 3.11. Take the expectation on both sides of the equality in Exercise 3.10 to find that

$$(48) \quad P\{\eta_1^t > s_1, \dots, \eta_k^t > s_k\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\}.$$

If all the numbers  $s_n$  except perhaps one are zero, it follows that

$$P\{\eta_n^t > s_n\} = P\{\eta_n > s_n\}, \quad n = 1, \dots, k,$$

so the random variables  $\eta_n^t$  have the same distribution as  $\eta_n$ . Inserting this back into the first equality, we obtain

$$P\{\eta_1^t > s_1, \dots, \eta_k^t > s_k\} = P\{\eta_1^t > s_1\} \cdots P\{\eta_k^t > s_k\},$$

so the  $\eta_n^t$  are independent.

To prove that the  $\eta_n^t$  are independent of  $N(t)$  integrate the formula in Exercise 3.10 over  $\{N(t) = n\}$  and multiply by  $P\{N(t) = n\}$  to get

$$P\{\eta_1^t > s_1, \dots, \eta_k^t > s_k, N(t) = n\} = P\{\eta_1 > s_1\} \cdots P\{\eta_k > s_k\} P\{N(t) = n\}.$$

But  $P\{\eta_n^t > s_n\} = P\{\eta_n > s_n\}$ , hence  $N(t)$  and the  $\eta_n^t$  are independent.

Because  $N(t+s) - N(t)$  can be defined in terms of  $\eta_1^t, \eta_2^t, \dots$  in the same way the original Poisson process  $N(t)$  is defined in terms of  $\eta_1, \eta_2, \dots$ , the result in Exercise 3.11 proves the theorem below.

**Theorem 3.8** (Poisson Process Stationarity). For any fixed  $t \geq 0$

$$N^t(s) = N(t+s) - N(t), \quad s \geq 0$$

is a Poisson process independent of  $N(t)$  with the same probability law as  $N(s)$ .

That is to say, for any  $s, t \geq 0$  the increment  $N(t+s) - N(t)$  is independent of  $N(t)$  and has the same probability distribution as  $N(s)$ . The assertion can be generalized to several increments, resulting in the following important theorem.

**Theorem 3.9** (Poisson Process Independent Increments). For any  $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$  the increments

$$N(t_1), N(t_2) - N(t_1), N(t_3) - N(t_2), \dots, N(t_n) - N(t_{n-1})$$

are independent and have the same probability distribution as

$$N(t_1), N(t_2 - t_1), N(t_3 - t_2), \dots, N(t_n - t_{n-1}).$$

PROOF. From definition 3.8 it follows immediately that each increment  $N(t_i) - N(t_{i-1})$  has the same distribution as  $N(t_i - t_{i-1})$  for  $i = 1, \dots, n$ .

It remains to prove independence. This can be done by induction. The case when  $n = 2$  is covered by definition 3.8. Now suppose that independence holds for  $n$  increments of a Poisson process for some  $n \geq 2$ . Take any sequence

$$0 \leq t_1 \leq t_2 \leq \cdots \leq t_n \leq t_{n+1}.$$

By the induction hypothesis

$$N(t_{n+1}) - N(t_n), \dots, N(t_2) - N(t_1)$$

are independent, since they can be regarded as increments of

$$N^{t_1}(s) = N(t_1 + s) - N(t_1),$$

which is a Poisson process by definition 3.8. By the same theorem these increments are independent of  $N(t_1)$ . It follows that the  $n + 1$  random variables

$$N(t_{n+1}) - N(t_n), \dots, N(t_2) - N(t_1), N(t_1)$$

are independent, completing the proof.

**DEFINITION 3.10 (Independent Increments).** We say that a stochastic process  $\xi(t)$ , where  $t \in T$ , has independent increments if

$$\xi(t_1) - \xi(t_0), \dots, \xi(t_n) - \xi(t_{n-1})$$

are independent for any  $t_0 < t_1 < \cdots < t_n$  such that  $t_0, t_1, \dots, t_n \in T$ .

**DEFINITION 3.11 (Stationary Increments).** A stochastic process  $\xi(t)$ , where  $t \in T$ , is said to have stationary increments if for any  $s, t \in T$  the probability distribution of  $\xi(t+h) - \xi(s+h)$  is the same for each  $h$  such that  $s+h, t+h \in T$ .

definition 3.9 implies that the Poisson process has stationary independent increments. The result in the next exercise is also a consequence of definition 3.9.

**EXERCISE 3.12.** Show that  $N(t) - \lambda t$  is a martingale with respect to the filtration  $\mathcal{F}_t$  generated by the family of random variables  $\{N(s) : s \in [0, t]\}$ .

**Hint:** Observe that  $N(t) - N(s)$  is independent of  $\mathcal{F}_s$  by definition 3.9.

**SOLUTION 3.12.** We need to verify conditions 1), 2), 3) of definition 3.3. Clearly,  $N(t) - \lambda t$  is  $\mathcal{F}_t$ -measurable. By Exercise 3.6

$$E(|N(t)|) = E(N(t)) = \lambda t < \infty,$$

which means that  $N(t)$  is integrable, and so is  $N(t) - \lambda t$ .

definition 3.9 implies that  $N(t) - N(s)$  is independent of  $\mathcal{F}_s$  for any  $0 \leq s \leq t$ , so

$$E(N(t) - N(s) \mid \mathcal{F}_s) = E(N(t) - N(s)) = E(N(t)) - E(N(s)) = \lambda t - \lambda s.$$

It follows that

$$E(N(t) - \lambda t \mid \mathcal{F}_s) = E(N(s) - \lambda s \mid \mathcal{F}_s) = N(s) - \lambda s,$$

completing the proof.

### 2.4. Various Exercises on the Poisson Process.

EXERCISE 3.13. Show that  $\xi_0 < \xi_1 < \xi_2 < \dots$  a.s.

**Hint:** What is the probability of the event  $\{\xi_{n-1} < \xi_n\} = \{\eta_n > 0\}$ ? What is the probability of the intersection of all such events?

SOLUTION 3.13. Since  $\eta_n = \xi_n - \xi_{n-1}$  and  $P\{\eta_n > 0\} = e^0 = 1$ ,

$$P\{\xi_0 < \xi_1 < \xi_2 < \dots\} = P\left(\bigcap_{n=1}^{\infty} \{\eta_n > 0\}\right) = 1.$$

Here we have used the property that if  $P(A_n) = 1$  for all  $n = 1, 2, \dots$ , then  $P(\bigcap_{n=1}^{\infty} A_n) = 1$ .

EXERCISE 3.14. Show that  $\lim_{n \rightarrow \infty} \xi_n = \infty$  a.s.

**Hint:** If  $\lim_{n \rightarrow \infty} \xi_n < \infty$ , then the sequence  $\eta_1, \eta_2, \dots$  of independent random variables, all having the same exponential distribution, must be bounded. What is the probability that such a sequence is bounded? Begin with computing the probability  $P\{\eta_1 \leq m, \dots, \eta_n \leq m\}$  for any fixed  $m > 0$ .

Although it is instructive to estimate  $P\{\lim_{n \rightarrow \infty} \xi_n < \infty\}$  in this way, there is a more elegant argument based on the strong law of large numbers. What does the law of large numbers tell us about the limit of  $\frac{\xi_n}{n}$  as  $n \rightarrow \infty$ ?

SOLUTION 3.14. Since

$$\lim_{n \rightarrow \infty} \xi_n = \sum_{n=1}^{\infty} \eta_n,$$

it follows that

$$\begin{aligned} \left\{ \lim_{n \rightarrow \infty} \xi_n < \infty \right\} &\subset \{ \eta_1, \eta_2, \dots \text{ is a bounded sequence} \} \\ (49) \qquad &= \bigcup_{m=1}^{\infty} \bigcap_{n=1}^{\infty} \{ \eta_n \leq m \}. \end{aligned}$$

Let us compute the probability of this event. Because  $\bigcap_{n=1}^N \{\eta_n \leq m\}$ ,  $N = 1, 2, \dots$  is a contracting sequence of events,

$$\begin{aligned} P\left(\bigcap_{n=1}^{\infty} \{\eta_n \leq m\}\right) &= \lim_{N \rightarrow \infty} P\left(\bigcap_{n=1}^N \{\eta_n \leq m\}\right) \\ (50) \qquad &= \lim_{N \rightarrow \infty} \prod_{n=1}^N P\{\eta_n \leq m\} \\ &= \lim_{N \rightarrow \infty} (1 - e^{-\lambda m})^N \\ &= 0. \end{aligned}$$

It follows that

$$\begin{aligned}
 (51) \quad P\left(\lim_{n \rightarrow \infty} \xi_n < \infty\right) &\leq P\left(\bigcup_{m=1}^{\infty} \bigcap_{n=1}^{\infty} \{\eta_n \leq m\}\right) \\
 &\leq \sum_{m=1}^{\infty} P\left(\bigcap_{n=1}^{\infty} \{\eta_n \leq m\}\right) \\
 &= 0,
 \end{aligned}$$

completing the proof.

While it is instructive to work through the above estimates, there exists a much more elegant argument. By the strong law of large numbers

$$\lim_{n \rightarrow \infty} \frac{\xi_n}{n} = \frac{1}{\lambda} \quad a.s.$$

Here  $\frac{1}{\lambda}$  is the expectation of each of the independent identically distributed random variables  $\eta_n$  (see Exercise 3.2). It follows that

$$\lim_{n \rightarrow \infty} \xi_n = \infty \quad a.s.,$$

as required.

EXERCISE 3.15. Show that  $\xi_n$  has absolutely continuous distribution with density

$$(52) \quad f_n(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}$$

with parameters  $n$  and  $\lambda$ . The density  $f_n(t)$  of the gamma distribution is shown in fig. 5 for  $n = 2, 4$  and  $\lambda = 1$ .

FIGURE 5. Density  $f_n(t)$  of the gamma distribution with parameters  $n = 2, \lambda = 1$  and  $n = 4, \lambda = 1$

**Hint:** Use the formula for  $P\{\xi_n > t\}$  in the proof of definition 3.7 to find the distribution function of  $\xi_n$ . Is this function differentiable? What is the derivative?

SOLUTION 3.15. In the proof of definition 3.7 it was shown that

$$P\{\xi_n > t\} = e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}$$

for  $t \geq 0$ , see ???. Therefore the distribution function

$$F_n(t) = P\{\xi_n \leq t\} = 1 - P\{\xi_n > t\} = e^{-\lambda t} \sum_{k=n}^{\infty} \frac{(\lambda t)^k}{k!}$$

of  $\xi_n$  is differentiable, the density  $f_n$  of  $\xi_n$  being

$$\begin{aligned}
 f_n(t) &= \frac{d}{dt} F_n(t) \\
 (53) \quad &= -\lambda e^{-\lambda t} \sum_{k=n}^{\infty} \frac{(\lambda t)^k}{k!} + \lambda e^{-\lambda t} \sum_{k=n}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!} \\
 &= \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}
 \end{aligned}$$

for  $t > 0$ , and clearly  $f_n(t) = 0$  for  $t \leq 0$ .

EXERCISE 3.16. Prove that

$$\lim_{t \rightarrow \infty} N(t) = \infty \quad a.s.$$

**Hint:** What is the limit of  $P\{N(k) \geq n\}$  as  $k \rightarrow \infty$ ? Can you express  $\{\lim_{t \rightarrow \infty} N(t) = \infty\}$  in terms of the sets  $\{N(k) \geq n\}$ ?

SOLUTION 3.16. Because  $N(t)$  has non-decreasing trajectories

$$\left\{ \lim_{t \rightarrow \infty} N(t) = \infty \right\} = \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} \{N(k) \geq n\}.$$

Also,  $\{N(k) \geq n\}, k = 1, 2, \dots$  is an expanding sequence of events and

$$\begin{aligned}
 (54) \quad P\{N(k) \geq n\} &= e^{-\lambda k} \sum_{i=n}^{\infty} \frac{(\lambda k)^i}{i!} \\
 &= 1 - e^{-\lambda k} \sum_{i=0}^{n-1} \frac{(\lambda k)^i}{i!} \rightarrow 1 \quad \text{as } k \rightarrow \infty.
 \end{aligned}$$

It follows that

$$P \left\{ \bigcup_{k=1}^{\infty} \{N(k) \geq n\} \right\} = 1,$$

so

$$P \left\{ \lim_{t \rightarrow \infty} N(t) = \infty \right\} = P \left\{ \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} \{N(k) \geq n\} \right\} = 1.$$

EXERCISE 3.17. Verify that

$$P\{N(t) \text{ is odd}\} = e^{-\lambda t} \sinh(\lambda t),$$

$$P\{N(t) \text{ is even}\} = e^{-\lambda t} \cosh(\lambda t).$$

**Hint:** What is the probability that  $N(t) = 2n + 1$ ? Compare this to the  $n$ -th term of the Taylor expansion of  $\sinh(\lambda t)$ .

SOLUTION 3.17. Since

$$\begin{aligned}
 \sinh(x) &= \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \\
 \cosh(x) &= \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!},
 \end{aligned}$$

we have

$$\begin{aligned}
 P\{N(t) \text{ is odd}\} &= \sum_{n=0}^{\infty} P\{N(t) = 2n+1\} \\
 (55) \qquad &= \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^{2n+1}}{(2n+1)!} \\
 &= e^{-\lambda t} \sinh(\lambda t),
 \end{aligned}$$

$$\begin{aligned}
 P\{N(t) \text{ is even}\} &= \sum_{n=0}^{\infty} P\{N(t) = 2n\} \\
 (56) \qquad &= \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^{2n}}{(2n)!} \\
 &= e^{-\lambda t} \cosh(\lambda t).
 \end{aligned}$$

EXERCISE 3.18. Show that

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \lambda \quad a.s.$$

if  $N(t)$  is a Poisson process with parameter  $\lambda$ .

**Hint:**  $N(n)$  is the sum of independent identically distributed random variables  $N(1), N(2) - N(1), \dots, N(n) - N(n-1)$ , so the strong law of large numbers can be applied to obtain the limit of  $N(n)/n$  as  $n \rightarrow \infty$ . Because  $N(t)$  is non-decreasing, the limit will not be affected if  $n$  is replaced by a continuous parameter  $t \geq 0$ .

SOLUTION 3.18. We can write

$$N(n) = N(1) + (N(2) - N(1)) + \dots + (N(n) - N(n-1)),$$

where  $N(1), N(2) - N(1), N(3) - N(2), \dots$  is a sequence of independent identically distributed random variables with expectation

$$E(N(1)) = E(N(2) - N(1)) = E(N(3) - N(2)) = \dots = \lambda.$$

By the strong law of large numbers

$$(6.8) \qquad \lim_{n \rightarrow \infty} \frac{N(n)}{n} = \lambda \quad a.s.$$

Now, if  $n \leq t \leq n+1$ , then  $N(n) \leq N(t) \leq N(n+1)$  and

$$\frac{N(n)}{n+1} \leq \frac{N(t)}{t} \leq \frac{N(n+1)}{n}.$$

By eq. (6.8) both sides tend to  $\lambda$  as  $n \rightarrow \infty$ , implying that

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \lambda \quad a.s.$$

### 3. Brownian Motion

Imagine a cloud of smoke in completely still air. In time, the cloud will spread over a large volume, the concentration of smoke varying in a smooth manner. However, if a single smoke particle is observed, its path turns out to be extremely rough due to frequent collisions with other particles. This exemplifies two aspects of the same phenomenon called diffusion: erratic particle trajectories at the microscopic level, giving rise to a very smooth behaviour of the density of the whole ensemble of particles. The Wiener process  $W(t)$  defined below is a mathematical device designed as a model of the motion of individual diffusing particles. In particular, its paths exhibit similar erratic behaviour to the trajectories of real smoke particles. Meanwhile, the density  $f_{W(t)}$  of the random variable  $W(t)$  is very smooth, given by the exponential function

$$(57) \quad f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}},$$

which is a solution of the diffusion equation

$$(58) \quad \frac{\partial f}{\partial t} = \frac{1}{2} \frac{\partial^2 f}{\partial x^2}$$

and can be interpreted as the density at time  $t$  of a cloud of smoke issuing from a single point source at time 0. The Wiener process  $W(t)$  is also associated with the name of the British botanist Robert Brown, who around 1827 observed the random movement of pollen particles in water. We shall study mainly the one-dimensional Wiener process, which can be thought of as the projection of the position of a smoke particle onto one of the axes of a coordinate system.

Apart from describing the motion of diffusing particles, the Wiener process is widely applied in mathematical models involving various noisy systems, for example, the behaviour of asset prices at the stock exchange. If the noise in the system is due to a multitude of independent random changes, then the Central Limit Theorem predicts that the net result will have the normal distribution, a property shared by the increments  $W(t) - W(s)$  of the Wiener process. This is one of the main reasons of the widespread use of  $W(t)$  in mathematical models.

#### 3.1. Definition and Basic Properties.

**DEFINITION 3.12** (Wiener Process / Brownian Motion). *The Wiener process (or Brownian motion) is a stochastic process  $W(t)$  with values in  $\mathbb{R}$  defined for  $t \in [0, \infty)$  such that*

- (1)  $W(0) = 0$  a.s.;
- (2) the sample paths  $t \mapsto W(t)$  are a.s. continuous;
- (3) for any finite sequence of times  $0 < t_1 < \dots < t_n$  and Borel sets  $A_1, \dots, A_n \subset \mathbb{R}$

$$\begin{aligned} & P\{W(t_1) \in A_1, \dots, W(t_n) \in A_n\} \\ &= \int_{A_1} \dots \int_{A_n} p(t_1, 0, x_1) p(t_2 - t_1, x_1, x_2) \dots \\ & \quad \dots p(t_n - t_{n-1}, x_{n-1}, x_n) dx_1 \dots dx_n, \end{aligned}$$

where

$$(6.5) \quad p(t, x, y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}}$$

defined for any  $x, y \in \mathbb{R}$  and  $t > 0$  is called the transition density.

A typical sample path of the Wiener process is shown in fig. 6.

FIGURE 6. A typical path of  $W(t)$

EXERCISE 3.19. Show that

$$f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$$

is the probability density of  $W(t)$  and find the expectation and variance of  $W(t)$ .

**Hint:** The density of  $W(t)$  can be obtained from condition 3) of definition 3.12 written for a single time  $t$  and a single Borel set. You will need the formula

$$\int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} dx = \sqrt{2\pi}$$

to compute the integrals in the expressions for the expectation and variance.

SOLUTION 3.19. Condition 3) of definition 3.12 implies that

$$f_{W(t)}(x) = p(t, 0, x)$$

is the density of  $W(t)$ . Therefore, integrating by parts, we can compute the expectation

$$(59) \quad \begin{aligned} E(W(t)) &= \int_{-\infty}^{+\infty} xp(t, 0, x) dx \\ &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} xe^{-\frac{x^2}{2t}} dx \\ &= -\frac{t}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} \frac{d}{dx} e^{-\frac{x^2}{2t}} dx \\ &= -\frac{t}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} \Big|_{-\infty}^{+\infty} = 0, \end{aligned}$$

and variance

$$(60) \quad \begin{aligned} E((W(t))^2) &= \int_{-\infty}^{+\infty} x^2 p(t, 0, x) dx \\ &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} x^2 e^{-\frac{x^2}{2t}} dx \\ &= -\frac{t}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} x \frac{d}{dx} e^{-\frac{x^2}{2t}} dx \\ &= -\frac{t}{\sqrt{2\pi t}} x e^{-\frac{x^2}{2t}} \Big|_{-\infty}^{+\infty} + \frac{t}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2t}} dx \\ &= 0 + \frac{t}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{u^2}{2}} du = t. \end{aligned}$$



We have used the substitution  $u = \frac{x}{\sqrt{t}}$  and the formula stated in the hint.

REMARK 3.1. The results of Exercise 3.19 mean that  $W(t)$  has the normal distribution with mean 0 and variance  $t$ .

EXERCISE 3.20. Show that

$$E(W(s)W(t)) = \min\{s, t\}.$$

**Hint:** The joint density of  $W(s)$  and  $W(t)$  will be needed. It can be found from condition 3) of definition 3.12 written for two times  $s$  and  $t$  and two Borel sets.

SOLUTION 3.20. Suppose that  $s < t$ . Condition 3) of definition 3.12 implies that the joint density of  $W(s)$  and  $W(t)$  is

$$f_{W(s), W(t)}(x, y) = p(s, 0, x)p(t - s, x, y).$$

It follows that

$$\begin{aligned} E(W(s)W(t)) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xyp(s, 0, x)p(t - s, x, y) dx dy \\ (61) \qquad &= \int_{-\infty}^{+\infty} xp(s, 0, x) \left( \int_{-\infty}^{+\infty} yp(t - s, x, y) dy \right) dx \\ &= \int_{-\infty}^{+\infty} x^2 p(s, 0, x) dx = s. \end{aligned}$$

This is because by the results of Exercise 3.19

$$\begin{aligned} \int_{-\infty}^{+\infty} yp(t - s, x, y) dy &= \int_{-\infty}^{+\infty} (x + u)p(t - s, x, x + u) du \\ (62) \qquad &= \int_{-\infty}^{+\infty} (x + u)p(t - s, 0, u) du \\ &= x \int_{-\infty}^{+\infty} p(t - s, 0, u) du + \int_{-\infty}^{+\infty} up(t - s, 0, u) du \\ &= x + 0 = x, \end{aligned}$$

and

$$\int_{-\infty}^{+\infty} x^2 p(s, 0, x) dx = s.$$

It follows that for arbitrary  $s, t \geq 0$

$$E(W(s)W(t)) = \min\{s, t\}.$$

EXERCISE 3.21. Show that

$$E(|W(t) - W(s)|^2) = |t - s|.$$

**Hint:** Expand the square and use the formula in Exercise 3.20.

SOLUTION 3.21. Suppose that  $s \leq t$ . Then by Exercise 3.20

$$\begin{aligned} (63) \qquad E(|W(t) - W(s)|^2) &= E(W(t)^2) - 2E(W(s)W(t)) + E(W(s)^2) \\ &= t - 2s + s = t - s. \end{aligned}$$

In general, for arbitrary  $s, t \geq 0$

$$E(|W(t) - W(s)|^2) = |t - s|.$$

EXERCISE 3.22. Compute the characteristic function  $E(\exp(i\lambda W(t)))$  for any  $\lambda \in \mathbb{R}$ .

**Hint:** Use the density of  $W(t)$  found in Exercise 3.19.

SOLUTION 3.22. Using the density  $f_{W(t)}(x) = p(t, 0, x)$  of  $W(t)$ , we compute

$$\begin{aligned} E(\exp(i\lambda W(t))) &= \int_{-\infty}^{+\infty} e^{i\lambda x} p(t, 0, x) dx \\ &= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} e^{i\lambda x} e^{-\frac{x^2}{2t}} dx \\ (64) \quad &= \frac{1}{\sqrt{2\pi t}} e^{-\frac{\lambda^2 t}{2}} \int_{-\infty}^{+\infty} e^{-\frac{(x-i\lambda t)^2}{2t}} dx \\ &= e^{-\frac{\lambda^2 t}{2}}. \end{aligned}$$

EXERCISE 3.23. Find  $E(W(t)^4)$ .

**Hint:** This can be done, for example, by expressing the expectation in terms of the density of  $W(t)$  and computing the resulting integral, or by computing the fourth derivative of the characteristic function of  $W(t)$  at 0. The second method is more efficient.

SOLUTION 3.23. Using the formula for the characteristic function of  $W(t)$  found in Exercise 3.22, we compute

$$\begin{aligned} E(W(t)^4) &= \left. \frac{d^4}{d\lambda^4} E(\exp(i\lambda W(t))) \right|_{\lambda=0} \\ (65) \quad &= \left. \frac{d^4}{d\lambda^4} e^{-\frac{1}{2}\lambda^2 t} \right|_{\lambda=0} \\ &= 3t^2. \end{aligned}$$

DEFINITION 3.13 ( $n$ -dimensional Wiener Process). We call  $W(t) = (W^1(t), \dots, W^n(t))$  an  $n$ -dimensional Wiener process if  $W^1(t), \dots, W^n(t)$  are independent  $\mathbb{R}$ -valued Wiener processes.

EXERCISE 3.24. For a two-dimensional Wiener process  $W(t) = (W^1(t), W^2(t))$  find the probability that  $|W(t)| < R$ , where  $R > 0$  and  $|x|$  is the Euclidean norm of  $x = (x^1, x^2)$  in  $\mathbb{R}^2$ , i.e.  $|x|^2 = (x^1)^2 + (x^2)^2$ .

**Hint:** Express the probability in terms of the joint density of  $W^1(t)$  and  $W^2(t)$ . Independence means that the joint density of  $W^1(t)$  and  $W^2(t)$  is the product of their respective densities, which are known from Exercise 3.19. It is convenient to use polar coordinates to compute the resulting integral over a disc.

**SOLUTION 3.24.** Since  $W^1(t), W^2(t)$  are independent, their joint density is the product of the densities of  $W^1(t)$  and  $W^2(t)$ . Therefore

$$\begin{aligned}
 P\{|W(t)| < R\} &= \int_{\{|x| < R\}} p(t, 0, x)p(t, 0, y) dx dy \\
 &= \frac{1}{2\pi t} \int_{\{|x| < R\}} e^{-\frac{x^2+y^2}{2t}} dx dy \\
 (66) \quad &= \frac{1}{2\pi t} \int_0^R \int_0^{2\pi} r e^{-\frac{r^2}{2t}} d\varphi dr \\
 &= - \int_0^R \frac{d}{dr} e^{-\frac{r^2}{2t}} dr \\
 &= 1 - e^{-\frac{R^2}{2t}}.
 \end{aligned}$$

We have used the polar coordinates  $r, \varphi$  to compute the integral.

### 3.2. Increments of Brownian Motion.

**Proposition 3.14** (Increments are Normal). *For any  $0 \leq s < t$  the increment  $W(t) - W(s)$  has the normal distribution with mean 0 and variance  $t - s$ .*

**PROOF.** By condition 3) of definition 3.12 the joint density of  $W(s), W(t)$  is

$$f_{W(s), W(t)}(x, y) = p(s, 0, x)p(t - s, x, y).$$

Hence, for any Borel set  $A$

$$\begin{aligned}
 P\{W(t) - W(s) \in A\} &= \int_{\{(x, y): y - x \in A\}} p(s, 0, x)p(t - s, x, y) dx dy \\
 &= \int_{-\infty}^{+\infty} p(s, 0, x) \left( \int_{\{y: y - x \in A\}} p(t - s, x, y) dy \right) dx \\
 &= \int_{-\infty}^{+\infty} p(s, 0, x) \left( \int_A p(t - s, x, x + u) du \right) dx \\
 (67) \quad &= \int_{-\infty}^{+\infty} p(s, 0, x) \left( \int_A p(t - s, 0, u) du \right) dx \\
 &= \int_A p(t - s, 0, u) du \int_{-\infty}^{+\infty} p(s, 0, x) dx \\
 &= \int_A p(t - s, 0, u) du.
 \end{aligned}$$

But  $f(u) = p(t - s, 0, u)$  is the density of the normal distribution with mean 0 and variance  $t - s$ , which proves the claim.

**Corollary 3.15.** *definition 3.14 implies that  $W(t)$  has stationary increments.*

**Proposition 3.16** (Independent Increments). *For any  $0 = t_0 \leq t_1 \leq \dots \leq t_n$  the increments*

$$W(t_1) - W(t_0), \dots, W(t_n) - W(t_{n-1})$$

*are independent.*

PROOF. From definition 3.14 we know that the increments of  $W(t)$  have the normal distribution. Because normally distributed random variables are independent if and only if they are uncorrelated, it suffices to verify that

$$E[(W(u) - W(t))(W(s) - W(r))] = 0$$

for any  $0 \leq r \leq s \leq t \leq u$ . But this follows immediately from Exercise 3.20:

$$\begin{aligned} E[(W(u) - W(t))(W(s) - W(r))] &= E(W(u)W(s)) - E(W(u)W(r)) \\ &\quad - E(W(t)W(s)) + E(W(t)W(r)) \\ (68) \qquad \qquad \qquad &= s - r - s + r \\ &= 0, \end{aligned}$$

as required.

**Corollary 3.17.** For any  $0 \leq s < t$  the increment  $W(t) - W(s)$  is independent of the  $\sigma$ -field  $\mathcal{F}_s = \sigma\{W(r) : 0 \leq r \leq s\}$ .

PROOF. By definition 3.16 the random variables  $W(t) - W(s)$  and  $W(r) - W(0) = W(r)$  are independent if  $0 \leq r \leq s \leq t$ . Because the  $\sigma$ -field  $\mathcal{F}_s$  is generated by such  $W(r)$ , it follows that  $W(t) - W(s)$  is independent of  $\mathcal{F}_s$ .

EXERCISE 3.25. Show that  $W(t)$  is a martingale with respect to the filtration  $\mathcal{F}_t$ .

**Hint:** Take advantage of the fact that  $W(t) - W(s)$  is independent of  $\mathcal{F}_s$  if  $s < t$ .

SOLUTION 3.25. For any  $0 \leq s < t$

$$\begin{aligned} E(W(t) \mid \mathcal{F}_s) &= E(W(t) - W(s) \mid \mathcal{F}_s) + E(W(s) \mid \mathcal{F}_s) \\ (69) \qquad \qquad &= E(W(t) - W(s)) + W(s) \\ &= W(s), \end{aligned}$$

since  $W(t) - W(s)$  is independent of  $\mathcal{F}_s$  by definition 3.17,  $W(s)$  is  $\mathcal{F}_s$ -measurable and  $E(W(t)) = E(W(s)) = 0$ .

EXERCISE 3.26. Show that  $|W(t)|^2 - t$  is a martingale with respect to the filtration  $\mathcal{F}_t$ .

**Hint:** Once again, use the fact that  $W(t) - W(s)$  is independent of  $\mathcal{F}_s$  if  $s < t$ .

SOLUTION 3.26. For any  $0 \leq s < t$

$$\begin{aligned} E(W(t)^2 \mid \mathcal{F}_s) &= E(|W(t) - W(s)|^2 \mid \mathcal{F}_s) + E(2W(t)W(s) \mid \mathcal{F}_s) \\ &\quad - E(W(s)^2 \mid \mathcal{F}_s) \\ (70) \qquad \qquad &= E(|W(t) - W(s)|^2) + 2W(s)E(W(t) \mid \mathcal{F}_s) - W(s)^2 \\ &= t - s + 2W(s)^2 - W(s)^2 \\ &= t - s + W(s)^2, \end{aligned}$$

since  $W(t) - W(s)$  is independent of  $\mathcal{F}_s$  and has the normal distribution with mean 0 and variance  $t - s$ ,  $W(s)$  is  $\mathcal{F}_s$ -measurable, and  $W(t)$  is a martingale. It follows that

$$E(W(t)^2 - t \mid \mathcal{F}_s) = W(s)^2 - s,$$

as required.

Let us state without proof the following useful characterization of the Wiener process in terms of its increments.

**Theorem 3.18** (Wiener Process Characterization). *A stochastic process  $W(t), t \geq 0$ , is a Wiener process if and only if the following conditions hold:*

- (1)  $W(0) = 0$  a.s.;
- (2) the sample paths  $t \mapsto W(t)$  are continuous a.s.;
- (3)  $W(t)$  has stationary independent increments;
- (4) the increment  $W(t) - W(s)$  has the normal distribution with mean 0 and variance  $t - s$  for any  $0 \leq s < t$ .

EXERCISE 3.27. Show that for any  $T > 0$

$$V(t) = W(t + T) - W(T)$$

is a Wiener process if  $W(t)$  is.

**Hint:** Are the increments of  $V(t)$  independent? What is their distribution? Does  $V(t)$  have continuous paths? Is it true that  $V(0) = 0$ ?

SOLUTION 3.27. For any  $0 \leq t_0 < t_1 < \dots < t_n$  the increments

$$V(t_n) - V(t_{n-1}), \dots, V(t_1) - V(t_0)$$

of  $V(t)$  are independent, since the increments

$$W(t_n + T) - W(t_{n-1} + T), \dots, W(t_1 + T) - W(t_0 + T)$$

of  $W(t)$  are independent. For any  $0 \leq s < t$  the increment  $V(t) - V(s)$  has the normal distribution with mean zero and variance  $t - s$ , since  $W(t + T) - W(s + T)$  does. Moreover, the paths  $t \mapsto V(t) = W(t + T) - W(T)$  are continuous and

$$V(0) = W(T) - W(T) = 0.$$

By definition 3.18  $V(t)$  is a Wiener process.

The Wiener process can also be characterized by its martingale properties. The following theorem is also given without proof.

**Theorem 3.19** (Lévy's Martingale Characterization). *Let  $W(t), t \geq 0$ , be a stochastic process and let  $\mathcal{F}_t = \sigma(W_s, s \leq t)$  be the filtration generated by it. Then  $W(t)$  is a Wiener process if and only if the following conditions hold:*

- (1)  $W(0) = 0$  a.s.;
- (2) the sample paths  $t \mapsto W(t)$  are continuous a.s.;
- (3)  $W(t)$  is a martingale with respect to the filtration  $\mathcal{F}_t$ ;
- (4)  $|W(t)|^2 - t$  is a martingale with respect to  $\mathcal{F}_t$ .

EXERCISE 3.28. Let  $c > 0$ . Show that  $V(t) = \frac{1}{c}W(c^2t)$  is a Wiener process if  $W(t)$  is.

**Hint:** Is  $V(t)$  a martingale? With respect to which filtration? Is  $|V(t)|^2 - t$  a martingale? Are the paths of  $V(t)$  continuous? Is it true that  $V(0) = 0$ ?

SOLUTION 3.28. It is clear that  $V(0) = \frac{1}{c}W(0) = 0$  a.s. and the paths  $t \mapsto V(t) = \frac{1}{c}W(c^2t)$  are a.s. continuous. We shall verify that  $V(t)$  and  $|V(t)|^2 - t$  are martingales with respect to the filtration

$$\begin{aligned} \mathcal{G}_t &= \sigma\{V(s) : 0 \leq s \leq t\} \\ &= \sigma\{W(c^2s) : 0 \leq s \leq t\} \\ &= \sigma\{W(s) : 0 \leq s \leq c^2t\} \\ &= \mathcal{F}_{c^2t}. \end{aligned} \tag{71}$$

Indeed, if  $s < t$ , then  $c^2s < c^2t$ , so

$$\begin{aligned}
 E(V(t) \mid \mathcal{G}_s) &= E\left(\frac{1}{c}W(c^2t) \mid \mathcal{F}_{c^2s}\right) \\
 (72) \qquad &= \frac{1}{c}E(W(c^2t) \mid \mathcal{F}_{c^2s}) \\
 &= \frac{1}{c}W(c^2s) = V(s),
 \end{aligned}$$

and

$$\begin{aligned}
 E(|V(t)|^2 - t \mid \mathcal{G}_s) &= E\left(\frac{1}{c^2}|W(c^2t)|^2 - t \mid \mathcal{F}_{c^2s}\right) \\
 (73) \qquad &= \frac{1}{c^2}E(|W(c^2t)|^2 - c^2t \mid \mathcal{F}_{c^2s}) \\
 &= \frac{1}{c^2}(|W(c^2s)|^2 - c^2s) \\
 &= |V(s)|^2 - s,
 \end{aligned}$$

since  $W(t)$  and  $|W(t)|^2 - t$  are martingales with respect to the filtration  $\mathcal{F}_t$ . It follows by Lévy's martingale characterization that  $V(t)$  is a Wiener process.

### 3.3. Sample Paths. Let

$$0 = t_0^n < t_1^n < \cdots < t_n^n = T,$$

where

$$t_i^n = \frac{iT}{n},$$

be a partition of the interval  $[0, T]$  into  $n$  equal parts. We denote by

$$\Delta_i^n W = W(t_{i+1}^n) - W(t_i^n)$$

the corresponding increments of the Wiener process  $W(t)$ .

EXERCISE 3.29. Show that

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} (\Delta_i^n W)^2 = T \quad \text{in } L^2.$$

**Hint:** You need to show that

$$\lim_{n \rightarrow \infty} E \left( \left[ \sum_{i=0}^{n-1} (\Delta_i^n W)^2 - T \right]^2 \right) = 0.$$

Use the independence of increments to simplify the expectation. What are the expectations of  $\Delta_i^n W$ ,  $(\Delta_i^n W)^2$  and  $(\Delta_i^n W)^4$ ?

SOLUTION 3.29. Since the increments  $\Delta_i^n W$  are independent and

$$E(\Delta_i^n W) = 0, \quad E((\Delta_i^n W)^2) = \frac{T}{n}, \quad E((\Delta_i^n W)^4) = \frac{3T^2}{n^2},$$

it follows that

$$\begin{aligned}
 E \left( \left[ \sum_{i=0}^{n-1} (\Delta_i^n W)^2 - T \right]^2 \right) &= E \left( \left[ \sum_{i=0}^{n-1} \left( (\Delta_i^n W)^2 - \frac{T}{n} \right) \right]^2 \right) \\
 &= \sum_{i=0}^{n-1} E \left[ \left( (\Delta_i^n W)^2 - \frac{T}{n} \right)^2 \right] \\
 &= \sum_{i=0}^{n-1} \left[ E((\Delta_i^n W)^4) - \frac{2T}{n} E((\Delta_i^n W)^2) + \frac{T^2}{n^2} \right] \\
 &= \sum_{i=0}^{n-1} \left[ \frac{3T^2}{n^2} - \frac{2T^2}{n^2} + \frac{T^2}{n^2} \right] = \frac{2T^2}{n} \rightarrow 0,
 \end{aligned}
 \tag{74}$$

as  $n \rightarrow \infty$ .

The next theorem on the variation of the paths of  $W(t)$  is a consequence of the result in Exercise 3.29. First, let us recall that the variation of a function is defined as follows.

**DEFINITION 3.20 (Variation).** *The variation of a function  $f : [0, T] \rightarrow \mathbb{R}$  is defined to be*

$$\limsup_{\Delta t \rightarrow 0} \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)|,$$

where  $t = (t_0, t_1, \dots, t_n)$  is a partition of  $[0, T]$ , i.e.  $0 = t_0 < t_1 < \dots < t_n = T$ , and where

$$\Delta t = \max_{i=0, \dots, n-1} |t_{i+1} - t_i|.$$

**Theorem 3.21 (Infinite Variation).** *The variation of the paths of  $W(t)$  is infinite a.s.*

**PROOF.** Consider the sequence of partitions  $t^n = (t_0^n, t_1^n, \dots, t_n^n)$  of  $[0, T]$  into  $n$  equal parts. Then

$$\sum_{i=0}^{n-1} |\Delta_i^n W|^2 \leq \left( \max_{i=0, \dots, n-1} |\Delta_i^n W| \right) \sum_{i=0}^{n-1} |\Delta_i^n W|.$$

Since the paths of  $W(t)$  are a.s. continuous on  $[0, T]$ ,

$$\lim_{n \rightarrow \infty} \left( \max_{i=0, \dots, n-1} |\Delta_i^n W| \right) = 0 \quad \text{a.s.}$$

By Exercise 3.29 there is a subsequence  $t^{n_k} = (t_0^{n_k}, t_1^{n_k}, \dots, t_{n_k}^{n_k})$  of partitions such that

$$\lim_{k \rightarrow \infty} \sum_{i=0}^{n_k-1} |\Delta_i^{n_k} W|^2 = T \quad \text{a.s.}$$

This is because every sequence of random variables convergent in  $L^2$  has a subsequence convergent a.s. It follows that

$$\lim_{k \rightarrow \infty} \sum_{i=0}^{n_k-1} |\Delta_i^{n_k} W| = \infty \quad \text{a.s.},$$

while

$$\lim_{k \rightarrow \infty} \Delta t^{n_k} = \lim_{k \rightarrow \infty} \frac{T}{n_k} = 0,$$

which proves the theorem.

definition 3.21 has important consequences for the theory of stochastic integrals presented in the next chapter. This is because an integral of the form

$$\int_0^T f(t) dW(t)$$

cannot be defined pathwise (that is, separately for each  $\omega \in \Omega$ ) as the Riemann-Stieltjes integral if the paths have infinite variation. It turns out that an intrinsically stochastic approach will be needed to tackle such integrals, see chapter 4.

EXERCISE 3.30. Show that  $W(t)$  is a.s. non-differentiable at  $t = 0$ .

**Hint:** By Exercise 3.28  $V_c(t) = \frac{1}{c}W(c^2t)$  is a Wiener process for any  $c > 0$ . Deduce that the probability

$$P \left\{ \frac{|W(t)|}{t} > cM \text{ for some } t \in [0, \frac{1}{c^2}] \right\}$$

is the same for each  $c > 0$ . What is the probability that the limit of  $\frac{|W(t)|}{t}$  exists as  $t \searrow 0$ , then?

SOLUTION 3.30. We claim that, with probability 1, for any positive integer  $n$  there is a  $t \in [0, \frac{1}{n^4}]$  such that  $\frac{|W(t)|}{t} > n$ . This condition implies that  $W(t)$  is not differentiable at  $t = 0$ .

Let us put

$$A_n = \left\{ \frac{|W(t)|}{t} > n \text{ for some } t \in [0, \frac{1}{n^4}] \right\}.$$

By Exercise 3.28

$$V_n(t) = \frac{1}{n^2}W(n^4t)$$

is a Brownian motion for any  $n$ . Therefore

$$\begin{aligned} P(A_n) &\geq P \left\{ \frac{|W(1/n^4)|}{1/n^4} > n \right\} \\ (75) \quad &= P \left\{ \frac{|V_n(1/n^4)|}{1/n^4} > n \right\} \\ &= P \{ |W(1)| > \frac{1}{n} \} \rightarrow 1 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Since  $A_1, A_2, \dots$  is a contracting sequence of events,

$$P \left( \bigcap_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} P(A_n) = 1,$$

which proves the claim.

EXERCISE 3.31. Show that for any  $t \geq 0$  the Wiener process  $W(t)$  is a.s. non-differentiable at  $t$ .

**Hint:**  $V_t(s) = W(s+t) - W(t)$  is a Wiener process for any  $t \geq 0$ .



**SOLUTION 3.31.** By Exercise 3.27  $V_t(s) = W(s+t) - W(t)$  is a Wiener process for any  $t \geq 0$ . Therefore, by Exercise 3.30  $V_t(s)$  is a.s. non-differentiable at  $s = 0$ . But this implies that  $W(t)$  is a.s. non-differentiable at  $t$ .

A weak point in the assertion in Exercise 3.31 is that for each  $t$  the event of measure 1 in which  $W(t)$  is non-differentiable at  $t$  may turn out to be different for each  $t \geq 0$ . The theorem below, which is presented without proof, shows that in fact the same event of measure 1 can be chosen for each  $t \geq 0$ . This is not a trivial conclusion because the set of  $t \geq 0$  is uncountable.

**Theorem 3.22.** *With probability 1 the Wiener process  $W(t)$  is non-differentiable at any  $t \geq 0$ .*

**3.4. Doob's Maximal  $L^2$  Inequality for Brownian Motion.** The inequality proved in this section is necessary to study the properties of stochastic integrals in the next chapter. It can be viewed as an extension of Doob's maximal  $L^2$  inequality in definition 1.2 to the case of continuous time. In fact, in the result below the Wiener process can be replaced by any square integrable martingale  $\xi(t), t \geq 0$  with a.s. continuous paths.

**Theorem 3.23** (Doob's Maximal  $L^2$  Inequality). *For any  $t > 0$*

$$(6.6) \quad E \left( \max_{s \leq t} |W(s)|^2 \right) \leq 4E|W(t)|^2.$$

**PROOF.** For  $t > 0$  and  $n \in \mathbb{N}$  we define

$$(6.7) \quad M_k^n = \left| W \left( \frac{kt}{2^n} \right) \right|, \quad 0 \leq k \leq 2^n.$$

Then, by Jensen's inequality,  $M_k^n, k = 0, \dots, 2^n$  is a non-negative square integrable submartingale with respect to the filtration  $\mathcal{F}_k^n = \mathcal{F}_{\frac{k+1}{2^n}}$ , so by definition 1.2

$$E \left( \max_{k \leq 2^n} |M_k^n|^2 \right) \leq 4E|M_{2^n}^n|^2 = 4E|W(t)|^2.$$

Since  $W(t)$  has a.s. continuous paths,

$$\lim_{n \rightarrow \infty} \max_{k \leq 2^n} |M_k^n|^2 = \max_{s \leq t} |W(s)|^2 \quad \text{a.s.}$$

Moreover, since  $M_k^n = M_{2k}^{n+1}$ , the sequence  $\sup_{k \leq 2^n} |M_k^n|^2, n \in \mathbb{N}$ , is increasing. Hence by the Lebesgue monotone convergence theorem  $\max_{s \leq t} |W(s)|^2$  is an integrable function and

$$E \left( \max_{s \leq t} |W(s)|^2 \right) = \lim_{n \rightarrow \infty} E \left( \max_{k \leq 2^n} |M_k^n|^2 \right) \leq 4E|W(t)|^2,$$

completing the proof.

### 3.5. Various Exercises on Brownian Motion.

**EXERCISE 3.32.** *Verify that the transition density  $p(t, x, y)$  satisfies the diffusion equation*

$$\frac{\partial p}{\partial t} = \frac{1}{2} \frac{\partial^2 p}{\partial y^2}.$$

**Hint:** Simply differentiate the expression eq. (6.5) for the transition density.

SOLUTION 3.32. *Differentiating*

$$p(t, x, y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}},$$

we obtain

$$\frac{\partial}{\partial t} p(t, x, y) = \frac{y^2 - 2yx + x^2 - t}{2t^2} p(t, x, y),$$

$$\frac{\partial}{\partial y} p(t, x, y) = \frac{x - y}{t} p(t, x, y),$$

$$\frac{\partial^2}{\partial y^2} p(t, x, y) = \frac{y^2 - 2yx + x^2 - t}{t^2} p(t, x, y),$$

so

$$\frac{\partial p}{\partial t} = \frac{1}{2} \frac{\partial^2 p}{\partial y^2},$$

as required.

EXERCISE 3.33. Show that  $Z(t) = -W(t)$  is a Wiener process if  $W(t)$  is.

**Hint:** Are the increments of  $Z(t)$  independent? How are they distributed? Are the paths of  $Z(t)$  continuous? Is it true that  $Z(0) = 0$ ?

SOLUTION 3.33. Clearly,  $Z(t) = -W(t)$  has a.s. continuous trajectories and  $Z(0) = -W(0) = 0$  a.s. If  $W(t)$  has stationary independent increments, then so does  $Z(t) = -W(t)$ . Finally,

$$Z(t) - Z(s) = -(W(t) - W(s))$$

has the same distribution as  $W(t) - W(s)$ , i.e. normal with mean 0 and variance  $t - s$ . By definition 3.18  $Z(t)$  is a Wiener process.

EXERCISE 3.34. Show that for any  $0 \leq s < t$

$$P\{W(t) \in A \mid W(s)\} = \int_A p(t - s, W(s), y) dy.$$

**Hint:** Write the conditional probability as the conditional expectation of  $1_A(W(t))$  given  $W(s)$ . Compute the conditional expectation by transforming the integral of  $1_A(W(t))$  over any event in the  $\sigma$ -field generated by  $W(s)$ . This can be done using the joint density of  $W(s)$  and  $W(t)$ . Refer to the chapter on conditional expectation if necessary.

SOLUTION 3.34. Let  $0 \leq s < t$ . Then

$$\begin{aligned} \int_{\{W(s) \in B\}} 1_A(W(t)) dP &= P\{W(s) \in B, W(t) \in A\} \\ &= \int_B \int_A p(s, 0, x) p(t - s, x, y) dx dy \\ (76) \quad &= \int_B \left( \int_A p(t - s, x, y) dy \right) p(s, 0, x) dx \\ &= \int_{\{W(s) \in B\}} \left( \int_A p(t - s, W(s), y) dy \right) dP, \end{aligned}$$

for any Borel set  $B \subset \mathbb{R}$ . It follows that

$$P\{W(t) \in A \mid W(s)\} = E(1_A(W(t)) \mid W(s)) = \int_A p(t - s, W(s), y) dy.$$

EXERCISE 3.35. Show that  $e^{W(t)}e^{-\frac{t}{2}}$  is a martingale. (It is called the exponential martingale.)

**Hint:** What is the expectation of  $e^{W(t)-W(s)}$  for  $s < t$ ? By independence it is equal to the conditional expectation of  $e^{W(t)-W(s)}$  given  $\mathcal{F}_s$ . This will give the martingale condition.

SOLUTION 3.35. We shall prove that  $e^{W(t)}e^{-\frac{t}{2}}$  is a martingale with respect to the filtration  $\mathcal{F}_t$ . Clearly, it is adapted to the filtration  $\mathcal{F}_t$ , since  $W(t)$  is. Let  $0 \leq s < t$ . Because  $W(t) - W(s)$  is independent of  $\mathcal{F}_s$  and  $W(s)$  is  $\mathcal{F}_s$ -measurable,

$$\begin{aligned} E(e^{W(t)} \mid \mathcal{F}_s) &= E(e^{W(t)-W(s)} e^{W(s)} \mid \mathcal{F}_s) \\ (77) \qquad &= e^{W(s)} E(e^{W(t)-W(s)} \mid \mathcal{F}_s) \\ &= e^{W(s)} E(e^{W(t)-W(s)}). \end{aligned}$$

The increment  $W(t) - W(s)$  has the normal distribution with mean 0 and variance  $t - s$ , so the expectation of  $e^{W(t)-W(s)}$  is equal to

$$\begin{aligned} E(e^{W(t)-W(s)}) &= \int_{-\infty}^{+\infty} e^x p(t-s, 0, x) dx \\ (78) \qquad &= e^{\frac{t-s}{2}} \int_{-\infty}^{+\infty} p(t-s, 0, x-t) dx \\ &= e^{\frac{t-s}{2}}. \end{aligned}$$

It follows that

$$E(e^{W(t)}e^{-\frac{t}{2}} \mid \mathcal{F}_s) = e^{W(s)}e^{-\frac{s}{2}}.$$

It also follows that  $e^{W(t)}e^{-\frac{t}{2}}$  is integrable. Therefore  $e^{W(t)}e^{-\frac{t}{2}}$  is a martingale.

EXERCISE 3.36. Compute  $E(W(s) \mid W(t))$  for  $0 \leq s < t$ .

**Hint:** You want to find a Borel function  $F$  such that  $E(W(s) \mid W(t)) = F(W(t))$ , i.e.

$$\int_{\{W(t) \in A\}} W(s) dP = \int_{\{W(t) \in A\}} F(W(t)) dP.$$

Either side of this equality can be transformed using the joint density of  $W(t)$  and  $W(s)$ .

SOLUTION 3.36. Let  $0 \leq s < t$ . We are looking for a Borel function  $F$  such that  $E(W(s) \mid W(t)) = F(W(t))$ , i.e.

$$\int_{\{W(t) \in A\}} W(s) dP = \int_{\{W(t) \in A\}} F(W(t)) dP$$

for any Borel set  $A$  in  $\mathbb{R}$ . The integral on the right-hand side can be written as

$$\int_{\{W(t) \in A\}} F(W(t)) dP = \int_A F(y) p(t, 0, y) dy,$$

and the integral on the left-hand side as

$$\int_{\{W(t) \in A\}} W(s) dP = \int_A \left( \int_{-\infty}^{+\infty} x p(s, 0, x) p(t-s, x, y) dx \right) dy,$$

using the expression for the joint density of  $W(s)$  and  $W(t)$  in Solution 3.20. Let us compute the inner integral:

$$(79) \quad \int_{-\infty}^{+\infty} xp(s, 0, x)p(t-s, x, y) dx = p(t, 0, y) \int_{-\infty}^{+\infty} xp\left(\frac{s(t-s)}{t}, \frac{s}{t}y, x\right) dx \\ = \frac{s}{t}yp(t, 0, y).$$

(To see that the first equality holds, just use formula eq. (6.5) for  $p(t, x, y)$ .) Therefore

$$\int_{\{W(t) \in A\}} W(s) dP = \int_A \frac{s}{t}yp(t, 0, y) dy.$$

It follows that  $F(y) = \frac{s}{t}y$ , i.e.

$$E(W(s) \mid W(t)) = \frac{s}{t}W(t).$$

## CHAPTER 4

### Itô Stochastic Calculus

#### 1. Itô Stochastic Integral: Definition

We shall follow a construction resembling that of the Riemann integral. First, the integral will be defined for a class of piecewise constant processes called random step processes. Then it will be extended to a larger class by approximation.

There are, however, at least two major differences between the Riemann and Itô integrals. One is the type of convergence. The approximations of the Riemann integral converge in  $\mathbb{R}$ , while the Itô integral will be approximated by sequences of random variables converging in  $L^2$ . The other difference is this. The Riemann sums approximating the integral of a function  $f : [0, T] \rightarrow \mathbb{R}$  are of the form

$$(80) \quad \sum_{j=0}^{n-1} f(s_j)(t_{j+1} - t_j),$$

where  $0 = t_0 < t_1 < \dots < t_n = T$  and  $s_j$  is an arbitrary point in  $[t_j, t_{j+1}]$  for each  $j$ . The value of the Riemann integral does not depend on the choice of the points  $s_j \in [t_j, t_{j+1}]$ . In the stochastic case the approximating sums will have the form

$$(81) \quad \sum_{j=0}^{n-1} f(s_j)(W(t_{j+1}) - W(t_j)).$$

It turns out that the limit of such approximations does depend on the choice of the intermediate points  $s_j$  in  $[t_j, t_{j+1}]$ . In the next exercise we take  $f(t) = W(t)$  and consider two different choices of intermediate points.

**EXERCISE 4.1.** Let  $0 = t_0^n < t_1^n < \dots < t_n^n = T$ , where  $t_j^n = \frac{jT}{n}$ , be a partition of the interval  $[0, T]$  into  $n$  equal parts. Find the following limits in  $L^2$ :

$$(82) \quad \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} W(t_j^n) (W(t_{j+1}^n) - W(t_j^n))$$

and

$$(83) \quad \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} W(t_{j+1}^n) (W(t_{j+1}^n) - W(t_j^n)).$$

**REMARK 4.1. *Hint:*** Apply Exercise 6.29. You will need to transform the sums to make this possible. The identities

$$(84) \quad a(b-a) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(a-b)^2,$$

$$(85) \quad b(b-a) = \frac{1}{2}(b^2 - a^2) + \frac{1}{2}(a-b)^2$$

may be of help.

SOLUTION 4.1. Using the first identity with  $a = W(t_j^n)$  and  $b = W(t_{j+1}^n)$ , we have

$$\begin{aligned} \sum_{j=0}^{n-1} W(t_j^n)(W(t_{j+1}^n) - W(t_j^n)) &= \frac{1}{2} \sum_{j=0}^{n-1} (W(t_{j+1}^n)^2 - W(t_j^n)^2) - \frac{1}{2} \sum_{j=0}^{n-1} (W(t_{j+1}^n) - W(t_j^n))^2 \\ &= \frac{1}{2}(W(T)^2 - W(0)^2) - \frac{1}{2} \sum_{j=0}^{n-1} (W(t_{j+1}^n) - W(t_j^n))^2 \\ &= \frac{1}{2}W(T)^2 - \frac{1}{2} \sum_{j=0}^{n-1} (W(t_{j+1}^n) - W(t_j^n))^2. \end{aligned}$$

The second term is half the quadratic variation of the Wiener process on  $[0, T]$ . As  $n \rightarrow \infty$ , this sum converges to  $T$  in  $L^2$ . Therefore,

$$\lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} W(t_j^n)(W(t_{j+1}^n) - W(t_j^n)) = \frac{1}{2}W(T)^2 - \frac{1}{2}T.$$

Similarly, using the second identity,

$$\begin{aligned} \sum_{j=0}^{n-1} W(t_{j+1}^n)(W(t_{j+1}^n) - W(t_j^n)) &= \frac{1}{2} \sum_{j=0}^{n-1} (W(t_{j+1}^n)^2 - W(t_j^n)^2) + \frac{1}{2} \sum_{j=0}^{n-1} (W(t_{j+1}^n) - W(t_j^n))^2 \\ &= \frac{1}{2}W(T)^2 + \frac{1}{2} \sum_{j=0}^{n-1} (W(t_{j+1}^n) - W(t_j^n))^2, \end{aligned}$$

which converges in  $L^2$  to

$$\lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} W(t_{j+1}^n)(W(t_{j+1}^n) - W(t_j^n)) = \frac{1}{2}W(T)^2 + \frac{1}{2}T.$$

The ambiguity resulting from different choices of the intermediate points  $s_j$  in each subinterval  $[t_j, t_{j+1}]$  can be removed by insisting that the approximations of the integrand should consist only of processes adapted to the underlying filtration  $\mathcal{F}_t$ . This amounts to taking  $s_j = t_j$  for each  $j$ . The choice is motivated by the interpretation of  $\mathcal{F}_t$ : the value of the approximation at  $t$  may depend only on what has happened up to time  $t$ , but not on any future events.

DEFINITION 4.1. We shall call  $f(t), t \geq 0$  a **random step process** if there is a finite sequence of numbers  $0 = t_0 < t_1 < \dots < t_n$  and square integrable random variables

$\eta_0, \eta_1, \dots, \eta_{n-1}$  such that

$$(86) \quad f(t) = \sum_{j=0}^{n-1} \eta_j \mathbf{1}_{[t_j, t_{j+1})}(t),$$

where  $\eta_j$  is  $\mathcal{F}_{t_j}$ -measurable for  $j = 0, 1, \dots, n-1$ . The set of random step processes will be denoted by  $M_{\text{step}}^2$ .

Observe that the assumption that the  $\eta_j$  are to be  $\mathcal{F}_{t_j}$ -measurable ensures that  $f(t)$  is adapted to the filtration  $\mathcal{F}_t$ . The assumption that the  $\eta_j$  are square integrable ensures that  $f(t)$  is square integrable for each  $t$ . Also,  $M_{\text{step}}^2$  is a vector space, that is,  $af + bg \in M_{\text{step}}^2$  for any  $f, g \in M_{\text{step}}^2$  and  $a, b \in \mathbb{R}$ .

**DEFINITION 4.2.** The stochastic integral of a random step process  $f \in M_{\text{step}}^2$  of the form (86) is defined by

$$(87) \quad I(f) = \sum_{j=0}^{n-1} \eta_j (W(t_{j+1}) - W(t_j)).$$

**Proposition 4.3.** For any random step process  $f \in M_{\text{step}}^2$  the stochastic integral  $I(f)$  is a square integrable random variable, i.e.  $I(f) \in L^2$ , such that

$$(88) \quad \mathbb{E}(|I(f)|^2) = \mathbb{E}\left(\int_0^\infty |f(t)|^2 dt\right).$$

**PROOF.** Let us denote the increment  $W(t_{j+1}) - W(t_j)$  by  $\Delta_j W$  and  $t_{j+1} - t_j$  by  $\Delta_j t$  for brevity. Then

$$(89) \quad \mathbb{E}(\Delta_j W) = 0 \quad \text{and} \quad \mathbb{E}((\Delta_j W)^2) = \Delta_j t.$$

First, we shall compute the expectation of

$$(90) \quad |I(f)|^2 = \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \eta_j \eta_k \Delta_j W \Delta_k W = \sum_{j=0}^{n-1} \eta_j^2 (\Delta_j W)^2 + 2 \sum_{k < j} \eta_j \eta_k \Delta_j W \Delta_k W.$$

Since  $\eta_j$  and  $\Delta_j W$  are independent,

$$(91) \quad \mathbb{E}(\eta_j^2 (\Delta_j W)^2) = \mathbb{E}(\eta_j^2) \mathbb{E}((\Delta_j W)^2) = \mathbb{E}(\eta_j^2) \Delta_j t.$$

If  $k < j$ , then  $\eta_j \eta_k \Delta_k W$  and  $\Delta_j W$  are independent, so

$$(92) \quad \mathbb{E}(\eta_j \eta_k \Delta_j W \Delta_k W) = \mathbb{E}(\eta_j \eta_k \Delta_k W) \mathbb{E}(\Delta_j W) = 0.$$

Therefore

$$(93) \quad \mathbb{E}(|I(f)|^2) = \sum_{j=0}^{n-1} \mathbb{E}(\eta_j^2) \Delta_j t.$$

It follows that  $I(f) \in L^2$ , since  $\eta_0, \eta_1, \dots, \eta_{n-1} \in L^2$ .

On the other hand,

$$(94) \quad |f(t)|^2 = \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \eta_j \eta_k \mathbf{1}_{[t_j, t_{j+1})}(t) \mathbf{1}_{[t_k, t_{k+1})}(t) = \sum_{j=0}^{n-1} \eta_j^2 \mathbf{1}_{[t_j, t_{j+1})}(t),$$

implying that

$$(95) \quad \mathbb{E} \left( \int_0^\infty |f(t)|^2 dt \right) = \sum_{j=0}^{n-1} \mathbb{E} (\eta_j^2) \Delta_j t.$$

This means that

$$(96) \quad \mathbb{E} (|I(f)|^2) = \mathbb{E} \left( \int_0^\infty |f(t)|^2 dt \right),$$

as required. □

EXERCISE 4.2. Verify that for any random step processes  $f, g \in M_{\text{step}}^2$

$$(97) \quad \mathbb{E}(I(f)I(g)) = \mathbb{E} \left( \int_0^\infty f(t)g(t) dt \right).$$

REMARK 4.2. **Hint:** Try to adapt the proof of Proposition 4.3. Use a common partition  $0 = t_0 < t_1 < \dots < t_n$  in which to represent both  $f$  and  $g$  in the form (86).

SOLUTION 4.2. Let  $f(t) = \sum_{j=0}^{n-1} \eta_j \mathbf{1}_{[t_j, t_{j+1})}(t)$  and  $g(t) = \sum_{j=0}^{n-1} \zeta_j \mathbf{1}_{[t_j, t_{j+1})}(t)$  be two random step processes represented with respect to a common partition  $0 = t_0 < t_1 < \dots < t_n$ . Then

$$I(f) = \sum_{j=0}^{n-1} \eta_j \Delta_j W, \quad I(g) = \sum_{k=0}^{n-1} \zeta_k \Delta_k W.$$

Their product is

$$I(f)I(g) = \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \eta_j \zeta_k \Delta_j W \Delta_k W = \sum_{j=0}^{n-1} \eta_j \zeta_j (\Delta_j W)^2 + \sum_{j \neq k} \eta_j \zeta_k \Delta_j W \Delta_k W.$$

Taking the expectation, for the diagonal terms ( $j = k$ ), we have

$$\mathbb{E}(\eta_j \zeta_j (\Delta_j W)^2) = \mathbb{E}(\eta_j \zeta_j) \mathbb{E}((\Delta_j W)^2) = \mathbb{E}(\eta_j \zeta_j) \Delta_j t,$$

since  $\eta_j \zeta_j$  is  $\mathcal{F}_{t_j}$ -measurable and  $(\Delta_j W)^2$  is independent of  $\mathcal{F}_{t_j}$ . For the off-diagonal terms, assume without loss of generality that  $k < j$ . Then  $\eta_j \zeta_k \Delta_k W$  is  $\mathcal{F}_{t_j}$ -measurable and  $\Delta_j W$  is independent of  $\mathcal{F}_{t_j}$  with  $\mathbb{E}(\Delta_j W) = 0$ . Thus,

$$\mathbb{E}(\eta_j \zeta_k \Delta_j W \Delta_k W) = \mathbb{E}(\eta_j \zeta_k \Delta_k W) \mathbb{E}(\Delta_j W) = 0.$$

Summing these up, we get

$$\mathbb{E}(I(f)I(g)) = \sum_{j=0}^{n-1} \mathbb{E}(\eta_j \zeta_j) \Delta_j t = \mathbb{E} \left( \int_0^\infty \sum_{j=0}^{n-1} \eta_j \zeta_j \mathbf{1}_{[t_j, t_{j+1})}(t) dt \right) = \mathbb{E} \left( \int_0^\infty f(t)g(t) dt \right).$$



EXERCISE 4.3. Show that  $I : M_{\text{step}}^2 \rightarrow L^2$  is a linear map, i.e. for any  $f, g \in M_{\text{step}}^2$  and any  $\alpha, \beta \in \mathbb{R}$

$$(98) \quad I(\alpha f + \beta g) = \alpha I(f) + \beta I(g).$$

REMARK 4.3. **Hint:** As in Exercise 4.2, use a common partition  $0 = t_0 < t_1 < \dots < t_n$  in which to represent both  $f$  and  $g$  in the form (86).

SOLUTION 4.3. Let  $f(t) = \sum_{j=0}^{n-1} \eta_j \mathbf{1}_{[t_j, t_{j+1})}(t)$  and  $g(t) = \sum_{j=0}^{n-1} \zeta_j \mathbf{1}_{[t_j, t_{j+1})}(t)$  be two random step processes using a common partition  $0 = t_0 < t_1 < \dots < t_n$ . The linear combination  $\alpha f + \beta g$  is also a random step process given by

$$(\alpha f + \beta g)(t) = \sum_{j=0}^{n-1} (\alpha \eta_j + \beta \zeta_j) \mathbf{1}_{[t_j, t_{j+1})}(t).$$

By Definition 4.2, the stochastic integral of  $\alpha f + \beta g$  is

$$\begin{aligned} I(\alpha f + \beta g) &= \sum_{j=0}^{n-1} (\alpha \eta_j + \beta \zeta_j) (W(t_{j+1}) - W(t_j)) \\ &= \alpha \sum_{j=0}^{n-1} \eta_j (W(t_{j+1}) - W(t_j)) + \beta \sum_{j=0}^{n-1} \zeta_j (W(t_{j+1}) - W(t_j)) \\ &= \alpha I(f) + \beta I(g). \end{aligned}$$

This confirms that  $I : M_{\text{step}}^2 \rightarrow L^2$  is a linear map.

The stochastic integral  $I(f)$  has been defined for any random step process  $f \in M_{\text{step}}^2$ . The next stage is to extend  $I$  to a larger class of processes by approximation. This larger class can be defined as follows.

DEFINITION 4.4. We denote by  $M^2$  the class of stochastic processes  $f(t), t \geq 0$  such that

$$(99) \quad \mathbb{E} \left( \int_0^\infty |f(t)|^2 dt \right) < \infty$$

and there is a sequence  $f_1, f_2, \dots \in M_{\text{step}}^2$  of random step processes such that

$$(100) \quad \lim_{n \rightarrow \infty} \mathbb{E} \left( \int_0^\infty |f(t) - f_n(t)|^2 dt \right) = 0.$$

In this case we shall say that the sequence of random step processes  $f_1, f_2, \dots$  approximates  $f$  in  $M^2$ .

DEFINITION 4.5. We call  $I(f) \in L^2$  the **Itô stochastic integral** (from 0 to  $\infty$ ) of  $f \in M^2$  if

$$(101) \quad \lim_{n \rightarrow \infty} \mathbb{E} (|I(f) - I(f_n)|^2) = 0$$

for any sequence  $f_1, f_2, \dots \in M_{\text{step}}^2$  of random step processes that approximates  $f$  in  $M^2$ , i.e. such that (100) is satisfied. We shall also write

$$(102) \quad \int_0^\infty f(t) dW(t)$$

in place of  $I(f)$ .

**Proposition 4.6.** *For any  $f \in M^2$  the stochastic integral  $I(f) \in L^2$  exists, is unique (as an element of  $L^2$ , i.e. to within equality a.s.) and satisfies*

$$(103) \quad \mathbb{E}(|I(f)|^2) = \mathbb{E}\left(\int_0^\infty |f(t)|^2 dt\right).$$

PROOF. *It will be convenient to write*

$$(104) \quad \|f\|_{M^2} = \sqrt{\mathbb{E}\left(\int_0^\infty |f(t)|^2 dt\right)} \quad \text{and} \quad \|\eta\|_{L^2} = \sqrt{\mathbb{E}(\eta^2)}$$

for any  $f \in M^2$  and  $\eta \in L^2$ . These are norms in  $M^2$  and  $L^2$ , respectively.

Let  $f_1, f_2, \dots \in M_{\text{step}}^2$  be a sequence of random step processes approximating  $f \in M^2$ , i.e. satisfying (100), which can be written as

$$(105) \quad \lim_{n \rightarrow \infty} \|f - f_n\|_{M^2} = 0.$$

We claim that  $I(f_1), I(f_2), \dots$  is a Cauchy sequence in  $L^2$ . Indeed, for any  $\varepsilon > 0$  there is an  $N$  such that  $\|f - f_n\|_{M^2} < \frac{\varepsilon}{2}$  for all  $n > N$ . By Proposition 4.3

$$(106) \quad \|I(f_m) - I(f_n)\|_{L^2} = \|I(f_m - f_n)\|_{L^2}$$

$$(107) \quad = \|f_m - f_n\|_{M^2}$$

$$(108) \quad \leq \|f - f_m\|_{M^2} + \|f - f_n\|_{M^2}$$

$$(109) \quad < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

for any  $m, n > N$ , which proves the claim.

Because  $L^2$  with the norm  $\|\cdot\|_{L^2}$  is a complete space (in fact a Hilbert space), every Cauchy sequence in  $L^2$  has a limit. It follows that  $I(f_1), I(f_2), \dots$  has a limit in  $L^2$  for any sequence  $f_1, f_2, \dots$  of random step processes approximating  $f$ . It remains to show that the limit is the same for all such sequences. Suppose that  $f_1, f_2, \dots$  and  $g_1, g_2, \dots$  are two sequences of random step processes approximating  $f$ . Then the interlaced sequence  $f_1, g_1, f_2, g_2, \dots$  approximates  $f$  too, so the sequence  $I(f_1), I(g_1), I(f_2), I(g_2), \dots$  has a limit in  $L^2$ . But then all subsequences of the latter sequence, in particular,  $I(f_1), I(f_2), \dots$  and  $I(g_1), I(g_2), \dots$  have the same limit, which we denote by  $I(f)$ . We have shown that

$$(110) \quad \lim_{n \rightarrow \infty} \|I(f) - I(f_n)\|_{L^2} = 0,$$

i.e. (101) holds for any sequence  $f_1, f_2, \dots$  of random step processes approximating  $f$ . Finally, by Proposition 4.3

$$(111) \quad \|I(f_n)\|_{L^2} = \|f_n\|_{M^2}$$

for each  $n$ , since the  $f_n$  are random step processes. By taking the limit as  $n \rightarrow \infty$  we obtain

$$(112) \quad \|I(f)\|_{L^2} = \|f\|_{M^2}.$$

But this is equality (103). □

EXERCISE 4.4. Show that for any  $f, g \in M^2$

$$(113) \quad \mathbb{E}(I(f)I(g)) = \mathbb{E}\left(\int_0^\infty f(t)g(t) dt\right).$$

REMARK 4.4. **Hint:** Write the left-hand side in terms of  $\mathbb{E}(|I(f) + I(g)|^2)$  and  $\mathbb{E}(|I(f) - I(g)|^2)$ , the right-hand side in terms of  $\mathbb{E}(\int_0^\infty |f(t) + g(t)|^2 dt)$  and  $\mathbb{E}(\int_0^\infty |f(t) - g(t)|^2 dt)$  and then use (103).

SOLUTION 4.4. Using the polarization identity for the inner product on  $L^2$ , we have

$$\mathbb{E}(I(f)I(g)) = \frac{1}{4} (\mathbb{E}|I(f) + I(g)|^2 - \mathbb{E}|I(f) - I(g)|^2).$$

By the linearity of the stochastic integral (which extends from step processes to  $M^2$  by the approximation definition),  $I(f) \pm I(g) = I(f \pm g)$ . Applying the Itô isometry (103) to  $I(f + g)$  and  $I(f - g)$ , we get

$$\begin{aligned} \mathbb{E}(I(f)I(g)) &= \frac{1}{4} \left( \mathbb{E}\left(\int_0^\infty |f(t) + g(t)|^2 dt\right) - \mathbb{E}\left(\int_0^\infty |f(t) - g(t)|^2 dt\right) \right) \\ &= \frac{1}{4} \mathbb{E}\left(\int_0^\infty (|f(t) + g(t)|^2 - |f(t) - g(t)|^2) dt\right). \end{aligned}$$

Since  $|f(t) + g(t)|^2 - |f(t) - g(t)|^2 = 4f(t)g(t)$ , the expression simplifies to

$$\mathbb{E}(I(f)I(g)) = \mathbb{E}\left(\int_0^\infty f(t)g(t) dt\right),$$

as required.

Having defined the Itô stochastic integral from 0 to  $\infty$ , we are now in a position to consider stochastic integrals over any finite time interval  $[0, T]$ .

DEFINITION 4.7. For any  $T > 0$  we shall denote by  $M_T^2$  the space of all stochastic processes  $f(t), t \geq 0$  such that

$$(114) \quad \mathbf{1}_{[0, T]} f \in M^2.$$

The Itô stochastic integral (from 0 to  $T$ ) of  $f \in M_T^2$  is defined by

$$(115) \quad I_T(f) = I(\mathbf{1}_{[0, T]} f).$$

We shall also write

$$(116) \quad \int_0^T f(t) dW(t)$$

in place of  $I_T(f)$ .

EXERCISE 4.5. Show that each random step process  $f \in M_{\text{step}}^2$  belongs to  $M_t^2$  for any  $t > 0$  and

$$(117) \quad I_t(f) = \int_0^t f(s) dW(s)$$

is a martingale.

REMARK 4.5. **Hint:** The stochastic integral of a random step process  $f$  is given by the sum (87). What is the conditional expectation of the  $j$ -th term of this sum given  $\mathcal{F}_s$  if  $s < t_j$ ? What is it when  $s \geq t_j$ ?

SOLUTION 4.5. A random step process  $f$  has the form  $f(t) = \sum_{j=0}^{n-1} \eta_j \mathbf{1}_{[t_j, t_{j+1})}(t)$ . For any  $t > 0$ ,  $\mathbf{1}_{[0, t)} f$  is also a random step process, so  $f \in M_t^2$ . To show that  $I_t(f)$  is a martingale, let  $s < t$ . We need to show  $\mathbb{E}(I_t(f) \mid \mathcal{F}_s) = I_s(f)$ . The stochastic integral  $I_t(f)$  can be written as

$$I_t(f) = \sum_{j=0}^{n-1} \eta_j (W(t_{j+1} \wedge t) - W(t_j \wedge t)).$$

Consider a single term  $X_j(t) = \eta_j (W(t_{j+1} \wedge t) - W(t_j \wedge t))$ . If  $s \geq t_{j+1}$ , then  $W(t_{j+1} \wedge t) - W(t_j \wedge t)$  is  $\mathcal{F}_{t_{j+1}}$ -measurable and thus  $\mathcal{F}_s$ -measurable. Also  $\eta_j$  is  $\mathcal{F}_{t_j} \subset \mathcal{F}_s$  measurable. So  $\mathbb{E}(X_j(t) \mid \mathcal{F}_s) = X_j(t) = X_j(s)$ . If  $s \leq t_j$ , then  $\eta_j$  is  $\mathcal{F}_{t_j}$ -measurable and  $W(t_{j+1} \wedge t) - W(t_j \wedge t)$  is independent of  $\mathcal{F}_{t_j}$ .

$$\begin{aligned} \mathbb{E}(X_j(t) \mid \mathcal{F}_s) &= \mathbb{E}(\mathbb{E}(X_j(t) \mid \mathcal{F}_{t_j}) \mid \mathcal{F}_s) \\ &= \mathbb{E}(\eta_j \mathbb{E}(W(t_{j+1} \wedge t) - W(t_j \wedge t) \mid \mathcal{F}_{t_j}) \mid \mathcal{F}_s) \\ &= \mathbb{E}(\eta_j \cdot 0 \mid \mathcal{F}_s) = 0. \end{aligned}$$

Since  $X_j(s) = 0$  for  $s \leq t_j$ , we have  $\mathbb{E}(X_j(t) \mid \mathcal{F}_s) = X_j(s)$ . If  $t_j < s < t_{j+1}$ , then

$$\begin{aligned} \mathbb{E}(X_j(t) \mid \mathcal{F}_s) &= \mathbb{E}(\eta_j (W(t_{j+1} \wedge t) - W(t_j))) \mid \mathcal{F}_s) \\ &= \eta_j \mathbb{E}(W(t_{j+1} \wedge t) - W(s) + W(s) - W(t_j) \mid \mathcal{F}_s) \\ &= \eta_j (0 + W(s) - W(t_j)) = X_j(s). \end{aligned}$$

In all cases  $\mathbb{E}(X_j(t) \mid \mathcal{F}_s) = X_j(s)$ . Summing over  $j$ , we obtain  $\mathbb{E}(I_t(f) \mid \mathcal{F}_s) = I_s(f)$ .

The processes for which the stochastic integral exists have been defined as those that can be approximated by random step processes. However, it is not always easy to check whether or not such an approximation exists. For practical purposes it is important to have a straightforward sufficient condition for a process to have a stochastic integral. In calculus there is a well-known result of this kind: the Riemann integral exists for any continuous function. Here is a theorem of this kind for the Itô integral.

**Theorem 4.8.** Let  $f(t), t \geq 0$  be a stochastic process with a.s. continuous paths adapted to the filtration  $\mathcal{F}_t$ . Then:

(1)  $f \in M^2$ , i.e. the Itô integral  $I(f)$  exists, whenever

$$(118) \quad \mathbb{E} \left( \int_0^\infty |f(t)|^2 dt \right) < \infty;$$

(2)  $f \in M_T^2$ , i.e. the Itô integral  $I_T(f)$  exists, whenever

$$(119) \quad \mathbb{E} \left( \int_0^T |f(t)|^2 dt \right) < \infty.$$

PROOF. (1) Suppose that  $f(t), t \geq 0$  is an adapted process with a.s. continuous paths. If (118) holds, then

$$(120) \quad f_n(t) = \begin{cases} n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f(s) ds & \frac{k}{n} \leq t < \frac{k+1}{n} \text{ for } k = 1, 2, \dots, n^2 - 1, \\ 0 & \text{otherwise,} \end{cases}$$

is a sequence of random step processes in  $M_{\text{step}}^2$ . Observe that for any  $k = 1, 2, \dots$

$$(121) \quad \int_{\frac{k}{n}}^{\frac{k+1}{n}} |f_n(t)|^2 dt = n \left| \int_{\frac{k-1}{n}}^{\frac{k}{n}} f(t) dt \right|^2 \leq \int_{\frac{k-1}{n}}^{\frac{k}{n}} |f(t)|^2 dt \quad a.s.$$

by Jensen's inequality. We claim that

$$(122) \quad \lim_{n \rightarrow \infty} \int_0^\infty |f(t) - f_n(t)|^2 dt = 0 \quad a.s.$$

This will imply that

$$(123) \quad \lim_{n \rightarrow \infty} \mathbb{E} \left( \int_0^\infty |f(t) - f_n(t)|^2 dt \right) = 0$$

by the dominated convergence theorem and condition (118) because

$$(124) \quad \int_0^\infty |f(t) - f_n(t)|^2 dt \leq 2 \int_0^\infty (|f(t)|^2 + |f_n(t)|^2) dt$$

$$(125) \quad \leq 4 \int_0^\infty |f(t)|^2 dt.$$

The last inequality follows, since

$$(126) \quad \int_0^\infty |f_n(t)|^2 dt \leq \int_0^\infty |f(t)|^2 dt \quad a.s.$$

for any  $n$ , by taking the sum from  $k = 0$  to  $\infty$  in (121). To verify the claim observe that

$$(127) \quad \int_0^\infty |f(t) - f_n(t)|^2 dt = \int_0^N |f(t) - f_n(t)|^2 dt + \int_N^\infty |f(t) - f_n(t)|^2 dt$$

$$(128) \quad \leq \int_0^N |f(t) - f_n(t)|^2 dt + 2 \int_N^\infty (|f(t)|^2 + |f_n(t)|^2) dt$$

$$(129) \quad \leq \int_0^N |f(t) - f_n(t)|^2 dt + 4 \int_{N-1}^\infty |f(t)|^2 dt \quad a.s.$$

The last inequality holds because

$$(130) \quad \int_N^\infty |f_n(t)|^2 dt \leq \int_{N-\frac{1}{n}}^\infty |f(t)|^2 dt \leq \int_{N-1}^\infty |f(t)|^2 dt \quad a.s.$$

for any  $n$  and  $N$ , by taking the sum from  $k = nN$  to  $\infty$  in (121). The claim follows because

$$(131) \quad \lim_{N \rightarrow \infty} \int_{N-1}^\infty |f(t)|^2 dt = 0 \quad a.s.$$

by (118) and

$$(132) \quad \lim_{n \rightarrow \infty} \int_0^N |f(t) - f_n(t)|^2 dt = 0 \quad a.s.$$

for any fixed  $N$  by the continuity of paths of  $f$ .

The above means that the sequence  $f_1, f_2, \dots \in M_{\text{step}}^2$  approximates  $f$  in the sense of Definition 4.4, so  $f \in M^2$ .

- (2) If  $f$  satisfies (119) for some  $T > 0$ , then  $\mathbf{1}_{[0,T]}f$  satisfies (118). Since  $f$  is adapted and has a.s. continuous paths,  $\mathbf{1}_{[0,T]}f$  is also adapted and its paths are a.s. continuous, except perhaps at  $T$ . But the lack of continuity at the single point  $T$  does not affect the argument in (1), so  $\mathbf{1}_{[0,T]}f \in M^2$ . This in turn implies that  $f \in M_T^2$ , completing the proof.

EXERCISE 4.6. Show that the Wiener process  $W(t)$  belongs to  $M_T^2$  for each  $T > 0$ .

REMARK 4.6. **Hint:** Apply part (2) of Theorem 4.8.

SOLUTION 4.6. The Wiener process  $W(t)$  is adapted to the filtration  $\mathcal{F}_t$  and has a.s. continuous paths. To apply Theorem 4.8(2), we check the integrability condition:

$$\mathbb{E} \left( \int_0^T |W(t)|^2 dt \right) = \int_0^T \mathbb{E}(W(t)^2) dt = \int_0^T t dt = \frac{1}{2}T^2 < \infty.$$

Thus,  $W \in M_T^2$  for each  $T > 0$ .

EXERCISE 4.7. Show that  $W(t)^2$  belongs to  $M_T^2$  for each  $T > 0$ .

REMARK 4.7. **Hint:** Once again, apply part (2) of Theorem 4.8.

SOLUTION 4.7. The process  $W(t)^2$  is adapted to  $\mathcal{F}_t$  and has a.s. continuous paths. We check the integrability condition:

$$\mathbb{E} \left( \int_0^T |W(t)^2|^2 dt \right) = \int_0^T \mathbb{E}(W(t)^4) dt = \int_0^T 3t^2 dt = T^3 < \infty.$$

Here we used the fact that for a Gaussian random variable  $W(t) \sim N(0, t)$ , the fourth moment is  $\mathbb{E}(W(t)^4) = 3(\mathbb{E}(W(t)^2))^2 = 3t^2$ . Thus,  $W^2 \in M_T^2$  for each  $T > 0$ .

The next theorem, which we shall state without proof, provides a characterization of  $M^2$  and  $M_T^2$ , i.e. a necessary and sufficient condition for a stochastic process  $f$  to belong to  $M^2$  or  $M_T^2$ . It involves the notion of a progressively measurable process.

DEFINITION 4.9. A stochastic process  $f(t), t \geq 0$  is called **progressively measurable** if for any  $t \geq 0$

$$(133) \quad (s, \omega) \mapsto f(s, \omega)$$

is a measurable function from  $[0, t] \times \Omega$  with the  $\sigma$ -field  $\mathcal{B}[0, t] \bar{\times} \mathcal{F}$  to  $\mathbb{R}$ . Here  $\mathcal{B}[0, t] \bar{\times} \mathcal{F}$  is the product  $\sigma$ -field on  $[0, t] \times \Omega$ , that is, the smallest  $\sigma$ -field containing all sets of the form  $A \times B$ , where  $A \subset [0, t]$  is a Borel set and  $B \in \mathcal{F}$ .

**Theorem 4.10.** (1) The space  $M^2$  consists of all progressively measurable stochastic processes  $f(t), t \geq 0$  such that

$$(134) \quad \mathbb{E} \left( \int_0^\infty |f(t)|^2 dt \right) < \infty.$$

(2) The space  $M_T^2$  consists of all progressively measurable stochastic processes  $f(t), t \geq 0$  such that

$$(135) \quad \mathbb{E} \left( \int_0^T |f(t)|^2 dt \right) < \infty.$$

## 2. Examples

According to Exercise 4.6, the Wiener process  $W(t)$  belongs to  $M_T^2$  for any  $T > 0$ . Therefore the stochastic integral in the next exercise exists.

EXERCISE 4.8. Verify the equality

$$(136) \quad \int_0^T W(t) dW(t) = \frac{1}{2}W(T)^2 - \frac{1}{2}T$$

by computing the stochastic integral from the definition, that is, by approximating the integrand by random step functions.

REMARK 4.8. **Hint:** It is convenient to use a partition of the interval  $[0, T]$  into  $n$  equal parts. The limit of the sums approximating the integral has been found in Exercise 4.1.

SOLUTION 4.8. Let  $0 = t_0^n < t_1^n < \dots < t_n^n = T$  be a partition of  $[0, T]$  into  $n$  equal parts. Consider the random step processes  $f_n(t) = \sum_{j=0}^{n-1} W(t_j^n) \mathbf{1}_{[t_j^n, t_{j+1}^n)}(t)$ . As  $n \rightarrow \infty$ ,  $f_n$  approximates  $W$  in  $M_T^2$  because  $W$  has continuous paths and is square integrable. The stochastic integral of  $f_n$  is

$$I(f_n) = \sum_{j=0}^{n-1} W(t_j^n)(W(t_{j+1}^n) - W(t_j^n)).$$

From Exercise 4.1, we know that the  $L^2$  limit of this sum is

$$\lim_{n \rightarrow \infty} I(f_n) = \frac{1}{2}W(T)^2 - \frac{1}{2}T.$$

By Definition 4.5, this limit is the Itô stochastic integral  $\int_0^T W(t) dW(t)$ .

EXERCISE 4.9. Verify the equality

$$(137) \quad \int_0^T t dW(t) = TW(T) - \int_0^T W(t) dt,$$

by computing the stochastic integral from the definition. (The integral on the right-hand side is understood as a Riemann integral defined pathwise, i.e. separately for each  $\omega \in \Omega$ .)

REMARK 4.9. **Hint:** You may want to use the same partition of  $[0, T]$  into  $n$  equal parts as in Solution 4.8. The sums approximating the stochastic integral can be transformed with the aid of the identity

$$(138) \quad c(b - a) = (db - ca) - b(d - c).$$

SOLUTION 4.9. Let  $t_j = \frac{jT}{n}$  for  $j = 0, \dots, n$ . Approximate the integrand  $f(t) = t$  by the random step process  $f_n(t) = \sum_{j=0}^{n-1} t_j \mathbf{1}_{[t_j, t_{j+1})}(t)$ . The stochastic integral is

$$I(f_n) = \sum_{j=0}^{n-1} t_j (W(t_{j+1}) - W(t_j)).$$

Using the identity  $c(b - a) = (db - ca) - b(d - c)$  with  $c = t_j, d = t_{j+1}, a = W(t_j), b = W(t_{j+1})$ , we get

$$\begin{aligned} I(f_n) &= \sum_{j=0}^{n-1} [(t_{j+1}W(t_{j+1}) - t_jW(t_j)) - W(t_{j+1})(t_{j+1} - t_j)] \\ &= (t_nW(t_n) - t_0W(t_0)) - \sum_{j=0}^{n-1} W(t_{j+1})\Delta_j t \\ &= TW(T) - \sum_{j=0}^{n-1} W(t_{j+1})\Delta_j t. \end{aligned}$$



As  $n \rightarrow \infty$ , the sum  $\sum_{j=0}^{n-1} W(t_{j+1})\Delta_j t$  converges pathwise (and in  $L^2$ ) to the Riemann integral  $\int_0^T W(t) dt$  because the paths of  $W(t)$  are continuous. Thus,

$$\int_0^T t dW(t) = TW(T) - \int_0^T W(t) dt.$$

EXERCISE 4.10. Show that  $W(t)^2$  belongs to  $M_T^2$  for each  $T > 0$  and verify the equality

$$(139) \quad \int_0^T W(t)^2 dW(t) = \frac{1}{3}W(T)^3 - \int_0^T W(t) dt,$$

where the integral on the right-hand side is a Riemann integral.

REMARK 4.10. **Hint:** As in the exercises above, it is convenient to use the partition of  $[0, T]$  into  $n$  equal parts. The identity

$$(140) \quad a^2(b-a) = \frac{1}{3}(b^3 - a^3) - a(b-a)^2 - \frac{1}{3}(b-a)^3$$

can be applied to transform the sums approximating the stochastic integral. You may also need the following identity:

$$(141) \quad (a^2 - b^2)^2 = (a-b)^4 + 4(a-b)^3b + 4(a-b)^2b^2.$$

SOLUTION 4.10. In Exercise 4.7 we showed that  $W^2 \in M_T^2$ . Let  $t_j = \frac{jT}{n}$  and  $a = W(t_j), b = W(t_{j+1})$ . The approximating sum is

$$\begin{aligned} I(f_n) &= \sum_{j=0}^{n-1} W(t_j)^2 (W(t_{j+1}) - W(t_j)) \\ &= \sum_{j=0}^{n-1} \left[ \frac{1}{3}(W(t_{j+1})^3 - W(t_j)^3) - W(t_j)(W(t_{j+1}) - W(t_j))^2 - \frac{1}{3}(W(t_{j+1}) - W(t_j))^3 \right] \\ &= \frac{1}{3}W(T)^3 - \sum_{j=0}^{n-1} W(t_j)(\Delta_j W)^2 - \frac{1}{3} \sum_{j=0}^{n-1} (\Delta_j W)^3. \end{aligned}$$

As  $n \rightarrow \infty$ ,  $\sum (\Delta_j W)^3 \rightarrow 0$  in  $L^2$  (and a.s. for a subsequence). For the middle term, we compare it to the Riemann integral  $\int_0^T W(t) dt \approx \sum W(t_j)\Delta_j t$ :

$$\begin{aligned} \mathbb{E} \left| \sum W(t_j)(\Delta_j W)^2 - \sum W(t_j)\Delta_j t \right|^2 &= \sum \mathbb{E}[W(t_j)^2((\Delta_j W)^2 - \Delta_j t)^2] \\ &= \sum \mathbb{E}(W(t_j)^2)\mathbb{E}((\Delta_j W)^2 - \Delta_j t)^2 \\ &= \sum t_j \cdot 2(\Delta_j t)^2 = 2\frac{T^2}{n^2} \sum \frac{jT}{n} \rightarrow 0. \end{aligned}$$

Thus  $\sum W(t_j)(\Delta_j W)^2 \rightarrow \int_0^T W(t) dt$  in  $L^2$ . Combining these results, we obtain

$$\int_0^T W(t)^2 dW(t) = \frac{1}{3}W(T)^3 - \int_0^T W(t) dt.$$

### 3. Properties of the Stochastic Integral

The basic properties of the Itô integral are summarized in the theorem below.

**Theorem 4.11.** *The following properties hold for any  $f, g \in M_t^2$ , any  $\alpha, \beta \in \mathbb{R}$ , and any  $0 \leq s < t$ :*

(1) **Linearity:**

$$(142) \quad \int_0^t (\alpha f(r) + \beta g(r)) dW(r) = \alpha \int_0^t f(r) dW(r) + \beta \int_0^t g(r) dW(r);$$

(2) **Isometry:**

$$(143) \quad \mathbb{E} \left( \left| \int_0^t f(r) dW(r) \right|^2 \right) = \mathbb{E} \left( \int_0^t |f(r)|^2 dr \right);$$

(3) **Martingale property:**

$$(144) \quad \mathbb{E} \left( \int_0^t f(r) dW(r) \mid \mathcal{F}_s \right) = \int_0^s f(r) dW(r).$$

PROOF. (1) If  $f$  and  $g$  belong to  $M_t^2$ , then  $\mathbf{1}_{[0,t]}f$  and  $\mathbf{1}_{[0,t]}g$  belong to  $M^2$ , so there are sequences  $f_1, f_2, \dots$  and  $g_1, g_2, \dots$  in  $M_{\text{step}}^2$  approximating  $\mathbf{1}_{[0,t]}f$  and  $\mathbf{1}_{[0,t]}g$ . It follows that  $\mathbf{1}_{[0,t]}(\alpha f + \beta g)$  can be approximated by  $\alpha f_1 + \beta g_1, \alpha f_2 + \beta g_2, \dots$ . By Exercise 4.3

$$(145) \quad I(\alpha f_n + \beta g_n) = \alpha I(f_n) + \beta I(g_n)$$

for each  $n$ . Taking the  $L^2$  limit on both sides of this equality as  $n \rightarrow \infty$ , we obtain

$$(146) \quad I(\mathbf{1}_{[0,t]}(\alpha f + \beta g)) = \alpha I(\mathbf{1}_{[0,t]}f) + \beta I(\mathbf{1}_{[0,t]}g),$$

which proves (1).

(2) This follows by approximating  $\mathbf{1}_{[0,t]}f$  by random step processes in  $M_{\text{step}}^2$  and using Proposition 4.3.

(3) If  $f$  belongs to  $M_t^2$ , then  $\mathbf{1}_{[0,t]}f$  belongs to  $M^2$ . Let  $f_1, f_2, \dots$  be a sequence of processes in  $M_{\text{step}}^2$  approximating  $\mathbf{1}_{[0,t]}f$ . By Exercise 4.5

$$(147) \quad \mathbb{E}(I(\mathbf{1}_{[0,t]}f_n) \mid \mathcal{F}_s) = I(\mathbf{1}_{[0,s]}f_n)$$

for each  $n$ . By taking the  $L^2$  limit of both sides of this equality as  $n \rightarrow \infty$ , we shall show that

$$(148) \quad \mathbb{E}(I(\mathbf{1}_{[0,t]}f) \mid \mathcal{F}_s) = I(\mathbf{1}_{[0,s]}f),$$

which is what needs to be proved. Indeed, observe that  $\mathbf{1}_{[0,s]}f_1, \mathbf{1}_{[0,s]}f_2, \dots$  is a sequence in  $M_{\text{step}}^2$  approximating  $\mathbf{1}_{[0,s]}f$ , so

$$(149) \quad I(\mathbf{1}_{[0,s]}f_n) \rightarrow I(\mathbf{1}_{[0,s]}f) \quad \text{in } L^2 \text{ as } n \rightarrow \infty.$$

Similarly,  $\mathbf{1}_{[0,t]}f_1, \mathbf{1}_{[0,t]}f_2, \dots$  is also a sequence in  $M_{\text{step}}^2$  approximating  $\mathbf{1}_{[0,t]}f$ , which implies that

$$(150) \quad I(\mathbf{1}_{[0,t]}f_n) \rightarrow I(\mathbf{1}_{[0,t]}f) \quad \text{in } L^2 \text{ as } n \rightarrow \infty.$$

The lemma below implies that

$$(151) \quad \mathbb{E}(I(\mathbf{1}_{[0,t]}f_n) \mid \mathcal{F}_s) \rightarrow \mathbb{E}(I(\mathbf{1}_{[0,t]}f) \mid \mathcal{F}_s) \quad \text{in } L^2 \text{ as } n \rightarrow \infty,$$

completing the proof.

**Lemma 4.12.** *If  $\xi$  and  $\xi_1, \xi_2, \dots$  are square integrable random variables such that  $\xi_n \rightarrow \xi$  in  $L^2$  as  $n \rightarrow \infty$ , then*

$$(152) \quad \mathbb{E}(\xi_n \mid \mathcal{G}) \rightarrow \mathbb{E}(\xi \mid \mathcal{G}) \quad \text{in } L^2 \text{ as } n \rightarrow \infty$$

for any  $\sigma$ -field  $\mathcal{G}$  on  $\Omega$  contained in  $\mathcal{F}$ .

PROOF. By Jensen's inequality, see Theorem 2.2,

$$(153) \quad |\mathbb{E}(\xi_n \mid \mathcal{G}) - \mathbb{E}(\xi \mid \mathcal{G})|^2 = |\mathbb{E}(\xi_n - \xi \mid \mathcal{G})|^2 \leq \mathbb{E}(|\xi_n - \xi|^2 \mid \mathcal{G}),$$

which implies that

$$(154) \quad \mathbb{E}(|\mathbb{E}(\xi_n \mid \mathcal{G}) - \mathbb{E}(\xi \mid \mathcal{G})|^2) \leq \mathbb{E}(\mathbb{E}(|\xi_n - \xi|^2 \mid \mathcal{G}))$$

$$(155) \quad = \mathbb{E}(|\xi_n - \xi|^2) \rightarrow 0$$

as  $n \rightarrow \infty$ . □

In the next theorem we consider the stochastic integral  $\int_0^t f(s) dW(s)$  as a function of the upper integration limit  $t$ . Similarly as for the Riemann integral, it is natural to ask if this is a continuous function of  $t$ . The answer to this question involves the notion of a modification of a stochastic process.

**DEFINITION 4.13.** *Let  $\xi(t)$  and  $\zeta(t)$  be stochastic processes defined for  $t \in T$ , where  $T \subset \mathbb{R}$ . We say that the processes are **modifications** (or versions) of one another if*

$$(156) \quad P\{\xi(t) = \zeta(t)\} = 1 \quad \text{for all } t \in T.$$

**REMARK 4.11.** *If  $T \subset \mathbb{R}$  is a countable set, then (156) is equivalent to the condition*

$$(157) \quad P\{\xi(t) = \zeta(t) \text{ for all } t \in T\} = 1.$$

However, this is not necessarily so if  $T$  is uncountable.

The following result is stated without proof.

**Theorem 4.14.** *Let  $f(s)$  be a process belonging to  $M_t^2$  and let*

$$(158) \quad \xi(t) = \int_0^t f(s) dW(s)$$

for every  $t \geq 0$ . Then there exists an adapted modification  $\zeta(t)$  of  $\xi(t)$  with a.s. continuous paths. This modification is unique up to equality a.s.

From now on we shall always identify  $\int_0^t f(s) dW(s)$  with the adapted modification having a.s. continuous paths. This convention works beautifully together with Theorem 4.8 whenever there is a need to show that a stochastic integral can be used as the integrand of another stochastic integral, i.e. belongs to  $M_T^2$  for  $T \geq 0$ . This is illustrated by the next exercise.

EXERCISE 4.11. *Show that*

$$(159) \quad \xi(t) = \int_0^t W(s) dW(s)$$

belongs to  $M_T^2$  for any  $T \geq 0$ .

REMARK 4.12. **Hint:** By Theorem 4.14,  $\xi(t)$  can be identified with an adapted modification having a.s. continuous trajectories. Because of this, it suffices to verify that  $\xi(t)$  satisfies condition (119) of Theorem 4.8.

SOLUTION 4.11. From Exercise 4.8, we have  $\xi(t) = \frac{1}{2}W(t)^2 - \frac{1}{2}t$ . This process is adapted and has a.s. continuous paths. We check the integrability condition (119):

$$\begin{aligned} \mathbb{E} \int_0^T |\xi(t)|^2 dt &= \int_0^T \mathbb{E} \left( \frac{1}{2}W(t)^2 - \frac{1}{2}t \right)^2 dt \\ &= \frac{1}{4} \int_0^T \mathbb{E}(W(t)^4 - 2tW(t)^2 + t^2) dt \\ &= \frac{1}{4} \int_0^T (3t^2 - 2t^2 + t^2) dt \\ &= \frac{1}{4} \int_0^T 2t^2 dt = \frac{1}{2} \cdot \frac{1}{3}T^3 = \frac{1}{6}T^3 < \infty. \end{aligned}$$

By Theorem 4.8,  $\xi \in M_T^2$  for any  $T \geq 0$ .

#### 4. Stochastic Differential and Itô Formula

Any continuously differentiable function  $x(t)$  such that  $x(0) = 0$  satisfies the formulae

$$(160) \quad x(T)^2 = 2 \int_0^T x(t) dx(t),$$

$$(161) \quad x(T)^3 = 3 \int_0^T x(t)^2 dx(t),$$

where  $dx(t)$  can simply be understood as a shorthand notation for  $x'(t)dt$ , the integrals on the right-hand side being Riemann integrals. Similar formulae have been obtained in Exercises 4.8 and 4.10 for the Wiener process:

$$(162) \quad W(T)^2 = \int_0^T dt + 2 \int_0^T W(t) dW(t),$$

$$(163) \quad W(T)^3 = 3 \int_0^T W(t) dt + 3 \int_0^T W(t)^2 dW(t).$$

Here the stochastic integrals resemble the corresponding expressions for a smooth function  $x(t)$ , but there are also the intriguing terms  $\int_0^T dt$  and  $3 \int_0^T W(t) dt$ . The formulae for  $W(T)^2$  and  $W(T)^3$  are examples of the much more general Itô formula, a crucial tool for transforming and computing stochastic integrals. Terms such as  $\int_0^T dt$  and  $3 \int_0^T W(t) dt$ , which have no analogues in the classical calculus of smooth functions, are a feature inherent in the Itô formula and referred to as the **Itô correction**. The class of processes appearing in the Itô formula is defined as follows.

**DEFINITION 4.15.** *A stochastic process  $\xi(t), t \geq 0$  is called an **Itô process** if it has a.s. continuous paths and can be represented as*

$$(164) \quad \xi(T) = \xi(0) + \int_0^T a(t) dt + \int_0^T b(t) dW(t) \quad a.s.,$$

where  $b(t)$  is a process belonging to  $M_T^2$  for all  $T > 0$  and  $a(t)$  is a process adapted to the filtration  $\mathcal{F}_t$  such that

$$(165) \quad \int_0^T |a(t)| dt < \infty \quad a.s.$$

for all  $T \geq 0$ . The class of all adapted processes  $a(t)$  satisfying (165) for some  $T > 0$  will be denoted by  $\mathcal{L}_T^1$ .

For an Itô process  $\xi$  it is customary to write (164) as

$$(166) \quad d\xi(t) = a(t) dt + b(t) dW(t)$$

and to call  $d\xi(t)$  the **stochastic differential** of  $\xi(t)$ . This is known as the Itô differential notation. It should be emphasized that the stochastic differential has no well-defined mathematical meaning on its own and should always be understood in the context of the rigorous equation (164). The Itô differential notation is an efficient way of writing this equation, rather than an attempt to give a precise mathematical meaning to the stochastic differential.

**EXAMPLE 4.1.** *The Wiener process  $W(t)$  satisfies*

$$(167) \quad W(T) = \int_0^T dW(t).$$

(The right-hand side is the stochastic integral  $I(f)$  of the random step process  $f = \mathbf{1}_{[0,T)}.$ ) This is an equation of the form (164) with  $a(t) = 0$  and  $b(t) = 1$ , which belong, respectively, to  $\mathcal{L}_T^1$  and  $M_T^2$  for any  $T \geq 0$ . It follows that the Wiener process is an Itô process.

EXAMPLE 4.2. *Every process of the form*

$$(168) \quad \xi(T) = \xi(0) + \int_0^T a(t) dt,$$

where  $a(t)$  is a process belonging to  $\mathcal{L}_T^1$  for any  $T \geq 0$ , is an Itô process. In particular, every deterministic process of this form, where  $a(t)$  is a deterministic integrable function, is an Itô process.

EXAMPLE 4.3. Since  $a(t) = 1$  and  $b(t) = 2W(t)$  belong, respectively, to the classes  $\mathcal{L}_T^1$  and  $M_T^2$  for each  $T \geq 0$ ,

$$(169) \quad W(T)^2 = \int_0^T dt + 2 \int_0^T W(t) dW(t)$$

is an Itô process; see Exercise 4.8. The last equation can also be written as

$$(170) \quad d(W(t)^2) = dt + 2W(t) dW(t),$$

providing a formula for the stochastic differential  $d(W(t)^2)$  of  $W(t)^2$ .

EXERCISE 4.12. Show that  $W(t)^3$  is an Itô process and find a formula for the stochastic differential  $d(W(t)^3)$ .

REMARK 4.13. **Hint:** Refer to Exercise 4.10.

SOLUTION 4.12. From Exercise 4.10, we have the equality

$$\int_0^T W(t)^2 dW(t) = \frac{1}{3}W(T)^3 - \int_0^T W(t) dt,$$

which can be rewritten as

$$W(T)^3 = 3 \int_0^T W(t) dt + 3 \int_0^T W(t)^2 dW(t).$$

This matches the form of an Itô process in (164) with  $a(t) = 3W(t)$  and  $b(t) = 3W(t)^2$ .  $a(t)$  is adapted and continuous, so it belongs to  $\mathcal{L}_T^1$ .  $b(t)$  belongs to  $M_T^2$  as shown in Exercise 4.7. Thus,  $W(t)^3$  is an Itô process with stochastic differential

$$d(W(t)^3) = 3W(t) dt + 3W(t)^2 dW(t).$$

EXERCISE 4.13. Show that  $tW(t)$  is an Itô process and find a formula for the stochastic differential  $d(tW(t))$ .

REMARK 4.14. **Hint:** Use Exercise 4.9.

SOLUTION 4.13. From Exercise 4.9, we have

$$\int_0^T t dW(t) = TW(T) - \int_0^T W(t) dt,$$

which can be rewritten as

$$TW(T) = \int_0^T W(t) dt + \int_0^T t dW(t).$$

This is in the form of an Itô process with  $a(t) = W(t)$  and  $b(t) = t$ .  $a(t)$  is adapted and continuous, so  $a \in \mathcal{L}_T^1$ .  $b(t)$  is deterministic and square-integrable on  $[0, T]$ , so  $b \in M_T^2$ . Thus,  $tW(t)$  is an Itô process and its stochastic differential is

$$d(tW(t)) = W(t) dt + t dW(t).$$

The above examples and exercises are particular cases of an extremely important general formula for transforming stochastic differentials established by Itô. To begin with, we shall state and prove a simplified version of the formula, followed by the general theorem. The proof of the simplified version captures the essential ingredients of the somewhat tedious general argument, which will be omitted. In fact, many of the essential ingredients of the proof are already present in the examples and exercises considered above.

**Theorem 4.16** (Itô formula, simplified version). *Suppose that  $F(t, x)$  is a real-valued function with continuous partial derivatives  $F'_t(t, x)$ ,  $F'_x(t, x)$  and  $F''_{xx}(t, x)$  for all  $t \geq 0$  and  $x \in \mathbb{R}$ . We also assume that the process  $F'_x(t, W(t))$  belongs to  $M_T^2$  for all  $T \geq 0$ . Then  $F(t, W(t))$  is an Itô process such that*

$$(171) \quad F(T, W(T)) - F(0, W(0)) = \int_0^T \left( F'_t(t, W(t)) + \frac{1}{2} F''_{xx}(t, W(t)) \right) dt + \int_0^T F'_x(t, W(t)) dW(t) \quad a.s.$$

In differential notation this formula can be written as

$$(172) \quad dF(t, W(t)) = \left( F'_t(t, W(t)) + \frac{1}{2} F''_{xx}(t, W(t)) \right) dt + F'_x(t, W(t)) dW(t).$$

REMARK 4.15. Compare the latter with the chain rule

$$(173) \quad dF(t, x(t)) = F'_t(t, x(t)) dt + F'_x(t, x(t)) dx(t)$$

for a smooth function  $x(t)$ , where  $dx(t)$  is understood as a shorthand notation for  $x'(t) dt$ . The additional term  $\frac{1}{2} F''_{xx}(t, W(t)) dt$  in (172) is called the **Itô correction**.

PROOF. First we shall prove the Itô formula under the assumption that  $F$  and the partial derivatives  $F'_x$  and  $F''_{xx}$  are bounded by some  $C > 0$ .

Consider a partition  $0 = t_0^n < t_1^n < \dots < t_n^n = T$ , where  $t_i^n = \frac{iT}{n}$ , of  $[0, T]$  into  $n$  equal parts. We shall denote the increments  $W(t_{i+1}^n) - W(t_i^n)$  by  $\Delta_i^n W$  and  $t_{i+1}^n - t_i^n$  by  $\Delta_i^n t$ . We shall also write  $W_i^n$  instead of  $W(t_i^n)$  for brevity. According to the Taylor formula, there is a point  $\tilde{W}_i^n$  in each interval  $[W(t_i^n), W(t_{i+1}^n)]$  and a point  $\tilde{t}_i^n$  in each interval  $[t_i^n, t_{i+1}^n]$  such that

$$\begin{aligned}
& F(T, W(T)) - F(0, W(0)) \\
&= \sum_{i=0}^{n-1} (F(t_{i+1}^n, W_{i+1}^n) - F(t_i^n, W_i^n)) \\
(174) \quad &= \sum_{i=0}^{n-1} (F(t_{i+1}^n, W_{i+1}^n) - F(t_i^n, W_{i+1}^n)) + \sum_{i=0}^{n-1} (F(t_i^n, W_{i+1}^n) - F(t_i^n, W_i^n)) \\
(175) \quad &= \sum_{i=0}^{n-1} F'_t(\tilde{t}_i^n, W_{i+1}^n) \Delta_i^n t + \sum_{i=0}^{n-1} F'_x(t_i^n, W_i^n) \Delta_i^n W + \frac{1}{2} \sum_{i=0}^{n-1} F''_{xx}(t_i^n, \tilde{W}_i^n) (\Delta_i^n W)^2 \\
(176) \quad &= \sum_{i=0}^{n-1} F'_t(\tilde{t}_i^n, W_{i+1}^n) \Delta_i^n t + \frac{1}{2} \sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n) \Delta_i^n t + \sum_{i=0}^{n-1} F'_x(t_i^n, W_i^n) \Delta_i^n W \\
(177) \quad &+ \frac{1}{2} \sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n) ((\Delta_i^n W)^2 - \Delta_i^n t) \\
(178) \quad &+ \frac{1}{2} \sum_{i=0}^{n-1} [F''_{xx}(t_i^n, \tilde{W}_i^n) - F''_{xx}(t_i^n, W_i^n)] (\Delta_i^n W)^2.
\end{aligned}$$

We shall deal separately with each sum in the last expression, splitting the proof into several steps.

**Step 1.** We claim that

$$(179) \quad \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} F'_t(\tilde{t}_i^n, W_{i+1}^n) \Delta_i^n t = \int_0^T F'_t(t, W(t)) dt \quad a.s.$$

This is because the paths of  $W(t)$  are a.s. continuous, and  $F'_t(t, x)$  is continuous as a function of two variables by assumption. Indeed, every continuous path of the Wiener process is bounded on  $[0, T]$ , i.e. there is an  $M > 0$ , which may depend on the path, such that

$$(180) \quad |W(t)| \leq M \quad \text{for all } t \in [0, T].$$

As a continuous function,  $F'_t(t, x)$  is uniformly continuous on the compact set  $[0, T] \times [-M, M]$  and  $W$  is uniformly continuous on  $[0, T]$ . It follows that

$$(181) \quad \lim_{n \rightarrow \infty} \sup_{i, t} |F'_t(\tilde{t}_i^n, W_{i+1}^n) - F'_t(t, W(t))| = 0 \quad a.s.,$$

where the supremum is taken over all  $i = 0, \dots, n-1$  and  $t \in [t_i^n, t_{i+1}^n]$ . By the definition of the Riemann integral this proves the claim.

**Step 2.** This is very similar to Step 1. By continuity

$$(182) \quad \lim_{n \rightarrow \infty} \sup_{i, t} |F''_{xx}(t_i^n, W_i^n) - F''_{xx}(t, W(t))| = 0 \quad a.s.,$$



where the supremum is taken over all  $i = 0, \dots, n-1$  and  $t \in [t_i^n, t_{i+1}^n]$ . By the definition of the Riemann integral

$$(183) \quad \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n) \Delta_i^n t = \int_0^T F''_{xx}(t, W(t)) dt \quad a.s.$$

**Step 3.** We shall verify that

$$(184) \quad \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} F'_x(t_i^n, W_i^n) \Delta_i^n W = \int_0^T F'_x(t, W(t)) dW(t) \quad \text{in } L^2.$$

If  $F'_x(t, x)$  is bounded by  $C > 0$ , then  $f(t) = F'_x(t, W(t))$  belongs to  $M_T^2$  by Theorem 4.8, and the sequence of random step processes

$$(185) \quad f_n = \sum_{i=0}^{n-1} F'_x(t_i^n, W_i^n) \mathbf{1}_{[t_i^n, t_{i+1}^n)} \in M_{\text{step}}^2$$

approximates  $f$ . Indeed, by continuity

$$(186) \quad \lim_{n \rightarrow \infty} |f_n(t) - f(t)|^2 = 0 \quad \text{for any } t \in [0, T], \quad a.s.$$

Because  $|f_n(t) - f(t)|^2 \leq 4C^2$ , it follows that

$$(187) \quad \lim_{n \rightarrow \infty} \int_0^T |f_n(t) - f(t)|^2 dt = 0 \quad a.s.$$

by Lebesgue's dominated convergence theorem. But  $\int_0^T |f_n(t) - f(t)|^2 dt \leq 4TC^2$ , so

$$(188) \quad \lim_{n \rightarrow \infty} \mathbb{E} \left( \int_0^T |f_n(t) - f(t)|^2 dt \right) = 0$$

again by Lebesgue's dominated convergence theorem. This shows that  $f_n$  approximates  $f$ , which in turn implies that  $I(f_n)$  tends to  $I(f)$  in  $L^2$ , concluding Step 3.

**Step 4.** If  $F''_{xx}$  is bounded by  $C > 0$ , then

$$(189) \quad \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n) ((\Delta_i^n W)^2 - \Delta_i^n t) = 0 \quad \text{in } L^2,$$

since

$$\begin{aligned}
(190) \quad & \mathbb{E} \left| \sum_{i=0}^{n-1} F''_{xx}(t_i^n, W_i^n) ((\Delta_i^n W)^2 - \Delta_i^n t) \right|^2 \\
&= \sum_{i=0}^{n-1} \mathbb{E} |F''_{xx}(t_i^n, W_i^n) ((\Delta_i^n W)^2 - \Delta_i^n t)|^2 \\
(191) \quad &= \sum_{i=0}^{n-1} \mathbb{E} |F''_{xx}(t_i^n, W_i^n)|^2 \mathbb{E} |(\Delta_i^n W)^2 - \Delta_i^n t|^2 \\
(192) \quad &\leq C^2 \sum_{i=0}^{n-1} \mathbb{E} |(\Delta_i^n W)^2 - \Delta_i^n t|^2 = 2C^2 \sum_{i=0}^{n-1} (\Delta_i^n t)^2 \\
(193) \quad &= 2C^2 \sum_{i=0}^{n-1} \frac{T^2}{n^2} = 2C^2 \frac{T^2}{n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.
\end{aligned}$$

The first equality above holds because for any  $i < j$

$$\begin{aligned}
(194) \quad & \mathbb{E} [F''_{xx}(t_i^n, W_i^n) ((\Delta_i^n W)^2 - \Delta_i^n t) F''_{xx}(t_j^n, W_j^n) ((\Delta_j^n W)^2 - \Delta_j^n t)] \\
(195) \quad &= \mathbb{E} [F''_{xx}(t_i^n, W_i^n) ((\Delta_i^n W)^2 - \Delta_i^n t) F''_{xx}(t_j^n, W_j^n)] \mathbb{E} [(\Delta_j^n W)^2 - \Delta_j^n t] \\
(196) \quad &= 0.
\end{aligned}$$

This is because the expressions in the last two square brackets are independent and the last expectation is equal to zero.

**Step 5.** By a similar continuity argument as in Steps 1 and 2

$$(197) \quad \lim_{n \rightarrow \infty} \sup_i |F''_{xx}(t_i^n, \tilde{W}_i^n) - F''_{xx}(t_i^n, W_i^n)| = 0 \quad a.s.,$$

where the supremum is taken over all  $i = 0, 1, \dots, n-1$ . Since  $\sum_{i=0}^{n-1} (\Delta_i^n W)^2 \rightarrow T$  in  $L^2$  as  $n \rightarrow \infty$ , there is a subsequence  $n_1 < n_2 < \dots$  such that

$$(198) \quad \sum_{i=0}^{n_k-1} (\Delta_i^{n_k} W)^2 \rightarrow T \quad a.s.$$

as  $k \rightarrow \infty$ . It follows that

$$(199) \quad \left| \sum_{i=0}^{n_k-1} \left( F''_{xx}(t_i^{n_k}, \tilde{W}_i^{n_k}) - F''_{xx}(t_i^{n_k}, W_i^{n_k}) \right) (\Delta_i^{n_k} W)^2 \right|$$

$$(200) \quad \leq \sup_i |F''_{xx}(t_i^{n_k}, \tilde{W}_i^{n_k}) - F''_{xx}(t_i^{n_k}, W_i^{n_k})| \sum_{i=0}^{n_k-1} (\Delta_i^{n_k} W)^2 \rightarrow 0 \quad a.s.$$

as  $k \rightarrow \infty$ .

In those steps above where  $L^2$  convergence was obtained, we also have convergence a.s. by taking a subsequence. This proves the Itô formula (171) under the assumption that the partial

derivatives  $F'_x(t, x)$  and  $F''_{xx}(t, x)$  are bounded. To complete the proof we need to remove this assumption.

Let  $F(t, x)$  be an arbitrary function satisfying the conditions of Theorem 4.16. For each positive integer  $n$  take a smooth function  $\varphi_n$  from  $\mathbb{R}$  to  $[0, 1]$  such that  $\varphi_n(x) = 1$  for any  $x \in [-n, n]$  and  $\varphi_n(x) = 0$  for any  $x \notin [-n - 1, n + 1]$ . Then

$$(201) \quad F_n(t, x) = \varphi_n(x)F(t, x)$$

also satisfies the conditions of Theorem 4.16 and has bounded partial derivatives  $(F_n)'_x(t, x)$  and  $(F_n)''_{xx}(t, x)$  for each  $n$ . Therefore, by the first part of the proof

$$(202) \quad \begin{aligned} & F_n(T, W(T)) - F_n(0, W(0)) \\ &= \int_0^T \left( (F_n)'_t(t, W(t)) + \frac{1}{2}(F_n)''_{xx}(t, W(t)) \right) dt + \int_0^T (F_n)'_x(t, W(t)) dW(t). \end{aligned}$$

Consider the expanding sequence of events

$$(203) \quad A_n = \left\{ \sup_{t \in [0, T]} |W(t)| < n \right\}.$$

Since  $F(t, x) = F_n(t, x)$  for every  $t \in [0, T]$  and  $x \in [-n, n]$ , it follows that (171) holds on  $A_n$ . It remains to show that

$$(204) \quad \lim_{n \rightarrow \infty} P(A_n) = 1$$

to prove that (171) holds a.s. But the latter is true because of Doob's maximal  $L^2$  inequality, Theorem 6.7, which implies that

$$(205) \quad n^2(1 - P(A_n)) = n^2 P \left\{ \sup_{t \in [0, T]} |W(t)| \geq n \right\}$$

$$(206) \quad \leq \mathbb{E} \left( \sup_{t \in [0, T]} |W(t)| \right)^2$$

$$(207) \quad \leq 4\mathbb{E}|W(T)|^2 = 4T,$$

completing the proof. □

EXAMPLE 4.4. For  $F(t, x) = x^2$  we have  $F'_t(t, x) = 0$ ,  $F'_x(t, x) = 2x$  and  $F''_{xx}(t, x) = 2$ . The Itô formula gives

$$(208) \quad d(W(t)^2) = dt + 2W(t) dW(t),$$

which is the same equality as in Exercise 4.8.

EXAMPLE 4.5. For  $F(t, x) = x^3$  we have  $F'_t(t, x) = 0$ ,  $F'_x(t, x) = 3x^2$  and  $F''_{xx}(t, x) = 6x$ . By the Itô formula we obtain the same equality

$$(209) \quad d(W(t)^3) = 3W(t) dt + 3W(t)^2 dW(t)$$

as in Exercise 4.10.

**EXERCISE 4.14** (Exponential martingale). *Show that the exponential martingale  $X(t) = e^{W(t)}e^{-t/2}$  is an Itô process and verify that it satisfies the equation*

$$(210) \quad dX(t) = X(t) dW(t).$$

**REMARK 4.16.** ***Hint:** Use the Itô formula with  $F(t, x) = e^x e^{-t/2}$ .*

**SOLUTION 4.14.** *Let  $F(t, x) = e^x e^{-t/2}$ . The partial derivatives are*

$$\begin{aligned} F'_t(t, x) &= -\frac{1}{2}e^x e^{-t/2} = -\frac{1}{2}F(t, x), \\ F'_x(t, x) &= e^x e^{-t/2} = F(t, x), \\ F''_{xx}(t, x) &= e^x e^{-t/2} = F(t, x). \end{aligned}$$

*Applying the simplified Itô formula (172) with  $W(t)$  as the process:*

$$\begin{aligned} dX(t) &= dF(t, W(t)) = \left( F'_t(t, W(t)) + \frac{1}{2}F''_{xx}(t, W(t)) \right) dt + F'_x(t, W(t)) dW(t) \\ &= \left( -\frac{1}{2}X(t) + \frac{1}{2}X(t) \right) dt + X(t) dW(t) \\ &= X(t) dW(t). \end{aligned}$$

*This shows that  $X(t)$  is an Itô process (since  $X(t)$  is adapted, continuous, and  $X(t) \in M_T^2$  due to the finiteness of exponential moments of Gaussian variables) and satisfies the given stochastic differential equation.*

As compared with the simplified version just proved, in the general Itô formula below  $W(t)$  will be replaced by an arbitrary Itô process  $\xi(t)$  such that

$$(211) \quad d\xi(t) = a(t) dt + b(t) dW(t),$$

where  $a$  belongs to  $\mathcal{L}_t^1$  and  $b$  to  $M_t^2$  for all  $t \geq 0$ . In the general case the proof will be omitted.

**Theorem 4.17** (Itô formula, general case). *Let  $\xi(t)$  be an Itô process as above. Suppose that  $F(t, x)$  is a real-valued function with continuous partial derivatives  $F'_t(t, x)$ ,  $F'_x(t, x)$  and  $F''_{xx}(t, x)$  for all  $t \geq 0$  and  $x \in \mathbb{R}$ . We also assume that the process  $b(t)F'_x(t, \xi(t))$  belongs to  $M_T^2$  for all  $T \geq 0$ . Then  $F(t, \xi(t))$  is an Itô process such that*

$$(212) \quad \begin{aligned} dF(t, \xi(t)) &= \left( F'_t(t, \xi(t)) + F'_x(t, \xi(t))a(t) + \frac{1}{2}F''_{xx}(t, \xi(t))b(t)^2 \right) dt \\ &\quad + F'_x(t, \xi(t))b(t) dW(t). \end{aligned}$$

A convenient way to remember the Itô formula is to write down the Taylor expansion for  $F(t, x)$  up to the terms with partial derivatives of order two, substituting  $\xi(t)$  for  $x$  and the expression on the right-hand side of (211) for  $d\xi(t)$ , and using the so-called Itô multiplication table

$$(213) \quad \begin{aligned} dt \, dt &= 0, & dt \, dW(t) &= 0, \\ dW(t) \, dt &= 0, & dW(t) \, dW(t) &= dt. \end{aligned}$$

This informal procedure gives

$$(214) \quad dF = F'_t dt + F'_x d\xi + \frac{1}{2} F''_{tt} dt \, dt + F''_{tx} dt \, d\xi + \frac{1}{2} F''_{xx} d\xi \, d\xi$$

$$(215) \quad = F'_t dt + F'_x (a \, dt + b \, dW)$$

$$(216) \quad + \frac{1}{2} F''_{tt} dt \, dt + F''_{tx} dt (a \, dt + b \, dW) + \frac{1}{2} F''_{xx} (a \, dt + b \, dW)(a \, dt + b \, dW)$$

$$(217) \quad = F'_t dt + F'_x (a \, dt + b \, dW) + \frac{1}{2} F''_{xx} b^2 dt$$

$$(218) \quad = \left( F'_t + F'_x a + \frac{1}{2} F''_{xx} b^2 \right) dt + F'_x b \, dW,$$

which is the expression in (212). Here we have omitted the arguments  $(t, \xi(t))$  and, respectively,  $(t)$  in all functions for brevity.

EXERCISE 4.15. *Applying the Itô formula to  $F(t, x) = x^n$ , show that*

$$(219) \quad dW(t)^n = \frac{n(n-1)}{2} W(t)^{n-2} dt + nW(t)^{n-1} dW(t).$$

REMARK 4.17. **Hint:** *This is a direct application of the Itô formula, but be careful with the assumptions, in particular make sure that  $nW(t)^{n-1}$  belongs to  $M_T^2$  for all  $T > 0$ .*

SOLUTION 4.15. *Let  $F(t, x) = x^n$ . The partial derivatives are*

$$F'_t = 0, \quad F'_x = nx^{n-1}, \quad F''_{xx} = n(n-1)x^{n-2}.$$

*Applying the simplified Itô formula (172):*

$$\begin{aligned} d(W(t)^n) &= \left( 0 + \frac{1}{2} n(n-1) W(t)^{n-2} \right) dt + nW(t)^{n-1} dW(t) \\ &= \frac{n(n-1)}{2} W(t)^{n-2} dt + nW(t)^{n-1} dW(t). \end{aligned}$$

*To ensure  $nW(t)^{n-1} \in M_T^2$ , we check the integrability condition:*

$$\mathbb{E} \int_0^T |nW(t)^{n-1}|^2 dt = n^2 \int_0^T \mathbb{E}(W(t)^{2n-2}) dt.$$

*Since  $W(t)$  is Gaussian, its moments of all orders are finite. Specifically,  $\mathbb{E}(W(t)^{2k}) = \frac{(2k)!}{2^k k!} t^k$ . Thus the integral is finite, and  $W(t)^n$  is a valid Itô process.*

EXERCISE 4.16 (Ornstein-Uhlenbeck process). Suppose that  $\alpha > 0$  and  $\sigma \in \mathbb{R}$  are fixed. Define  $Y(t), t \geq 0$  to be an adapted modification of the Itô integral

$$(220) \quad Y(t) = \sigma e^{-\alpha t} \int_0^t e^{\alpha s} dW(s)$$

with a.s. continuous paths. Show that  $Y(t)$  satisfies

$$(221) \quad dY(t) = -\alpha Y(t) dt + \sigma dW(t).$$

REMARK 4.18. **Hint:**  $Y(t) = F(t, \xi(t))$  with  $\xi(t) = \sigma \int_0^t e^{\alpha s} dW(s)$  and  $F(t, x) = e^{-\alpha t} x$ .

SOLUTION 4.16. Let  $\xi(t) = \sigma \int_0^t e^{\alpha s} dW(s)$ . Then  $\xi(t)$  is an Itô process with  $d\xi(t) = 0 dt + \sigma e^{\alpha t} dW(t)$ , so  $a(t) = 0$  and  $b(t) = \sigma e^{\alpha t}$ . Let  $F(t, x) = e^{-\alpha t} x$ . The partial derivatives are

$$F'_t = -\alpha e^{-\alpha t} x = -\alpha F, \quad F'_x = e^{-\alpha t}, \quad F''_{xx} = 0.$$

Using the general Itô formula (212) for  $Y(t) = F(t, \xi(t))$ :

$$\begin{aligned} dY(t) &= \left( F'_t + F'_x a + \frac{1}{2} F''_{xx} b^2 \right) dt + F'_x b dW(t) \\ &= (-\alpha e^{-\alpha t} \xi(t) + e^{-\alpha t} \cdot 0 + 0) dt + e^{-\alpha t} (\sigma e^{\alpha t}) dW(t) \\ &= -\alpha Y(t) dt + \sigma dW(t). \end{aligned}$$

This confirms that  $Y(t)$  satisfies the given stochastic differential equation.

## 5. Stochastic Differential Equations

This section will be devoted to stochastic differential equations of the form

$$(222) \quad d\xi(t) = f(\xi(t)) dt + g(\xi(t)) dW(t).$$

Solutions will be sought in the class of Itô processes  $\xi(t)$  with a.s. continuous paths. As in the theory of ordinary differential equations, we need to specify an initial condition

$$(223) \quad \xi(0) = \xi_0.$$

Here  $\xi_0$  can be a fixed real number or, in general, a random variable. Being an Itô process,  $\xi(t)$  must be adapted to the filtration  $\mathcal{F}_t$  of  $W(t)$ , so  $\xi_0$  must be  $\mathcal{F}_0$ -measurable.

EXAMPLE 4.6. The stochastic differential equation

$$(224) \quad dX(t) = X(t) dW(t)$$

was used as a motivation for developing Itô stochastic calculus at the beginning of the present chapter. In Exercise 4.14 it was verified that the exponential martingale

$$(225) \quad X(t) = e^{W(t)} e^{-t/2}$$

satisfies (224). It also satisfies the initial condition  $X(0) = 1$ . This is an example of a linear stochastic differential equation. For the solution of a general equation of this type with an arbitrary initial condition, see Exercise 4.20.

EXAMPLE 4.7. In Exercise 4.16 it was shown that the Ornstein-Uhlenbeck process

$$(226) \quad Y(t) = \sigma e^{-\alpha t} \int_0^t e^{\alpha s} dW(s)$$

satisfies the stochastic differential equation

$$(227) \quad dY(t) = -\alpha Y(t) dt + \sigma dW(t)$$

with initial condition  $Y(0) = 0$ . This is an example of an inhomogeneous linear stochastic differential equation. See Exercise 4.17 for a solution with an arbitrary initial condition.

DEFINITION 4.18. An Itô process  $\xi(t), t \geq 0$  is called a **solution** of the initial value problem

$$(228) \quad d\xi(t) = f(\xi(t)) dt + g(\xi(t)) dW(t),$$

$$(229) \quad \xi(0) = \xi_0$$

if  $\xi_0$  is an  $\mathcal{F}_0$ -measurable random variable, the processes  $f(\xi(t))$  and  $g(\xi(t))$  belong, respectively, to  $\mathcal{L}_T^1$  and  $M_T^2$ , and

$$(230) \quad \xi(T) = \xi_0 + \int_0^T f(\xi(t)) dt + \int_0^T g(\xi(t)) dW(t) \quad a.s.$$

for all  $T \geq 0$ .

REMARK 4.19. In view of this definition, the notion of a stochastic differential equation is a fiction. In fact, only stochastic integral equations of the form (230) have a rigorous mathematical meaning. However, it proves convenient to use stochastic differentials informally and talk of stochastic differential equations to draw on the analogy with ordinary differential equations. This analogy will be employed to solve some stochastic differential equations later on in this section.

The existence and uniqueness theorem below resembles that in the theory of ordinary differential equations, where it is also crucial for the right-hand side of the equation to be Lipschitz continuous as a function of the solution.

**Theorem 4.19.** Suppose that  $f$  and  $g$  are Lipschitz continuous functions from  $\mathbb{R}$  to  $\mathbb{R}$ , i.e. there is a constant  $C > 0$  such that for any  $x, y \in \mathbb{R}$

$$(231) \quad |f(x) - f(y)| \leq C|x - y|,$$

$$(232) \quad |g(x) - g(y)| \leq C|x - y|.$$

Moreover, let  $\xi_0$  be an  $\mathcal{F}_0$ -measurable square integrable random variable. Then the initial value problem

$$(233) \quad d\xi(t) = f(\xi(t)) dt + g(\xi(t)) dW(t),$$

$$(234) \quad \xi(0) = \xi_0$$

has a solution  $\xi(t), t \geq 0$  in the class of Itô processes. The solution is unique in the sense that if  $\eta(t), t \geq 0$  is another Itô process satisfying (233) and (234), then the two processes are identical a.s., that is,

$$(235) \quad P\{\xi(t) = \eta(t) \text{ for all } t \geq 0\} = 1.$$

PROOF (Outline). Let us fix  $T > 0$ . We are looking for a process  $\xi \in M_T^2$  such that

$$(236) \quad \xi(s) = \xi_0 + \int_0^s f(\xi(t)) dt + \int_0^s g(\xi(t)) dW(t) \quad \text{a.s.}$$

for all  $s \in [0, T]$ . Once we have shown that such a  $\xi \in M_T^2$  exists, to obtain a solution to the stochastic differential equation (233) with initial condition (234) it suffices to take a modification of  $\xi$  with a.s. continuous paths, which exists by Theorem 4.14.

To show that a solution to the stochastic integral equation (236) exists we shall employ the Banach fixed point theorem in  $M_T^2$  with the norm

$$(237) \quad \|\xi\|_\lambda^2 = \mathbb{E} \int_0^T e^{-\lambda t} |\xi(t)|^2 dt,$$

which turns  $M_T^2$  into a complete normed vector space. The number  $\lambda > 0$  should be chosen large enough, see below. To apply the fixed point theorem define  $\Phi : M_T^2 \rightarrow M_T^2$  by

$$(238) \quad \Phi(\xi)(s) = \xi_0 + \int_0^s f(\xi(t)) dt + \int_0^s g(\xi(t)) dW(t)$$

for any  $\xi \in M_T^2$  and  $s \in [0, T]$ . We claim that  $\Phi$  is a strict contraction, i.e.

$$(239) \quad \|\Phi(\xi) - \Phi(\zeta)\|_\lambda \leq \alpha \|\xi - \zeta\|_\lambda$$

for some  $\alpha < 1$  and all  $\xi, \zeta \in M_T^2$ . Then, by the Banach theorem,  $\Phi$  has a unique fixed point  $\xi = \Phi(\xi)$ . This is the desired solution to (236).

It remains to verify that  $\Phi$  is indeed a strict contraction. It suffices to show that the two maps  $\Phi_1$  and  $\Phi_2$ , where

$$(240) \quad \Phi_1(\xi)(s) = \int_0^s f(\xi(t)) dt, \quad \Phi_2(\xi)(s) = \int_0^s g(\xi(t)) dW(t),$$

are strict contractions with contracting constants  $\alpha_1$  and  $\alpha_2$  such that  $\alpha_1 + \alpha_2 < 1$ . For  $\Phi_1$  this follows from the Lipschitz continuity of  $f$ . For  $\Phi_2$  we need to use the Lipschitz continuity of  $g$  and the isometry property of the Itô integral. Let us mention just one essential step in the latter case. For any  $\xi, \zeta \in M_T^2$



$$(241) \quad \|\Phi_2(\xi) - \Phi_2(\zeta)\|_\lambda^2 = \mathbb{E} \int_0^T e^{-\lambda s} \left| \int_0^s [g(\xi(t)) - g(\zeta(t))] dW(t) \right|^2 ds$$

$$(242) \quad = \mathbb{E} \int_0^T e^{-\lambda s} \int_0^s |g(\xi(t)) - g(\zeta(t))|^2 dt ds$$

$$(243) \quad \leq C^2 \mathbb{E} \int_0^T e^{-\lambda s} \int_0^s |\xi(t) - \zeta(t)|^2 dt ds$$

$$(244) \quad = C^2 \mathbb{E} \int_0^T \left( \int_t^T e^{-\lambda s} e^{\lambda t} ds \right) e^{-\lambda t} |\xi(t) - \zeta(t)|^2 dt$$

$$(245) \quad \leq \frac{C^2}{\lambda} \mathbb{E} \int_0^T e^{-\lambda t} |\xi(t) - \zeta(t)|^2 dt = \frac{C^2}{\lambda} \|\xi - \zeta\|_\lambda^2,$$

since  $\int_t^T e^{-\lambda s} e^{\lambda t} ds = \frac{1}{\lambda}(1 - e^{-\lambda(T-t)}) \leq \frac{1}{\lambda}$ . Here  $C$  is the Lipschitz constant of  $g$ . If  $\lambda > C^2/\varepsilon$ , then  $\Phi_2$  is a strict contraction with contracting constant  $\leq \varepsilon$ .

There remain some technical points to be settled, but the main idea of the proof is shown above.  $\square$

EXERCISE 4.17. Find a solution of the stochastic differential equation

$$(246) \quad dX(t) = -\alpha X(t) dt + \sigma dW(t)$$

with initial condition  $X(0) = x_0$ , where  $x_0$  is an arbitrary real number. Show that the solution is unique.

REMARK 4.20. **Hint:** Use the substitution  $Y(t) = e^{\alpha t} X(t)$ .

SOLUTION 4.17. Let  $Y(t) = e^{\alpha t} X(t)$ . By the general Itô formula with  $F(t, x) = e^{\alpha t} x$ :

$$\begin{aligned} dY(t) &= (\alpha e^{\alpha t} X(t) + e^{\alpha t}(-\alpha X(t)) + 0) dt + e^{\alpha t} \sigma dW(t) \\ &= \sigma e^{\alpha t} dW(t). \end{aligned}$$

Integrating from 0 to  $t$ :

$$Y(t) - Y(0) = \sigma \int_0^t e^{\alpha s} dW(s).$$

Since  $Y(0) = e^0 X(0) = x_0$ , we have

$$e^{\alpha t} X(t) = x_0 + \sigma \int_0^t e^{\alpha s} dW(s),$$

which gives the solution

$$X(t) = x_0 e^{-\alpha t} + \sigma e^{-\alpha t} \int_0^t e^{\alpha s} dW(s).$$

Uniqueness follows from Theorem 4.19 because the coefficients  $f(x) = -\alpha x$  and  $g(x) = \sigma$  are Lipschitz continuous ( $|f(x) - f(y)| = \alpha|x - y|$  and  $|g(x) - g(y)| = 0$ ).

A linear stochastic differential equation has the general form

$$(247) \quad dX(t) = aX(t) dt + bX(t) dW(t),$$

where  $a$  and  $b$  are real numbers. In particular, for  $a = 0$  and  $b = 1$  we obtain the stochastic differential equation  $dX(t) = X(t) dW(t)$  in Example 4.6. The solution to the initial value problem for any linear stochastic differential equation can be found by exploiting the analogy with ordinary differential equations, as presented in the exercises below.

EXERCISE 4.18. Suppose that  $w(t), t \geq 0$  is a deterministic real-valued function of class  $C^1$  such that  $w(0) = 0$ . Solve the ordinary differential equation

$$(248) \quad dx(t) = ax(t) dt + bx(t) dw(t),$$

with initial condition  $x(0) = x_0$  to find that

$$(249) \quad x(t) = x_0 e^{at+bw(t)}.$$

(We write  $dw(t)$  in place of  $w'(t) dt$  to emphasize the analogy with stochastic differential equations.)

REMARK 4.21. **Hint:** The variables can be separated:

$$(250) \quad \frac{dx(t)}{x(t)} = (a + bw'(t)) dt.$$

SOLUTION 4.18. The ordinary differential equation can be written as

$$\frac{dx(t)}{dt} = (a + bw'(t))x(t).$$

Separating variables, we have

$$\int \frac{dx}{x} = \int (a + bw'(t)) dt.$$

Integrating from 0 to  $t$ :

$$\ln |x(t)| - \ln |x(0)| = at + b(w(t) - w(0)).$$

Given  $w(0) = 0$  and  $x(0) = x_0$ , this simplifies to

$$\ln \left| \frac{x(t)}{x_0} \right| = at + bw(t),$$

which implies

$$x(t) = x_0 e^{at+bw(t)}.$$

By analogy with the deterministic solution (249), let us consider a process defined by

$$(251) \quad X(t) = X_0 e^{at+bW(t)}$$

for any  $t \geq 0$ , where  $W(t)$  is a Wiener process.

EXERCISE 4.19. Show that  $X(t)$  defined by (251) is a solution of the linear stochastic differential equation

$$(252) \quad dX(t) = \left( a + \frac{b^2}{2} \right) X(t) dt + bX(t) dW(t),$$

with initial condition  $X(0) = X_0$ .

REMARK 4.22. **Hint:** Use the Itô formula with  $F(t, x) = e^{at+bx}$ .

SOLUTION 4.19. Let  $F(t, x) = X_0 e^{at+bx}$ . The partial derivatives are

$$\begin{aligned} F'_t &= aX_0 e^{at+bx} = aF, \\ F'_x &= bX_0 e^{at+bx} = bF, \\ F''_{xx} &= b^2 X_0 e^{at+bx} = b^2 F. \end{aligned}$$

Applying the simplified Itô formula (172) for  $X(t) = F(t, W(t))$ :

$$\begin{aligned} dX(t) &= \left( F'_t + \frac{1}{2} F''_{xx} \right) dt + F'_x dW(t) \\ &= \left( aX(t) + \frac{1}{2} b^2 X(t) \right) dt + bX(t) dW(t) \\ &= \left( a + \frac{b^2}{2} \right) X(t) dt + bX(t) dW(t). \end{aligned}$$

The initial condition is  $X(0) = X_0 e^{a \cdot 0 + bW(0)} = X_0$ . This verifies that  $X(t)$  is a solution to the given SDE.

EXERCISE 4.20. Show that the linear stochastic differential equation

$$(253) \quad dX(t) = aX(t) dt + bX(t) dW(t)$$

with initial condition  $X(0) = X_0$  has a unique solution given by

$$(254) \quad X(t) = X_0 e^{\left(a - \frac{b^2}{2}\right)t + bW(t)}.$$

REMARK 4.23. **Hint:** Apply the result of Exercise 4.19 with suitably redefined constants.

SOLUTION 4.20. We want to find a solution for  $dX(t) = aX(t) dt + bX(t) dW(t)$ . According to Exercise 4.19, the process  $X(t) = X_0 e^{\tilde{a}t + bW(t)}$  satisfies

$$dX(t) = \left( \tilde{a} + \frac{b^2}{2} \right) X(t) dt + bX(t) dW(t).$$

To match the given equation, we set  $a = \tilde{a} + \frac{b^2}{2}$ , which gives  $\tilde{a} = a - \frac{b^2}{2}$ . Substituting this  $\tilde{a}$  into the expression for  $X(t)$ :

$$X(t) = X_0 e^{\left(a - \frac{b^2}{2}\right)t + bW(t)}.$$

Uniqueness follows from Theorem 4.19 because the coefficients  $f(x) = ax$  and  $g(x) = bx$  are Lipschitz continuous ( $|ax - ay| = |a||x - y|$  and  $|bx - by| = |b||x - y|$ ).

Having solved the general linear stochastic differential equation (247), let us consider an example of a non-linear stochastic differential equation. Once again, we begin with a deterministic problem.

EXERCISE 4.21. Suppose that  $w(t), t \geq 0$  is a deterministic real-valued function of class  $C^1$  such that  $w(0) = 0$ . Solve the ordinary differential equation

$$(255) \quad dx(t) = \sqrt{1 + x(t)^2} dt + \sqrt{1 + x(t)^2} dw(t)$$

with initial condition  $x(0) = x_0$ .

REMARK 4.24. **Hint:** The variables in this differential equation can be separated.

SOLUTION 4.21. The ordinary differential equation can be written as

$$\frac{dx(t)}{dt} = \sqrt{1 + x(t)^2} (1 + w'(t)).$$

Separating variables, we have

$$\int \frac{dx}{\sqrt{1 + x^2}} = \int (1 + w'(t)) dt.$$

Integrating from 0 to  $t$ :

$$[\sinh^{-1} z]_{x_0}^{x(t)} = [s + w(s)]_0^t.$$

Since  $w(0) = 0$ , this gives

$$\sinh^{-1} x(t) - \sinh^{-1} x_0 = t + w(t).$$

Let  $C = \sinh^{-1} x_0$ . Then  $\sinh^{-1} x(t) = C + t + w(t)$ , which implies

$$x(t) = \sinh(C + t + w(t)).$$

EXERCISE 4.22. Show that the process defined by

$$(256) \quad X(t) = \sinh(C + t + W(t)),$$

where  $W(t)$  is a Wiener process and  $C = \sinh^{-1} X_0$ , is a solution of the stochastic differential equation

$$(257) \quad dX(t) = \left( \sqrt{1 + X(t)^2} + \frac{1}{2} X(t) \right) dt + \left( \sqrt{1 + X(t)^2} \right) dW(t)$$

with initial condition  $X(0) = X_0$ .

REMARK 4.25. **Hint:** Use the Itô formula with  $F(t, x) = \sinh(t + x)$ .

SOLUTION 4.22. Let  $F(t, x) = \sinh(C + t + x)$ . The partial derivatives are

$$F'_t(t, x) = \cosh(C + t + x),$$

$$F'_x(t, x) = \cosh(C + t + x),$$

$$F''_{xx}(t, x) = \sinh(C + t + x) = F(t, x).$$

Applying the simplified Itô formula (172) to  $X(t) = F(t, W(t))$ :

$$\begin{aligned} dX(t) &= \left( F'_t + \frac{1}{2} F''_{xx} \right) dt + F'_x dW(t) \\ &= \left( \cosh(C + t + W(t)) + \frac{1}{2} \sinh(C + t + W(t)) \right) dt + \cosh(C + t + W(t)) dW(t). \end{aligned}$$

Since  $X(t) = \sinh(C + t + W(t))$ , we have  $\sqrt{1 + X(t)^2} = \sqrt{1 + \sinh^2(C + t + W(t))} = \cosh(C + t + W(t))$ . Substituting these into the expression for  $dX(t)$ , we obtain

$$dX(t) = \left( \sqrt{1 + X(t)^2} + \frac{1}{2}X(t) \right) dt + \sqrt{1 + X(t)^2} dW(t).$$

The initial condition is  $X(0) = \sinh(C + 0 + W(0)) = \sinh(C) = \sinh(\sinh^{-1} X_0) = X_0$ . This verifies that  $X(t)$  is a solution to the given SDE.

We shall conclude this chapter with an example of a stochastic differential equation which does not satisfy the assumptions of Theorem 4.19. It turns out that the solution may fail to exist for all times  $t \geq 0$ . This is a familiar phenomenon in ordinary differential equations. However, stochastic differential equations add a new effect, which does not even make sense in the deterministic case: the maximum time of existence of the solution, called the **explosion time** may be a (non-constant) random variable, in fact a stopping time.

EXAMPLE 4.8. Consider the stochastic differential equation

$$(258) \quad dX(t) = X(t)^3 dt + X(t)^2 dW(t).$$

Then

$$(259) \quad X(t) = \frac{1}{1 - W(t)}$$

is a solution, which can be verified, at least formally, by using the Itô formula with  $F(t, x) = \frac{1}{1-x}$ . The solution  $X(t)$  exists only up to the first hitting time

$$(260) \quad \tau = \inf\{t \geq 0 : W(t) = 1\}.$$

This is the explosion time of  $X(t)$ . Observe that

$$(261) \quad \lim_{t \nearrow \tau} X(t) = \infty.$$

Strictly speaking, the Itô formula stated in Theorem 4.17 does not cover this case, since  $F(t, x) = \frac{1}{1-x}$  has a singularity at  $x = 1$ . Definition 4.18 does not apply either, as it requires the solution  $X(t)$  to be defined for all  $t \geq 0$ . Suitable extensions of the Itô formula and the definition of a solution are required to study stochastic differential equations involving explosions. However, to prevent an explosion of this book, we have to refer the interested reader to a further course in stochastic analysis.